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(54) **GENETICALLY MODIFIED YEAST AND
FERMENTATION PROCESSES FOR THE
PRODUCTION OF XYLITOL**

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(57)

ABSTRACT

Disclosed herein are genetically engineered yeast cells capable of producing xylitol. The engineered yeast cell may comprise an exogenous polynucleotide sequence encoding a xylitol-phosphate dehydrogenase (XPDH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs: 12-15, 28-31, or 33.

Specification includes a Sequence Listing.

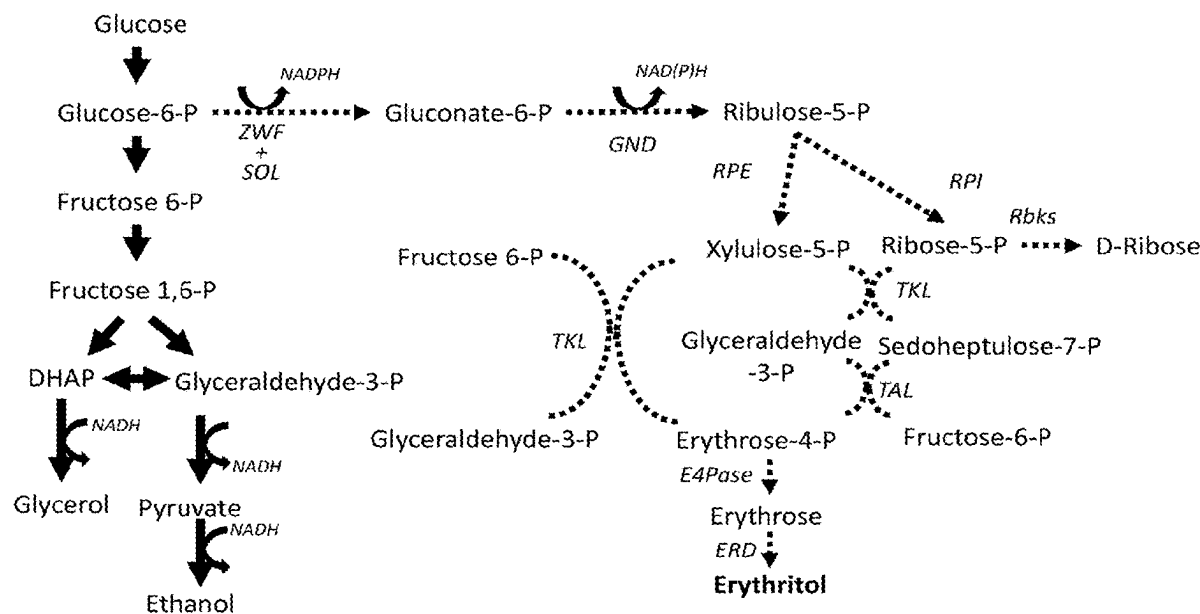


FIG. 1

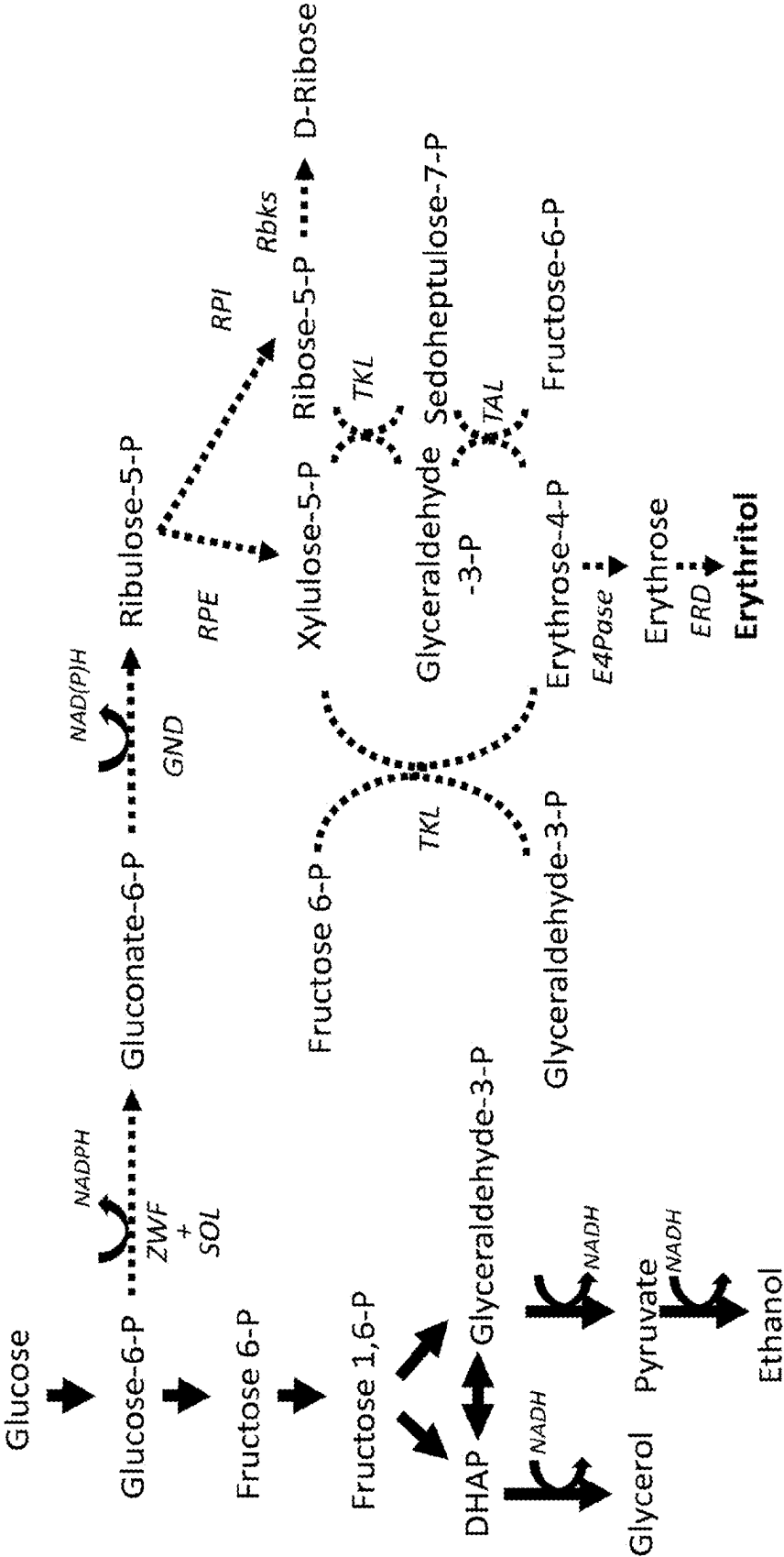


FIG. 2

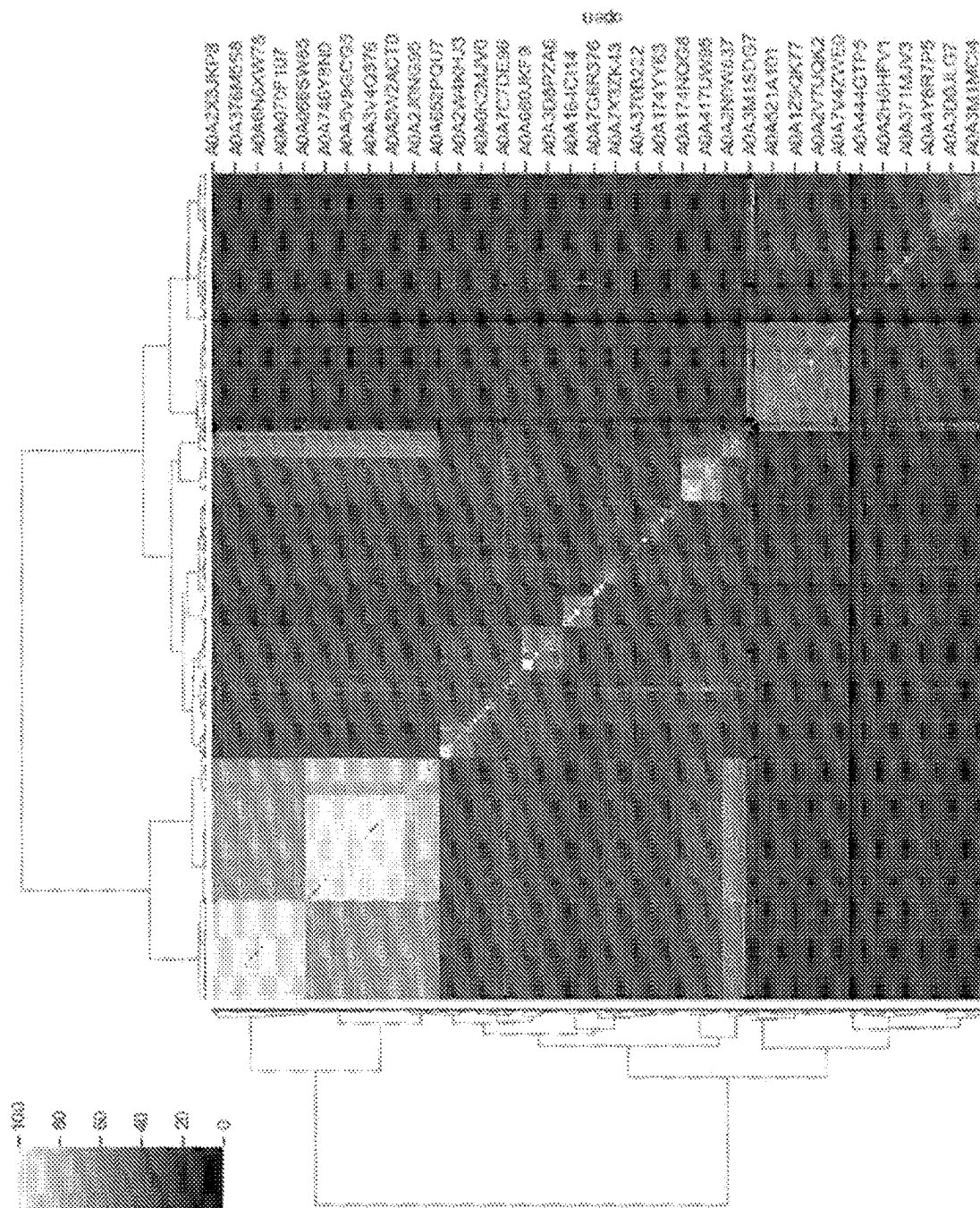


FIG. 3

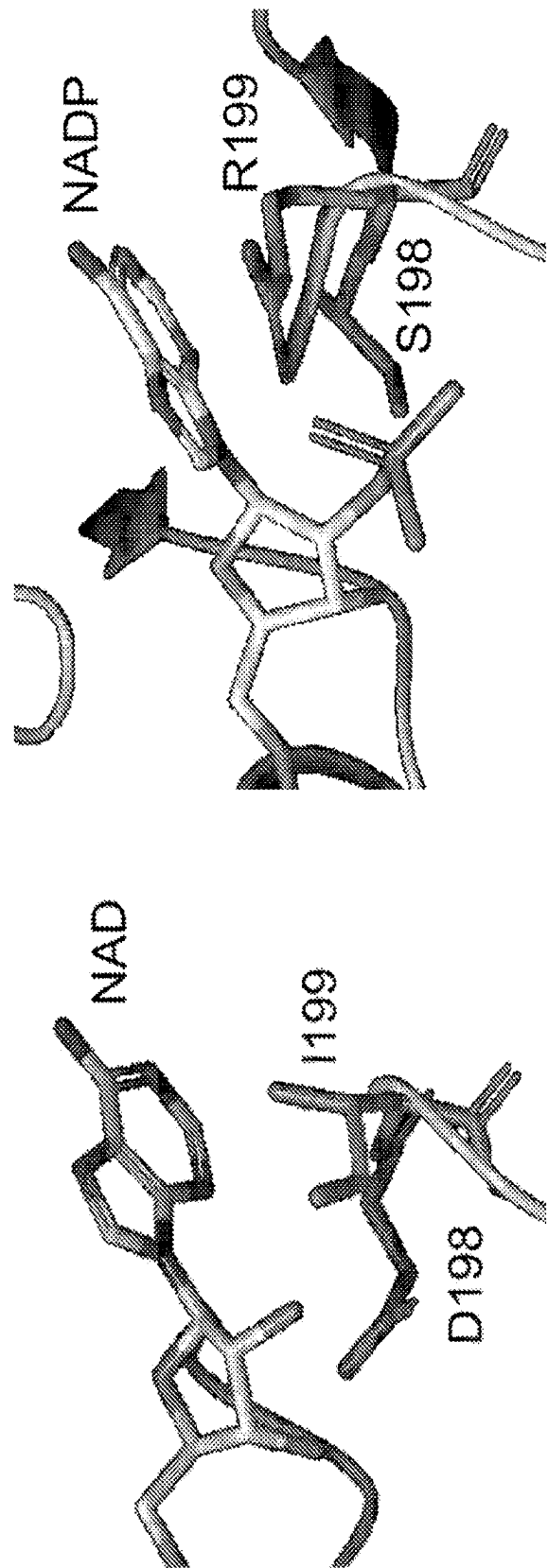


FIG. 4

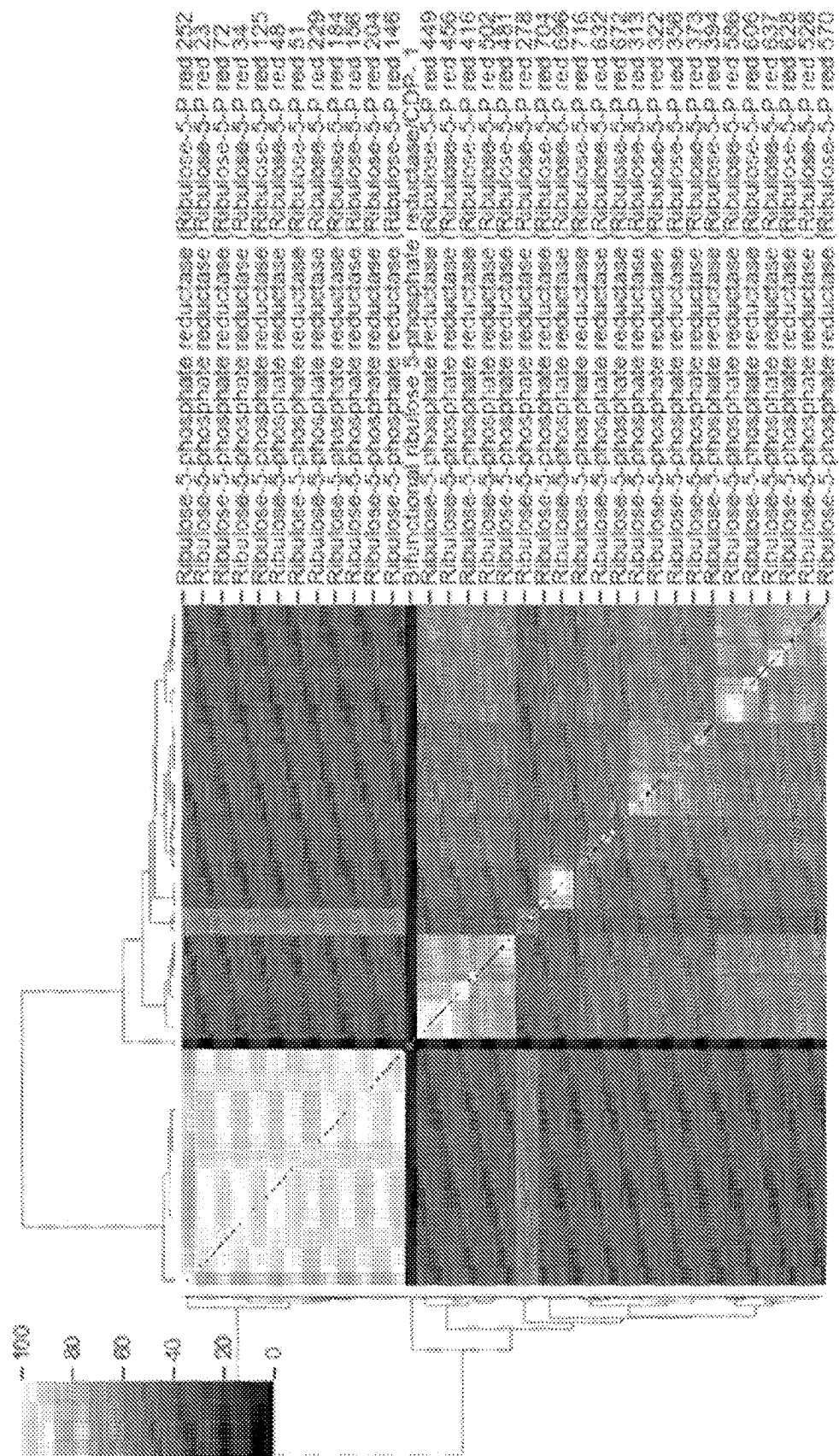


FIG. 5

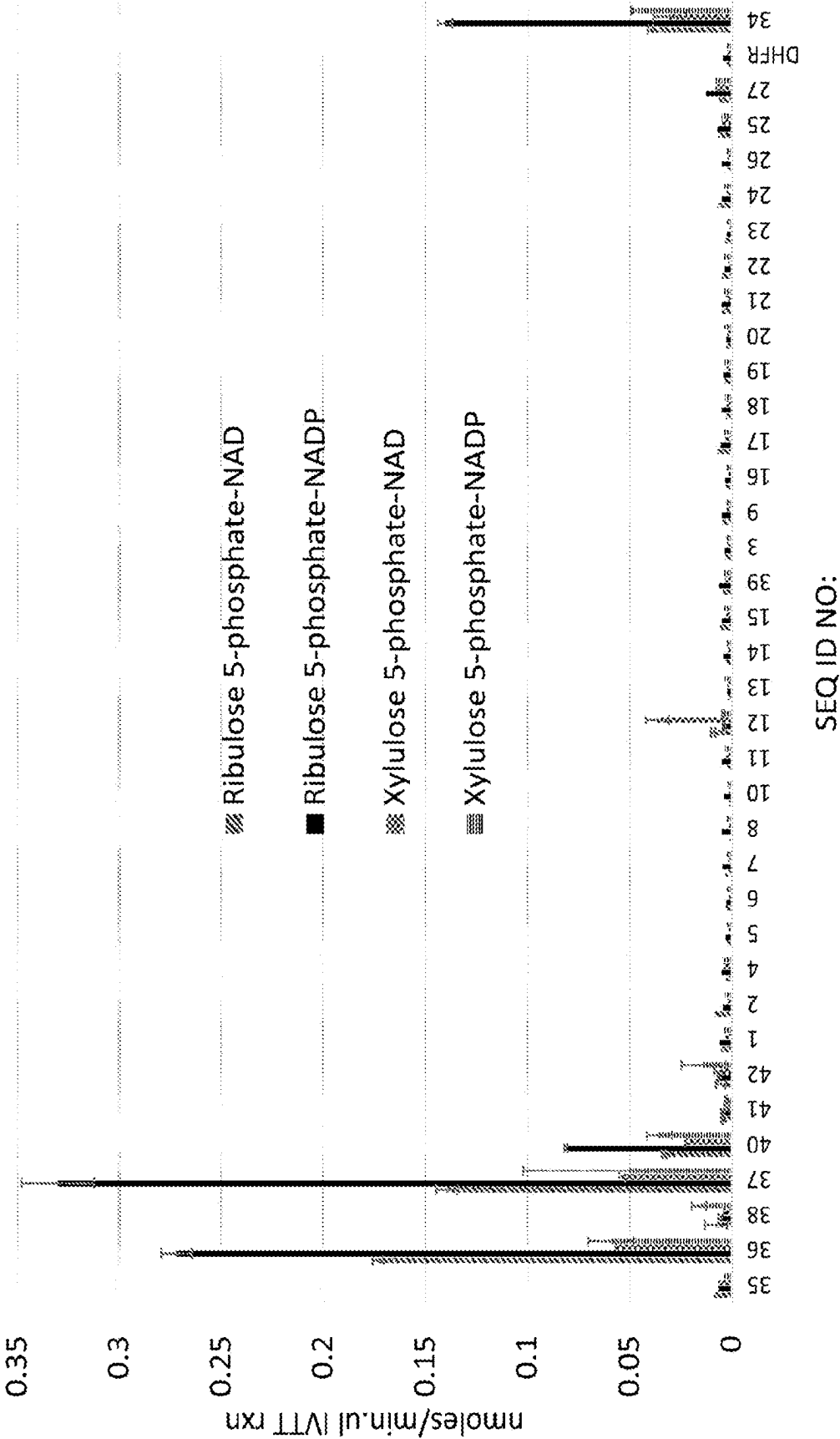


FIG. 6

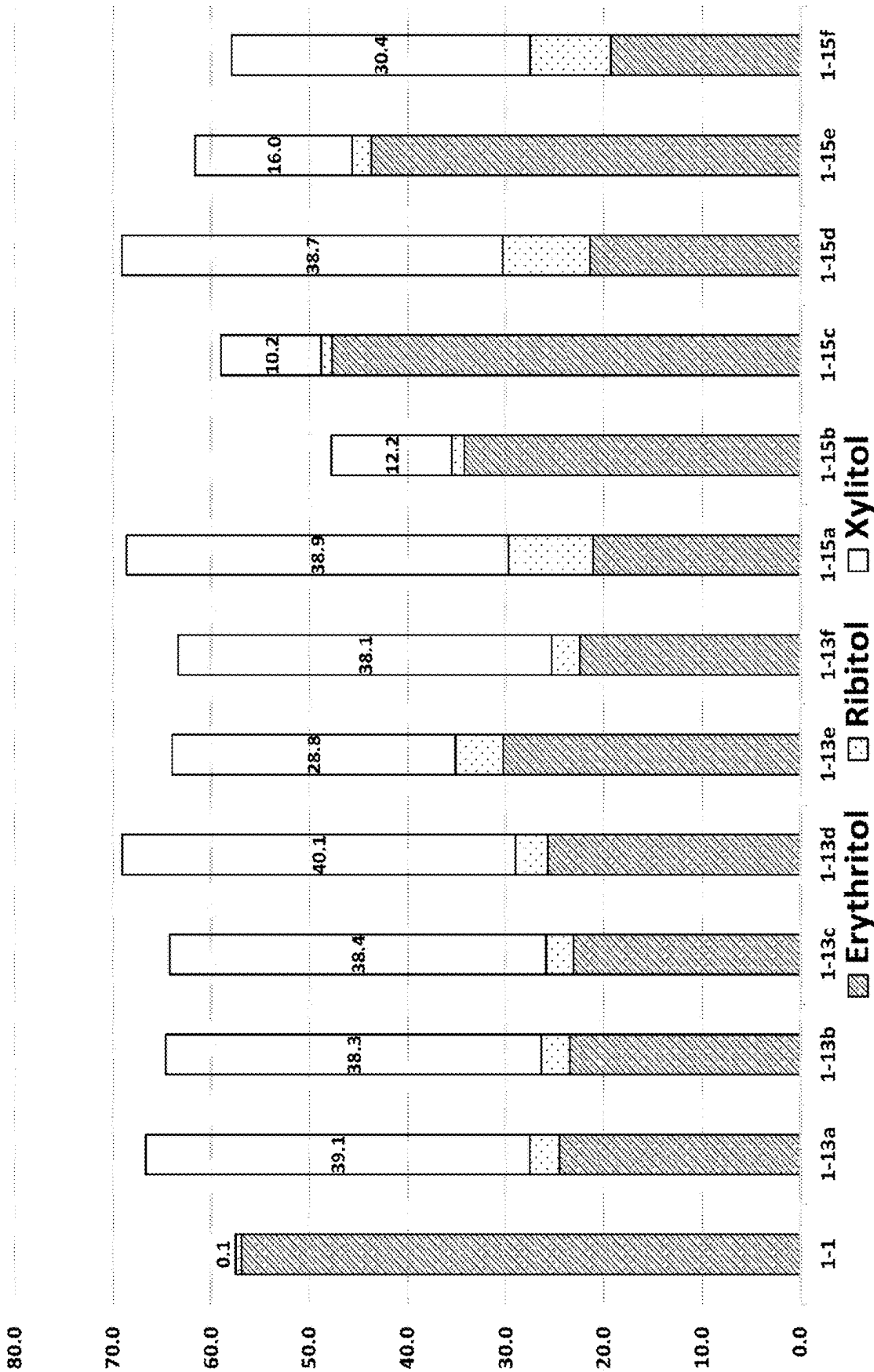


FIG. 7

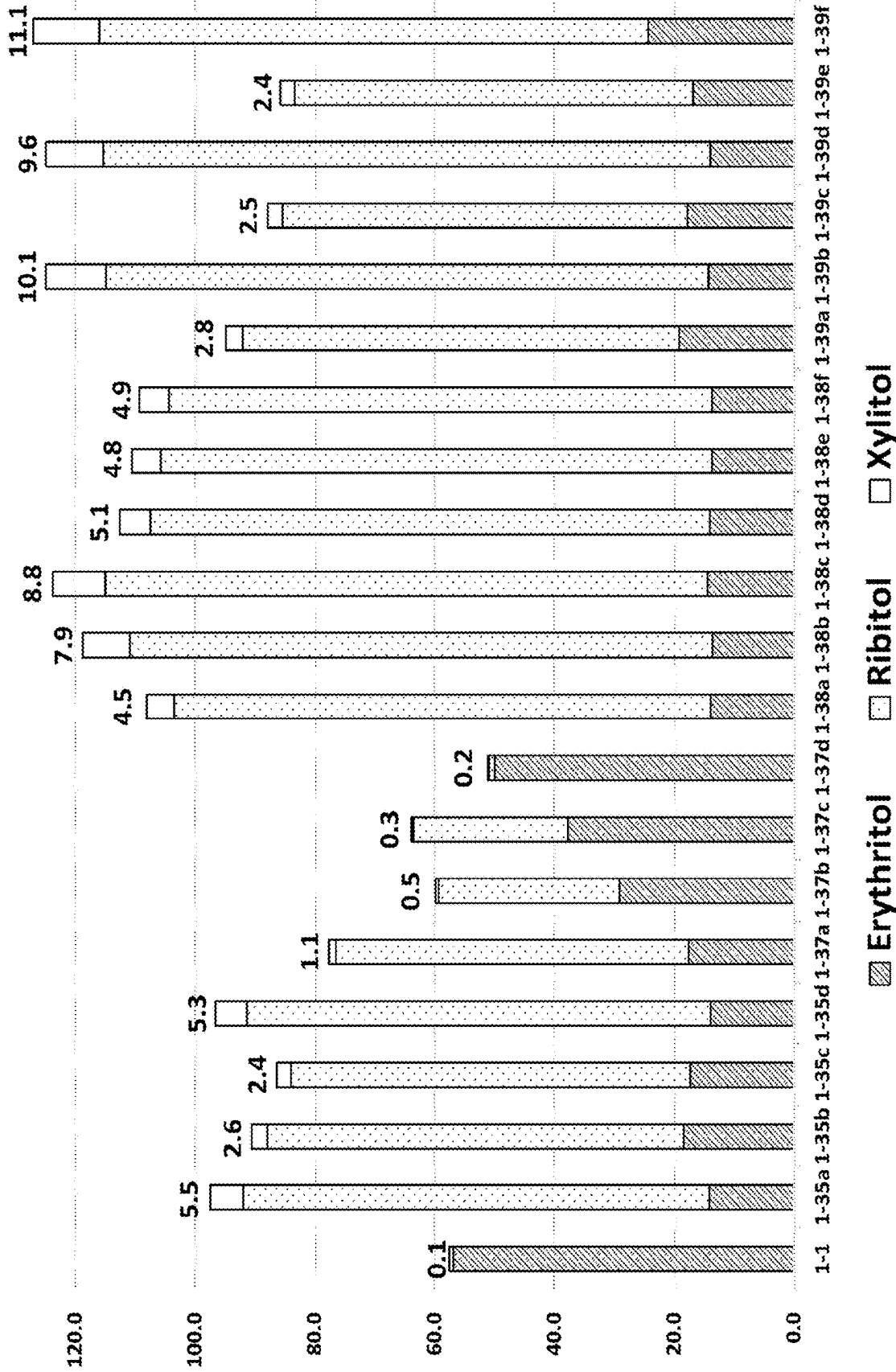


FIG. 8

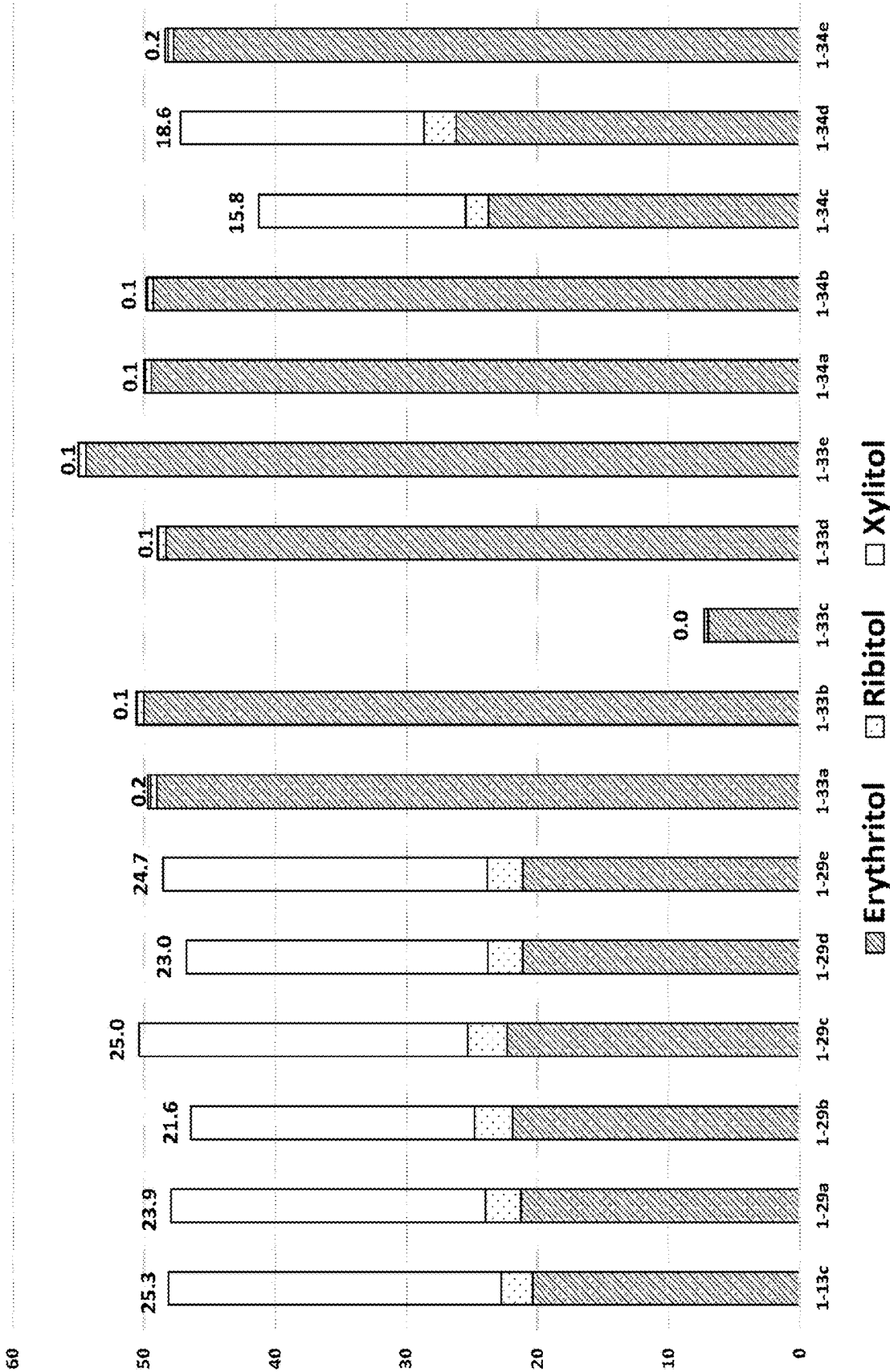


FIG. 9

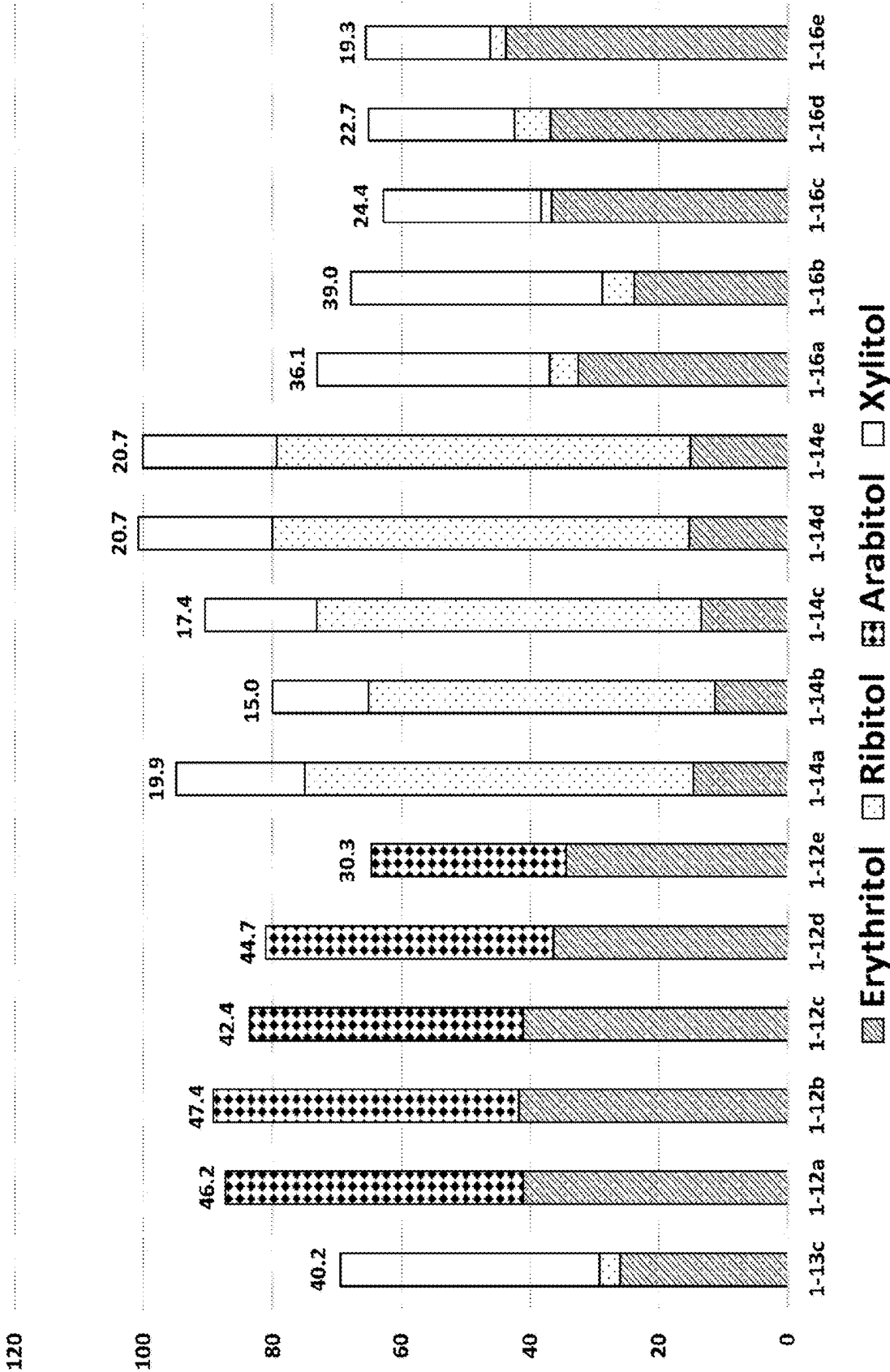


FIG. 10

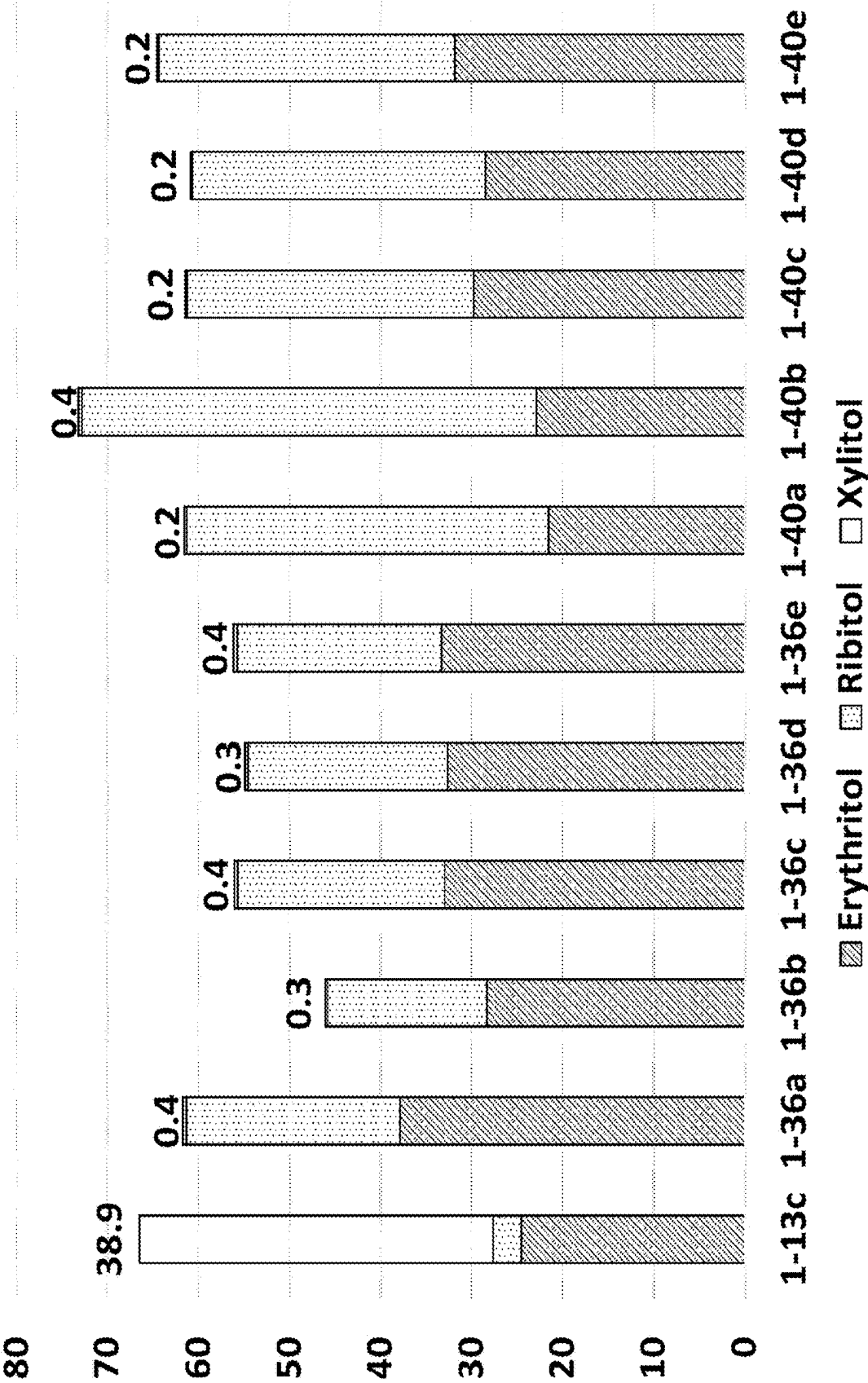
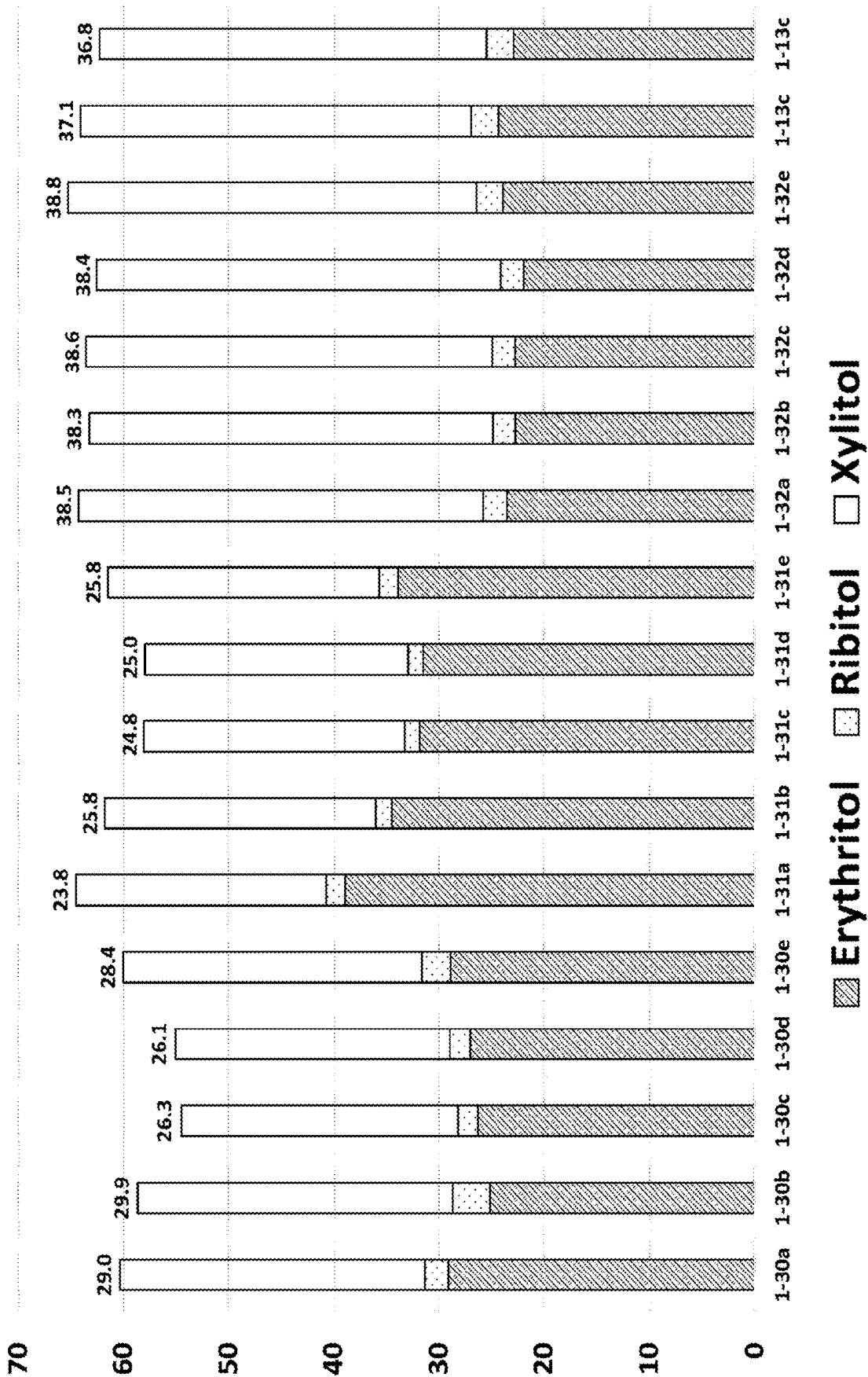


FIG. 11



GENETICALLY MODIFIED YEAST AND FERMENTATION PROCESSES FOR THE PRODUCTION OF XYLITOL

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/364,375, filed May 9, 2022, which is incorporated by reference herein in its entirety.

REFERENCE TO A SEQUENCE LISTING SUBMITTED VIA PATENT CENTER

[0002] The content of the Sequence Listing XML file of the sequence listing named "PT-1349-WO-PCT.xml" which is 448,927 bytes in size created on May 4, 2023 and electronically submitted via Patent Center herewith the application is incorporated by reference in its entirety.

BACKGROUND

[0003] Xylitol is a low-calorie sweetener used as a food additive and sugar substitute. Commonly used in drug, dietary supplement, confectionary, and toothpaste compositions, xylitol has also been associated with anticariogenic properties when used in chewing gums. Traditional methods of xylitol production, including chemically catalyzed hydrogenation of xylose hydrolyzed from biomass extracted xylan, are both monetarily and environmentally costly. These methods require high temperatures and pressures, large amounts of water, and metal catalysts that must be mined. In contrast, fermentation processes have been used commercially at large scale to produce other organic molecules, such as ethanol, citric acid, lactic acid, and the like, and may offer a cost effective and sustainable alternative to traditional xylitol processing methods.

[0004] Accordingly, provided herein are genetically modified yeast and fermentation methods for the production of xylitol while reducing production of erythritol.

SUMMARY

[0005] The present disclosure provides a genetically engineered yeast cell capable of producing xylitol, the engineered yeast cell comprising an exogenous polynucleotide sequence encoding a xylitol-phosphate dehydrogenase (XPDH) enzyme. The XPDH enzyme may comprise a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs: 12-15, 28-31, and 33. The yeast cell may be an osmotolerant yeast cell. The yeast cell may be a cell of the subphylum Ustilaginomycotina. The yeast cell may be selected from the group consisting of *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*, *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Moniliella*, *Ustilaginomycetes*, *Trichosporon*, *Yarrowia lipolytica*, *Penicillium*, *Torula*, *Pichia*, *Candida*, *Candida magnoliae*, and *Aureobasidium*. The yeast cell may be a yeast cell of the genus.

[0006] The disclosure also provides a genetically engineered *Moniliella* cell capable of producing xylitol, the engineered *Moniliella* cell comprising an exogenous polynucleotide sequence encoding a xylitol-phosphate dehydrogenase (XPDH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%,

at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs: 12-15, 28-31, and 33.

[0007] The XPDH enzyme may have a sequence at least 85% identical to at least one of SEQ ID NOs: 12-15, 28-31, and 33 or to at least one of SEQ ID NOs: 14, 15, 28, or 31. The XPDH enzyme may have a sequence at least 90% identical to at least one of SEQ ID NOs: 12-15, 28-31, and 33 or to at least one of SEQ ID NOs: 14, 15, 28, or 31.

[0008] The engineered cell described herein may be a *Moniliella pollinis* cell. The yeast cell may be capable of producing xylitol at a titer of at least 20, 30, 50, 75, or 100 g/L when used in a fermentation process in the presence of dextrose at 35° for 96 hours. Erythritol production by the yeast cell may be reduced relative to erythritol production in an equivalent yeast cell lacking the exogenous polynucleotide sequence. The exogenous polynucleotide sequence may be integrated into the genome of the yeast cell at a loci selected from the ER1 locus, the ER3 locus, the PDC1 locus, the pyrF locus, the TRP3 locus, the gpdIIA locus, and the gpdIIB locus. The exogenous polynucleotide sequence may be operably linked to a heterologous or artificial promoter. The promoter may be a constitutive promoter. The promoter may be selected from the group consisting of pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:86), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:130), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:132), translational elongation factor 1 promoter (TEFp; SEQ ID NO:133), modified TEFp (SEQ ID NO:131), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:134), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:135), enolase promoter (ENO1p; SEQ ID NO:136), asparagine synthetase promoter (ASNSp; SEQ ID NO:137), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:138), and RPL16B (SEQ ID NO:139).

[0009] The disclosure provides a method for producing xylitol, the method comprising contacting a substrate comprising dextrose with an engineered yeast cell described herein, wherein fermentation of the substrate by the engineered yeast produces xylitol. The disclosure also provides a method for producing xylitol, the method comprising contacting a substrate comprising dextrose with an engineered yeast cell comprising an exogenous polynucleotide sequence encoding a xylitol-phosphate dehydrogenase (XPDH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs: 12-15, 28-31, and 33, wherein fermentation of the substrate by the engineered yeast produces xylitol. The fermentation temperature may be at or between 25° C. to 45° C., 30° C. to 40° C., or 32° C. to 37° C. The volumetric oxygen uptake rate (OUR) may be between 0.5 to 40, 1 to 35, 2 to 30, 3 to 25, 4 to 20, or 5 to 15 mmol O₂/(L·h). Xylitol may be produced at a rate of at least 0.1, 0.2, 0.3, 0.4, 0.5, 0.75, or at least 1.0 g L⁻¹ h⁻¹. Xylitol production may be at least at least 10, 20, 30, 40, 50, 75, or 100 g/L when the fermentation is run at 35° C. for 96 hours. Erythritol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence. Erythritol production may be less than 50, 40, 30, or less than 20 g/L when the fermentation is run at 35° C. for 96 hours. Glycerol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell

lacking the exogenous polynucleotide sequence. Ethanol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

BRIEF DESCRIPTION OF THE FIGURES

[0010] This patent or application contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawings will be provided by the Office upon request and the payment of the necessary fee.

[0011] The drawings illustrate generally, by way of example, but not by way of limitation, various aspects discussed herein.

[0012] FIG. 1 shows the native pentose phosphate pathway (dotted lines and arrows) and the native glycolysis pathways (solid lines and arrows) in *Moniliella pollinis*.

[0013] FIG. 2 shows diversity in the galactitol-1-phosphate-5-dehydrogenase (G1PDH)/xylitol-phosphate dehydrogenase (XPDH) sequence space.

[0014] FIG. 3 shows the structural characteristics of the NAD or NADP binding pocket located +23 amino acids from the characteristic GXGXXG motif (SEQ ID NO:133) of XPDH enzymes.

[0015] FIG. 4 shows diversity in the ribulose-5-phosphate reductase sequence space.

[0016] FIG. 5 shows in vitro activity of TarJ and XPDH enzymes as outlined in Example 3.

[0017] FIG. 6 shows erythritol, ribitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-1, 1-13a-f, and 1-15a-f as outlined in Example 5. Data labels report the concentration (g/L) of xylitol.

[0018] FIG. 7 shows erythritol, ribitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-1, 1-35a-d, 1-37a-d, 1-38a-f, and 1-39a-f as outlined in Example 5. Data labels report the concentration (g/L) of xylitol.

[0019] FIG. 8 shows erythritol, ribitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-13c, 1-29a-e, 1-33a-e, and 1-34a-e as outlined in Example 6. Data labels report the concentration (g/L) of xylitol.

[0020] FIG. 9 shows erythritol, ribitol, arabitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-13c, 1-12a-e, 1-14a-e, and 1-16a-e as outlined in Example 8. Data labels report the concentration (g/L) of xylitol (strains 1-13c, 1-14a-e, and 1-16a-e) or arabitol (strains 1-12a-e).

[0021] FIG. 10 shows erythritol, ribitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-13c, 1-36a-e, and 1-40a-e as outlined in Example 9. Data labels report the concentration (g/L) of xylitol.

[0022] FIG. 11 shows erythritol, ribitol, and xylitol metabolite concentrations (g/L) at 96 hours of shake flask fermentations of strains 1-30a-e, 1-31a-e, 1-32a-e, and 1-13c as outlined in Example 10. Data labels report the concentration (g/L) of xylitol.

DETAILED DESCRIPTION

[0023] Reference will now be made in detail to certain aspects of the disclosed subject matter, examples of which

are illustrated in part in the accompanying drawings. While the disclosed subject matter will be described in conjunction with the enumerated claims, it will be understood that the exemplified subject matter is not intended to limit the claims to the disclosed subject matter.

[0024] In this document, the terms “a,” “an,” or “the” are used to include one or more than one unless the context clearly dictates otherwise. The term “or” is used to refer to a nonexclusive “or” unless otherwise indicated. All publications, patents, and patent documents referred to in this document are incorporated by reference herein in their entirety, as though individually incorporated by reference. In the event of inconsistent usages between this document and those documents so incorporated by reference, the usage in the incorporated reference should be considered supplementary to that of this document; for irreconcilable inconsistencies, the usage in this document controls.

[0025] Values expressed in a range format should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range were explicitly recited. For example, a range of “about 0.1% to about 5%” or “about 0.1% to 5%” should be interpreted to include not just about 0.1% to about 5%, but also the individual values (e.g., 1%, 2%, 3%, and 4%) and the sub-ranges (e.g., 0.1% to 0.5%, 1.1% to 2.2%, 3.3% to 4.4%) within the indicated range. The statement “about X to Y” has the same meaning as “about X to about Y,” unless indicated otherwise. Likewise, the statement “about X, Y, or about Z” has the same meaning as “about X, about Y, or about Z,” unless indicated otherwise.

[0026] Unless expressly stated, ppm (parts per million), percentage, and ratios are on a by weight basis. Percentage on a by weight basis is also referred to as wt % or % (wt) below.

[0027] This disclosure relates to various recombinant cells engineered to produce xylitol. In general, the recombinant cells described herein have an active pentose phosphate pathway and are characterized by expression of an exogenous xylitol-phosphate dehydrogenase (XPDH) enzyme. The disclosure further provides fermentation methods for the production of xylitol from dextrose using the genetically engineered cells described herein.

[0028] In general, recombinant cells described herein are yeast cells. As used herein, “yeast” refers to eukaryotic single celled microorganisms classified as members of the fungus kingdom. Yeast are unicellular organisms which evolved from multicellular ancestors with some species retaining multicellular characteristics such as forming strings of connected budding cells known as pseudo hyphae or false hyphae. Yeast cells may also be referred to in the art as yeast-like cells, and as used herein “yeast cell” encompasses both yeast and yeast-like cells. Suitable yeast and yeast-like host cells for modification may include, but are not limited to, *Saccharomyces cerevisiae*, *Komagataella* sp., *Kluyveromyces* (e.g., *Kluyveromyces lactis*, *Kluyveromyces marxianus*), *Yarrowia lipolytica*, *Issatchenkia orientalis*, *Pichia galeiformis*, *Pichia* sp. YB-4149 (NRRL designation), *Pichia pastoris*, *Candida* (e.g., *Candida magnoliae*, *Candida ethanolica*), *Pichia deserticola*, *Pichia membranifadens*, *Pichia fermentans*, *Aspergillus*, *Trichoderma*, *Myceliophthora thermophila*, *Moniliella* (e.g., *Moniliella pollinis*), *Pfaffia*, *Yamadazyma*, *Hansenula*, *Pichia kudriavzevii*,

Trichosporonoides (e.g., *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*), *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Penicillium*, and *Torula*. An ordinarily skilled artisan would understand the requirements for selection of a suitable yeast cell, and recombinant yeast cells of the present disclosure are not limited to those expressly recited herein. Methods for genetic engineering of yeast cells are known and described in the art and a skilled artisan would understand the methods necessary to transform and engineer a suitable yeast cell.

[0029] A suitable yeast cell may be a cell of the phylum Basidiomycota and the subphylum Ustilaginomycotina. Suitable yeast of the subphylum Ustilaginomycotina include, but are not limited to, *Ustilago* (e.g., *U. cynodontis*, *U. maydis*, *U. sphaerogena*, *U. cordal*, *U. scitaminea*, *U. coicis*, *U. syntherismae*, *U. esculenta*, *U. neglecta*, *U. crus-galli*, *Ustilago avenae*), *Sporisorium* (e.g., *Sporisorium exsertum*), *Moniliella* (e.g., *M. pollinis*, *M. tomentosa*, *M. acetoabutans*, *M. fonsecae*, *M. madida*, *M. megachiliensis*, *M. oedocephalis*, *M. nigrescens*), and *Pseudozyma* (e.g., *Pseudozyma tsukubaensis*), and *Trichosporonoides* (e.g., *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*). Yeast of the subphylum Ustilaginomycotina have been known and described in the art as potential production organisms for valuable chemicals such as itaconate, malate, succinate, mannitol, and erythritol and other valuable biotechnological applications. See, for example, Geiser et al. (Prospecting the biodiversity of the fungal family Ustilaginaceae for the production of value-added chemicals,” *Fungal Biol Biotechnol*, 2014, 1:2), Feldbrugge et al., (“The biotechnological use and potential of plant pathogenic smut fungi,” *Appl Microbiol Biotechnol*, 2013, 97 (8): 3253-65), Guevarra et al., (“Accumulation of itaconic, 2-hydroxyparaconic, itatartaric, and malic acids by strains of the genus *Ustilago*, *Agric. Biol. Chem.*, 1990, 54 (9), 2353-2358), and Moon et al., (“Biotechnological production of erythritol and its applications,” *Appl Microbiol Biotechnol*, 2010, 86:1017-1025).

[0030] A suitable yeast cell will have an active pentose phosphate pathway that produces ribulose-5-phosphate. As used herein “active pentose phosphate pathway” refers to expression of one or more functional enzymes which, together, convert glucose-6-phosphate, NADP⁺ or NAD⁺, and water to NADPH or NADH, CO₂, and ribulose-5-phosphate. Continuing in a non-oxidative phase, the pathway may also produce other pentose (i.e., 5-carbon) sugars. For example, the pentose phosphate pathway may produce ribulose-5-phosphate, ribose-5-phosphate, xylulose-5-phosphate, fructose 6-phosphate, combinations thereof, and the like, depending on the enzymatic activities present. The active pentose phosphate pathway may be native to the yeast cell or it may be introduced into the yeast cell by genetic engineering.

[0031] The yeast cell may be an osmotolerant yeast cell. As used herein, “osmotolerant” refers to a yeast capable of growth and reproduction under conditions of high osmolarity, such as at least 10% (w/v), at least 20% (w/v), at least 30% (w/v), at least 40% (w/v), at least 50% (w/v), or at least 60% (w/v) glucose and/or at least 6% (w/v), at least 10% (w/v), at least 12% (w/v), at least 13% (w/v), at least 15% (w/v) sodium chloride. Species and strains of osmotolerant yeast are known and described in the art, including many species of yeast used in industrial fermentation processes. Likewise, methods for assaying yeast osmotolerance are

known and described in the art. See, for example, Tiwari, S., et al., (“Nectar yeast community of tropical flowering plants and assessment of their osmotolerance and xylitol-producing potential,” *Current Microbiology*, 2022, 79:28).

[0032] The recombinant yeast cell may be a recombinant *Moniliella* cell, for example, a *Moniliella pollinis* cell. FIG. 1 shows the predicted native pentose phosphate and glycolysis pathways in *Moniliella pollinis*. *Moniliella* has previously been used in the fermentation production of erythritol and methods for genetically modifying and fermenting *Moniliella* are known and described in the art. See, for example, Li et al. (“Methods for genetic transformation of filamentous fungi,” 2017, *Microb Cell Fact*, 16:168).

[0033] Various plasmids and methods for transformation of *Moniliella* are also described in the Examples below. For example, *Moniliella* may be transformed using a bipartite polynucleotide sequence(s) in which, following recombination, the exogenous polynucleotide of interest is integrated at the specified locus and the selection marker is expressible within the cell. Suitable selection markers are known and used in the art. The selectable marker may include, but is not limited to, *amdS* (for example broken into a 3' portion, SEQ ID NO: 167, and a 5' portion, SEQ ID NO: 174), G418 resistance gene (for example broken into a 3' portion, SEQ ID NO: 172, and a 5' portion, SEQ ID NO: 175), zeocin resistance gene (for example broken into a 3' portion, SEQ ID NO: 168, and a 5' portion, SEQ ID NO: 169), nourseothricin N-acetyl transferase (NAT) (for example broken into a 3' portion, SEQ ID NO: 171, and a 5' portion, SEQ ID NO: 170), and invertase gene (SUC2) (for example a 3' portion of SEQ ID NO: 173 and a 5' portion of SEQ ID NO: 176).

[0034] The recombinant cells described herein include one or more exogenous polynucleotide sequences encoding one or more exogenous polypeptides that, when expressed improve the fermentation of glucose to ribitol by the recombinant cells.

[0035] The terms “glucose” and “dextrose” are used interchangeably herein and refer to D-glucose except where expressly indicated otherwise.

[0036] As used herein, “exogenous” refers to genetic material or an expression product thereof that originates from outside of the host organism. For example, the exogenous genetic material or expression product thereof can be a modified form of genetic material native to the host organism, it can be derived from another organism, it can be a modified form of a component derived from another organism, or it can be a synthetically derived component. For example, a *K. lactis* invertase gene is exogenous when introduced into *S. cerevisiae*.

[0037] As used herein, “native” refers to genetic material or an expression product thereof that is found, apart from individual-to-individual mutations which do not affect function or expression, within the genome of wild-type cells of the host cell. For the purposes of this application, the *Moniliella pollinis* cell “*Moniliella tomentosa* var *pollinis* TCV364” described in U.S. Pat. No. 6,440,712, which is incorporated herein by reference in its entirety, and deposited under the Budapest Treaty at BCCM/MUCL (Belgian Coordinated Collections of Micro-organisms/Mycothèque de l'Université Catholique de Louvain by Eridania Béghin Say, Vilvoorde R&D Centre, Havenstraat 84, B-1800 Vilvoorde) on Mar. 28, 1997 under number MUCL40385, is considered the wild-type *Moniliella pollinis* cell.

[0038] As used herein, the terms “polypeptide” and “peptide” are used interchangeably and refer to the collective primary, secondary, tertiary, and quaternary amino acid sequences and structure necessary to give the recited macromolecule its function and properties. As used herein, “enzyme” or “biosynthetic pathway enzyme” refer to a protein that catalyzes a chemical reaction. The recitation of any particular enzyme, either independently or as part of a biosynthetic pathway is understood to include the co-factors, co-enzymes, and metals necessary for the enzyme to properly function. A summary of the amino acids and their three and one letter symbols as understood in the art is presented in Table 1. The amino acid name, three letter symbol, and one letter symbol are used interchangeably herein.

TABLE 1

Amino Acid three and one letter symbols		
Amino Acid	Three letter symbol	One letter symbol
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

[0039] Variants or sequences having substantial identity or homology with the polypeptides described herein can be utilized in the practice of the disclosed recombinant cells, compositions, and methods. Such sequences can be referred to as variants or modified sequences. That is, a polypeptide sequence can be modified yet still retain the ability to exhibit the desired activity. Generally, the variant or modified sequence may include greater than about 45%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% sequence identity with the wild-type, naturally occurring polypeptide sequence, or with a variant polypeptide as described herein.

[0040] As used herein, the phrases “% sequence identity,” “% identity,” and “percent identity,” are used interchangeably and refer to the percentage of residue matches between at least two amino acid sequences or at least two nucleic acid sequences aligned using a standardized algorithm. Methods of amino acid and nucleic acid sequence alignment are well-known. Sequence alignment and generation of sequence identity include global alignments and local alignments which are carried out using computational approaches. An alignment can be performed using BLAST (National Center for Biological Information (NCBI) Basic Local Alignment Search Tool) version 2.2.31 software with default parameters. Amino acid % sequence identity between amino acid sequences can be determined using standard protein BLAST with the following default param-

eters: Max target sequences: 100; Short queries: Automatically adjust parameters for short input sequences; Expect threshold: 10; Word size: 6; Max matches in a query range: 0; Matrix: BLOSUM62; Gap Costs: (Existence: 11, Extension: 1); Compositional adjustments: Conditional compositional score matrix adjustment; Filter: none selected; Mask: none selected. Nucleic acid % sequence identity between nucleic acid sequences can be determined using standard nucleotide BLAST with the following default parameters: Max target sequences: 100; Short queries: Automatically adjust parameters for short input sequences; Expect threshold: 10; Word size: 28; Max matches in a query range: 0; Match/Mismatch Scores: 1, -2; Gap costs: Linear; Filter: Low complexity regions; Mask: Mask for lookup table only. A sequence having an identity score of XX % (for example, 80%) with regard to a reference sequence using the NCBI BLAST version 2.2.31 algorithm with default parameters is considered to be at least XX % identical or, equivalently, have XX % sequence identity to the reference sequence.

[0041] Polypeptide or polynucleotide sequence identity may be measured over the length of an entire defined polypeptide sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined polypeptide sequence, for instance, a fragment of at least 15, at least 20, at least 30, at least 40, at least 50, at least 70 or at least 150 contiguous residues. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

[0042] The polypeptides disclosed herein may include “variant” polypeptides, “mutants,” and “derivatives thereof.” As used herein the term “wild-type” is a term of the art understood by skilled persons and means the typical form of a polypeptide as it occurs in nature as distinguished from variant or mutant forms. As used herein, a “variant,” “mutant,” or “derivative” refers to a polypeptide molecule having an amino acid sequence that differs from a reference protein or polypeptide molecule. A variant or mutant may have one or more insertions, deletions, or substitutions of an amino acid residue relative to a reference molecule.

[0043] The amino acid sequences of the polypeptide variants, mutants, derivatives, or fragments as contemplated herein may include conservative amino acid substitutions relative to a reference amino acid sequence. For example, a variant, mutant, derivative, or fragment polypeptide may include conservative amino acid substitutions relative to a reference molecule. “Conservative amino acid substitutions” are those substitutions that are a substitution of an amino acid for a different amino acid where the substitution is predicted to interfere least with the properties of the reference polypeptide. In other words, conservative amino acid substitutions substantially conserve the structure and the function of the reference polypeptide. Conservative amino acid substitutions generally maintain (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a beta sheet or alpha helical conformation, (b) the charge and/or hydrophobicity of the molecule at the site of the substitution, and/or (c) the bulk of the side chain.

[0044] As used herein, terms “polynucleotide,” “polynucleotide sequence,” and “nucleic acid sequence,” and “nucleic acid,” are used interchangeably and refer to a

sequence of nucleotides or any fragment thereof. These phrases also refer to DNA or RNA of natural or synthetic origin, which may be single-stranded or double-stranded and may represent the sense or the antisense strand. The DNA polynucleotides may be a cDNA (e.g., coding DNA) or a genomic DNA sequence (e.g., including both introns and exons).

[0045] A polynucleotide is said to encode a polypeptide if, in its native state or when manipulated by methods known to those skilled in the art, it can be transcribed and/or translated to produce the polypeptide or a fragment thereof. The anti-sense strand of such a polynucleotide is also said to encode the sequence.

[0046] Those of skill in the art understand the degeneracy of the genetic code and that a variety of polynucleotides can encode the same polypeptide. In some aspects, the polynucleotides (e.g., polynucleotides encoding an XPDH polypeptide) may be codon-optimized for expression in a particular cell including, without limitation, a plant cell, bacterial cell, fungal cell, or animal cell. While polypeptides encoded by polynucleotide sequences found in various species are disclosed herein any polynucleotide sequences may be used which encodes a desired form of the polypeptides described herein. Thus, non-naturally occurring sequences may be used. These may be desirable, for example, to enhance expression in heterologous expression systems of polypeptides or proteins. Computer programs for generating degenerate coding sequences are available and can be used for this purpose. Pencil, paper, the genetic code, and a human hand can also be used to generate degenerate coding sequences.

[0047] The recombinant cells described herein may include deletions or disruptions in one or more native genes. The phrase “deletion or disruption” refers to the status of a native gene in the recombinant cell that has either a completely eliminated coding region (deletion) or a modification of the gene, its promoter, or its terminator (such as by a deletion, insertion, or mutation) so that the gene no longer produces an active expression product, produces severely reduced quantities of the expression product (e.g., at least a 75% reduction or at least a 90% reduction) or produces an expression product with severely reduced activity (e.g., at least 75% reduced or at least 90% reduced). The deletion or disruption can be achieved by genetic engineering methods, forced evolution, mutagenesis, RNA interference (RNAi), and/or selection and screening. The native gene to be deleted or disrupted may be replaced with an exogenous nucleic acid of interest for the expression of an exogenous gene product (e.g., polypeptide, enzyme, and the like).

[0048] The recombinant cells described herein may include one or more genetic modifications in which an exogenous nucleic acid is integrated into the genome of the host cell. One of skill in the art know how to select suitable loci in a yeast genome for integration of the exogenous nucleic acid. Suitable integration loci may include, but are not limited to, the PDC1, GPDI, CYB2A, CYB2B, g4240, YMR226, MDHB, ATO2, Adh9091, Adh1202, ADE2, ADH2556, GAL6, MDH1, SCW11, ER1, ER3, pyrF, TRP3, gpdIIA, and gpdIIB loci. For example, in a *M. pollinis* host cells, suitable interaction loci may include, but are not limited to, the ER1 locus (defined as the locus flanked by SEQ ID NO:85 and SEQ ID NO:162), the ER3 locus (defined as the locus flanked by SEQ ID NO: 155 and SEQ ID NO:165), the PDC1 locus (defined as the locus flanked

by SEQ ID NO: 152 and SEQ ID NO:164), the pyrF locus (defined as the locus flanked by SEQ ID NO: 153 and SEQ ID NO: 163), the TRP3 locus (defined as the locus flanked by SEQ ID NO:156 and SEQ ID NO: 159), the gpdIIA locus (defined as the locus flanked by SEQ ID NO:157 and SEQ ID NO: 161); and the gpdIIB locus (defined as the locus flanked by SEQ ID NO: 158 and SEQ ID NO: 166). The exogenous nucleic acid may also be integrated in an intergenic region or other location in the host cell genome not specifically specified herein. Other suitable integration loci may be determined by one of skill in the art. Furthermore, one of skill in the art would recognize how to use sequences to design primers to verify correct gene integration at the chosen locus.

[0049] The recombinant cell may have one or more copies of a given exogenous nucleic acid sequence integrated in a host chromosome(s) and replicated together with the chromosome(s) into which it has been integrated. For example, the yeast cell may be transformed with nucleic acid construct including a polynucleotide sequence encoding for a polypeptide described herein and the polynucleotide sequence encoding for the polypeptide may be integrated in one or more copies in a host chromosome(s). The recombinant cell may include multiple copies (two or more) of a given polynucleotide sequence encoding a polypeptide described herein. The recombinant cell may have one, two, three, four, five, six, seven, eight, nine, ten, or more copies of a polynucleotide sequence encoding a polypeptide described herein integrated into the genome. The multiple copies of said polynucleotide sequence may all be incorporated at a single locus or may be incorporated at multiple loci.

[0050] The recombinant cells described herein are capable of producing xylitol and include an exogenous polynucleotide sequence encoding a xylitol-phosphate dehydrogenase (XPDH) enzyme. The exogenous polynucleotide sequence may be an exogenous xylitol-phosphate dehydrogenase (XPDH) gene. Following conversion of xylulose 5-phosphate to xylitol 5-phosphate, it is believed a native phosphatase enzyme removes the phosphate to produce xylitol.

[0051] A “xylitol-phosphate dehydrogenase gene” and an “XPDH gene” are used interchangeably herein and refer to any gene or polynucleotide that encodes a polypeptide with xylitol-phosphate dehydrogenase activity. As used herein, “xylitol-phosphate dehydrogenase activity” refer to the ability to catalyze the conversion of xylulose-5-phosphate and NADPH or NADH to xylitol 5-phosphate and NADP⁺ or NAD⁺. The XPDH gene may be derived from any suitable source. For example, the XPDH gene may be derived from *Clostridium difficile*, *Lactobacillus rhamnosus*, *Bacillus halodurans*, *Alkalihalobacillus ligniniphilus*, *Jeotgalibacillus soli*, *Heyndrickxia sporothermodurans*, *Clostridium fungisolvans*, or *Neobacillus cucumis*. The XPDH gene may encode an amino acid at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, or 100% sequence identity to the amino acid sequence of at least one of SEQ ID NOs: 12-15, 28-32, or 33. The XPDH gene may encode an amino acid at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99%, or 100% sequence identity to the amino acid sequence of at least one of SEQ ID NOs: 14, 15, 28, or 31.

[0052] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Clostridium*

difficile gene encoding the amino acid of SEQ ID NO:12. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO: 12.

[0053] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Clostridium difficile* gene encoding the amino acid of SEQ ID NO:13. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO: 13.

[0054] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Lactobacillus rhamnosus* gene encoding the amino acid of SEQ ID NO:14. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO: 14.

[0055] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Bacillus halodurans* gene encoding the amino acid of SEQ ID NO:15. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:15.

[0056] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Alkalihalobacillus ligniniphilus* gene encoding the amino acid of SEQ ID NO:28. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:28.

[0057] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Jeotgalibacillus soli* gene encoding the amino acid of SEQ ID NO:29. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:29.

[0058] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Heyndrickxia sporothermodurans* gene encoding the amino acid of SEQ ID NO:30. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:30.

[0059] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Clostridium fungisolvans* gene encoding the amino acid of SEQ ID NO:31. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:31.

[0060] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Neobacillus*

cucumis gene encoding the amino acid of SEQ ID NO:33. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:33.

[0061] The exogenous polynucleotides in the recombinant cells described herein may be under the control of a promoter. For example, the exogenous nucleic acid may be operably linked to a heterologous or artificial promoter. Suitable promoters are known and described in the art. Promoters may include, but are not limited to, pyruvate decarboxylase promoter (PDC), translation elongation factor 2 promoter (TEF2), SED1, alcohol dehydrogenase 1A promoter (ADH1), hexokinase 2 promoter (HXK2), FLO5 promoter, pyruvate kinase 1 promoter (PYK1p; SEQ ID NO: 86); 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:130); glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:132); translational elongation factor 1 promoter (TEFp; SEQ ID NO:133); modified TEFp (SEQ ID NO:131); phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:134); 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:135); enolase promoter (ENO1p; SEQ ID NO:136); asparagine synthetase promoter (ASNSp; SEQ ID NO:137); 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO: 138); and RPL16B (SEQ ID NO:139).

[0062] The exogenous nucleic acids in the recombinant cells described herein may be under the control of a terminator. For example, the exogenous nucleic acid may be operably linked to a heterologous or artificial terminator. Suitable terminators are known and described in the art. Terminators may include, but are not limited to, GAL10 terminator, PDC terminator, transaldolase terminator (TAL) 6PGD terminator (6PGDt; SEQ ID NO:140); ASNS terminator (ASNSt; SEQ ID NO: 141); ENO1 terminator (ENO1t; SEQ ID NO:142); hexokinase 1 terminator (HXK1t; SEQ ID NO: 143); PGK1 terminator (PGK1t; SEQ ID NO:144); PGM1 terminator (PGM1t; SEQ ID NO: 145); PYK1 terminator (PYK1t; SEQ ID NO:146); RPLA terminator (RPLAt; SEQ ID NO: 147); transaldolase 1 terminator (TAL1t; SEQ ID NO:148); TDH3 terminator (TDH3t; SEQ ID NO: 149); translation elongation factor 2 terminator (TEF2t; SEQ ID NO:150); and triosephosphate isomerase 1 terminator (TPI1t; SEQ ID NO:151).

[0063] A promoter or terminator is “operably linked” to a given polynucleotide (e.g., a gene) if its position in the genome or expression cassette relative to said polynucleotide is such that the promoter or terminator, as the case may be, performs its transcriptional control function.

[0064] The polypeptides described herein may be provided as part of a construct. As used herein, the term “construct” refers to recombinant polynucleotides including, without limitation, DNA and RNA, which may be single-stranded or double-stranded and may represent the sense or the antisense strand. Recombinant polynucleotides are polynucleotides formed by laboratory methods that include polynucleotide sequences derived from at least two different natural sources or they may be synthetic. Constructs thus may include new modifications to endogenous genes introduced by, for example, genome editing technologies. Constructs may also include recombinant polynucleotides created using, for example, recombinant DNA methodologies. The construct may be a vector including a promoter oper-

ably linked to the polynucleotide encoding a polypeptide as described herein. As used herein, the term “vector” refers to a polynucleotide capable of transporting another polynucleotide to which it has been linked. The vector may be a plasmid, which refers to a circular double-stranded DNA loop into which additional DNA segments may be integrated.

[0065] The disclosure also provides fermentation methods for the production of xylitol using the recombinant cells described herein. The fermentation methods include the step of fermenting a substrate using the genetically engineered yeasts described herein to product xylitol. The fermentation method can include additional steps, as would be understood by a person skilled in the art. Non-limiting examples of additional process steps include maintaining the temperature of the fermentation broth within a predetermined range, adjusting the pH during fermentation, and isolating the xylitol from the fermentation broth. The fermentation process may be a fully aerobic process.

[0066] The fermentation method can be run using a suitable fermentation substrate. The substrate of the fermentation method can include glucose, sucrose, galactose, mannose, molasses, xylose, fructose, hydrolysates of starch, lignocellulosic hydrolysates, or a combination thereof. One skilled in the art will recognize what fermentation substrate is suitable for a given fermentation organism and system.

[0067] The fermentation process can be run under various conditions. The fermentation temperature, i.e., the temperature of the fermentation broth during processing, may be ambient temperature. Alternatively, or additionally, the fermentation temperature may be maintained within a predetermined range. For example, the fermentation temperature can be maintained in the range of 25° C. to 45° C., 30° C. to 40° C., or 32° C. to 37° C., preferably about 35° C. However, a skilled artisan will recognize that the fermentation temperature is not limited to any specific range or temperature recited herein and may be modified as appropriate.

[0068] The fermentation process can be run within certain oxygen uptake rate (OUR) ranges. The volumetric OUR of the fermentation process can be in the range of 0.5 to 40, 1 to 35, 2 to 30, 3 to 25, 4 to 20, or 5 to 15 mmol O₂/(L·h). In some embodiments, the specific OUR can be in the range of 0.05 to 10, 0.1 to 8, 0.15 to 5, 0.2 to 1, or 0.3 to 0.75 mmol O₂/(g cell dry weight· h). However, the volumetric or specific OURs of the fermentation process are not limited to any specific rates or ranges recited herein.

[0069] The fermentation process can be run at various cell concentrations. In some embodiments, the cell dry weight at the end of fermentation can be 5 to 40, 8 to 30, or 10 to 20 g cell dry weight/L. Further, the pitch density or pitching rate of the fermentation process can vary. In some embodiments, the pitch density can be 0.05 to 11, 0.1 to 10, or 0.25 to 8 g cell dry weight/L.

[0070] The initial dextrose concentration of the fermentation may be at least 100, 200, 250, 300, 350, or at least 400 g/L dextrose. The initial dextrose concentration may be between 100 to 400, 150 to 350, or 250 to 325 g/L.

[0071] The fermentation process can be associated with various characteristics, such as, but not limited to, fermentation production rate, pathway fermentation yield, final titer, and peak fermentation rate. These characteristics can be affected by the selection of the yeast and/or genetic modification of the yeast used in the fermentation process.

These characteristics can be affected by adjusting the fermentation process conditions. These characteristics can be adjusted via a combination of yeast selection or modification and the selection of fermentation process conditions.

[0072] The xylitol production rate of the process may be at least at least 0.2, 0.3, 0.5, 0.75, or at least 1.0 g L⁻¹ h⁻¹. The xylitol mass yield of the process may be at least 55 percent, at least 65 percent, at least 70 percent, at least 75 percent, at least 80 percent, or at least 85 percent. The final xylitol titer of the process may be at least 20, 30, 50, 75, or 100 g/L.

[0073] The fermentation process can be run as a dextrose-fed batch. Further, the fermentation process can be a batch process, continuous process, or semi-continuous process, as would be understood by a person skilled in the art.

EXAMPLES

[0074] The invention is further described in detail by reference to the following experimental examples. These examples are provided for purposes of illustration only and are not intended to be limiting unless otherwise specified. Thus, the invention should in no way be construed as being limited to the following examples, but rather should be construed to encompass any and all variations which become evident as a result of the teaching provided herein.

Example 1-Xylitol-Phosphate Dehydrogenase Diversity

[0075] Roughly three thousand galactitol-1-phosphate-5-dehydrogenase (G1PDH)/xylitol-phosphate dehydrogenase (XPDH) enzyme sequences were obtained from Uniprot and analyzed. FIG. 2 illustrates the natural sequence diversity for this set of sequences. This set is diverse, with ~25% of the enzymes having no homologue more than 75% identical. As these enzymes tend to prefer NAD to NADP as a cofactor, the cofactor binding preferences of the homologs were assessed in a manner similar to that described by Duax et al., (“Rational proteomics I. Fingerprinting identification and cofactor specificity in the short-chain oxidoreductase (SCOR) enzyme family,” *Proteins*, 2003, 53 (4): 931-943). Cofactor binding pockets were identified by proximity to the Rossmann fold (+23 to +30 amino acids from the GXGXXG motif (SEQ ID NO: 129)) and scored on the basis of total charge in an 8-residue window. The top 8 candidates that were predicted to use NADP were selected for further characterization, along with 4 candidates predicted to use NAD, and 3 controls.

[0076] Upon further review of the structural characteristics of the predicted binding pocket for factors that may influence cofactor preference, an important aspartate residue was identified. See FIG. 3. The polypeptide of SEQ ID NO:34 and substitutions thereof were used to construct a structural homology model to predict cofactor binding pocket confirmations FIG. 3 shows the C-terminal end of the penultimate β -strand on the outside of the Rossmann Fold domain. Without wishing to be bound by any particular theory, it is predicted that enzymes in which the first residue in this region (residue 198 relative to SEQ ID NO:34) is an aspartate and the second residue (residue 199 relative to SEQ ID NO:34) is a large hydrophobic amino acid (for example, isoleucine) will prefer an NAD cofactor due to the hydrogen bonding of the aspartate to the hydroxyl groups of the NAD ribose. However, enzymes in which that first

residue (residue 198 relative to SEQ ID NO:34) is an alanine, glycine, or serine and the second residue (residue 199 relative to SEQ ID NO:34) is lysine or arginine will prefer an NADP cofactor as the positive charge on the lysine or arginine residue will interact with the negative charge of the phosphate of the NADP and the smaller residue in the first position allows space in the binding pocket for said phosphate. Based on this analysis, 12 additional enzymes were selected for their predicted preference for NADP. Finally, 6 additional enzymes with sequence similarity to active XPDH enzymes were selected for testing.

Example 2-TarJ' Diversity

[0077] Roughly eight hundred ribulose 5-phosphate reductase sequences were obtained from Uniprot and analyzed. FIG. 3 illustrates the natural sequence diversity for this set of sequences. Overall, the diversity in this set is low, as only 10% of the enzymes have no sequence similarity more than 75% identical. As these enzymes tend to prefer

NADP to NAD as a cofactor, no scoring was performed, and the sequences were simply aligned in Geneious (ClustalW, default settings). Eight enzymes were selected for further analysis based on sequence similarity.

Example 3-In Vitro Enzyme Assays

[0078] Polynucleotides encoding suspected XPDH homologs (Table 2) or TarJ' homologs (Table 3) were cloned into a vector containing a T7 promoter and terminator for cell-free protein expression (New England Biolabs, PUR-Express® In Vitro Protein Synthesis). Cell-free synthesized proteins were analyzed for activity on four substrates (ribulose 5-phosphate, xylulose 5-phosphate, ribulose, and xylulose) with either NADP or NAD cofactors. Seven enzymes (XPDH of SEQ ID NOs: 12 and 34, TarJ' of SEQ ID NOs: 36, 37, 38, 40, and 42), were able to catalyze the reduction of either ribulose 5-phosphate or xylulose 5-phosphate (FIG. 5) but not the reduction of xylulose or ribulose (data not shown).

TABLE 2

XPDH Homologs		
Source Organism	Gene Name/Accession Numbers	Polypeptide SEQ ID NO:
<i>Mariniphaga sediminis</i>	D1164_16045	1
<i>Candidatus Dormibacteraeota bacterium</i>	DLM67_26945	2
<i>Candidatus Falkowbacteria bacterium</i>	COT93_01110	3
<i>Acidobacteria bacterium</i>	DMG39_01315	4
<i>Spirochaetaceae bacterium</i>	DCP56_03760	5
<i>Candidatus Hydrogenedentes bacterium</i>	ENW01_11985	6
<i>Rhizobium</i> sp.	E6Q76_20070	7
<i>Citrobacter freundii</i>	C2U38_22630	8
<i>Escherichia coli</i>	gatD	9
<i>Escherichia coli</i> (strain K12)	rspB	10
<i>Lactobacillus salivarius</i> cp400	LSCP400_11731	11
<i>Clostridium difficile</i>	WP_004454628.1	12
<i>Clostridium difficile</i>	WP_011860836.1	13
<i>Lactobacillus rhamnosus</i>	AAT02414.1	14
<i>Bacillus halodurans</i>	WP_010896369.1	15
<i>Lachnospiraceae bacterium</i>	A0A3C1WRV0_9FIRM	16
<i>Faecalibacterium</i> sp. OF03-6AC	DXA66_03105	17
<i>Agathobacter ruminis</i>	CSX02_08830	18
<i>Alicyclobacillus acidoterrestris</i>	AAC03nite_18160	19
<i>Ruminococcus</i> sp. KGMB03662	FFK04_01895	20
<i>Clostridiaceae bacterium</i>	D3Z58_08610	21
<i>Thermoplasmata archaeon</i>	A0A497HL05	22
<i>Lachnospiraceae bacterium</i>	A0A3B918G1	23
<i>Prevotella marshii</i> DSM 16973 = JCM 13450	E0NTD1	24
<i>Blautia</i> sp. OM05-6	A0A417S5N4	25
<i>Candidatus Bathyarchaeota archaeon</i>	A0A7C4RGE8	26
<i>Lachnospiraceae bacterium</i>	A0A7U9REN4	27

TABLE 3

TarJ' Homologs		
Source Organism	Gene Name/Accession Numbers	Polypeptide SEQ ID NO:
<i>Staphylococcus aureus</i>	AMV81374.1	34
<i>Staphylococcus aureus</i> subsp. <i>aureus</i> 71193	ST398NM01_0269	35
<i>Eubacterium ventriosum</i>	DW918_	36
<i>Pradoshia</i> sp. D12	F7984_17520	37
<i>Lactobacillus plantarum</i> EGD-AQ4	N692_02285	38
uncultured <i>Ruminococcus</i> sp.	DPQ25_03780	39
<i>Kandleria vitulina</i>	SAMN04487759_1396	40
<i>Lactiplantibacillus plantarum</i>	C7M34_02724 C7M35_00347	41
	C7M36_02765 C7M47_01038	

TABLE 3-continued

TarJ' Homologs		
Source Organism	Gene Name/ Accession Numbers	Polypeptide SEQ ID NO:
<i>Lachnospiraceae bacterium</i>	FET70_01115 LPJSA22_01709	
	Nizo2802_1951	
	IMSAGC019_01831	42

Example 4-Genetically Modified *Moniliella pollinis* Strains

[0079] Strain 1-1 is the *Moniliella pollinis* host strain “*Moniliella tomentosa* var *pollinis* TCV364” described in U.S. Pat. No. 6,440,712, which is incorporated herein by reference in its entirety, and deposited under the Budapest Treaty at BCCM/MUCL (Belgian Coordinated Collections of Micro-organisms/Mycothèque de l'Université Catholique de Louvain by *Eridania* Béghin Say, Vilvoorde R&D Centre, Havenstraat 84, B-1800 Vilvoorde) on Mar. 28, 1997 under number MUCL40385. Table 4 below lists various *Moniliella pollinis* strains, including information on the parent strain, the sequence with which the parent strain was transformed, and characterizations of the expression cassette (s) contained on the transformed sequence. Each “XPDH/TarJ' Homolog Expression Cassette” contained, in order, a 5' ER1 flanking sequence (SEQ ID NO:85), a MpPYK1 promoter (SEQ ID NO:86), a gene encoding the indicated XPDH or TarJ' homolog (one of SEQ ID NOs: 87-128), a Mp6PGD terminator (SEQ ID NO:140), and a 5' portion of a G418 resistance gene expression cassette (SEQ ID NO:175). Each “Selectable Marker Cassette” contained, in order, a 3' portion of a G418 resistance gene expression cassette (SEQ ID NO:172), an MpTEF2 terminator (SEQ ID NO:150), and a 3' ER1 flanking sequence (SEQ ID NO:160). Upon bipartite transformation with both the XPDH/TarJ' Homolog Expression Cassette and the Selectable Marker Cassette, the two cassettes recombine for integration of both the nucleotide sequence encoding the XPDH or TarJ' homolog and the G418 resistance marker at the ER1 locus.

[0080] The indicated *Moniliella pollinis* parent strain was transformed with the indicated sequence(s) by first protoplasting the parent strain by adding an enzyme mixture containing 0.6M MgSO₄, 7.5 g/L driselase, and 12.5 g/L *Trichoderma harzianum* lysing enzyme to a mycelial pellet of the parent strain. Protoplasts were then pelleted, washed with 0.6M MgSO₄, and resuspended in STC medium (0.6M sucrose, 50 mM CaCl₂, 10 mM Tris-HCl, pH 7.5). Bipartite transformations were prepared by adding 100 µg single stranded salmon sperm DNA and 1.5 to 5 µg each of the 5' and 3' DNA transformation fragments (3-10 µg total; see

Table 4 for list of fragments) to approximately 200 µL protoplast mixture (10⁸ cells/mL). 1 mL 50% PEG in STC medium was then added to the salmon sperm DNA, transformation DNA, and protoplast mixture and the resulting combination was incubated for 15 minutes at room temperature. Following incubation, recovery broth (0.4M sucrose, 1 g/L yeast extract, 1 g/L malt extract, 10 g/L glucose, pH 4.5) was added to the mixture and incubated at 27° C., 100 rpm, for 16 to 24 hours. Following the incubation, protoplasts were pelleted by centrifugation and resuspended in 1 mL PBS.

[0081] The resuspended protoplasts were plated on PDA+250 mg/L geneticin (G418) selection plates and incubated at 30-35° C. for at least 2-4 days until transformants grow. Resulting transformants were evaluated by colony PCR for integration of the indicated sequence. A PCR verified isolate was then designated as the indicated strain number. In some instances, more than one PCR verified isolate, e.g., “sister” isolates, are indicated by letters following the strain number. For example, strain 1-2 has 5 sister isolates, strains 1-2a, 1-2b, 1-2c, 1-2d, and 1-2e.

[0082] For example, Strain 1-1 was transformed with SEQ ID NO:43 and SEQ ID NO:44. SEQ ID NO: 43 contains (i) 3' flanking DNA for targeted chromosomal integration into the ER1 locus (SEQ ID NO:162), and (ii) a 3' portion of the G418 resistance gene selectable marker (SEQ ID NO: 172). SEQ ID NO:44 contains (i) an expression cassette for the XPDH homolog from *M. sediminis*, SEQ ID NO:87 encoding the amino acid sequence of SEQ ID NO:1, under the control of the PYK1 promoter of SEQ ID NO:86 and the PGD terminator of SEQ ID NO:140; (ii) 5' flanking DNA for targeted chromosomal integration into the ER1 locus (SEQ ID NO:85); and (iii) a 5' portion of the G418 resistance gene selectable marker (SEQ ID NO: 175). Transformants were selected on PDA+250 mg/L geneticin (G418) selection plates and incubated at 30-35° C. for at least 2 days until transformants grow. Resulting transformants were streaked for single colony isolation on PDA+geneticin (G418) plates and single colonies were selected. Selected colonies were evaluated by colony PCR for integration of the indicated sequence. PCR verified isolates were designated strains 1-2a, 1-2b, 1-2c, 1-2d, and 1-2e.

TABLE 4

Strain	Parent Strain	XPDH/TarJ' Homolog Expression Cassette			Selectable Marker Cassette
		Transformation SEQ ID NO:	Nucleotide SEQ ID NO:	Encoded Polypeptide SEQ ID NO:	Transformation SEQ ID NO:
1-2a-e	1-1	44	87	1	43
1-3a-e	1-1	45	88	2	43
1-4a-e	1-1	46	89	3	43
1-5a-e	1-1	47	90	4	43
1-6a-e	1-1	48	91	5	43

TABLE 4-continued

Strain	Parent Strain	XPDH/TarJ' Homolog Expression Cassette			Selectable Marker Cassette
		Transformation SEQ ID NO:	Nucleotide SEQ ID NO:	Encoded Polypeptide SEQ ID NO:	Transformation SEQ ID NO:
1-7a-e	1-1	49	92	6	43
1-8a-d	1-1	50	93	7	43
1-9a-e	1-1	51	94	8	43
1-10a-e	1-1	52	95	9	43
1-11a-e	1-1	53	96	10	43
1-12a-e	1-1	54	97	11	43
1-13a-f	1-1	55	98	12	43
1-14a-e	1-1	56	99	13	43
1-15a-f	1-1	57	100	14	43
1-16a-e	1-1	58	101	15	43
1-17a-e	1-1	59	102	16	43
1-18a-e	1-1	60	103	17	43
1-19a-e	1-1	61	104	18	43
1-20a-e	1-1	62	105	19	43
1-21a-e	1-1	63	106	20	43
1-22a-e	1-1	64	107	21	43
1-23a-e	1-1	65	108	22	43
1-24a-e	1-1	66	109	23	43
1-25a-e	1-1	67	110	24	43
1-26a-e	1-1	68	111	25	43
1-27a-e	1-1	69	112	26	43
1-28a-e	1-1	70	113	27	43
1-29a-e	1-1	71	114	28 (<i>Alkalihalobacillus ligniniphilus</i> XPDH WP_017729640.1)	43
1-30a-e	1-1	72	115	29 (<i>Jeotgalibacillus soli</i> XPDH WP_041086875.1)	43
1-31a-e	1-1	73	116	30 (<i>Heyndrickxia sporothermodurans</i> XPDH WP_066235909.1)	43
1-32a-e	1-1	74	117	31 (<i>Clostridium fungisolvans</i> XPDH WP_183276691.1)	43
1-33a-e	1-1	75	118	32 (<i>Alkalihalobacillus nanhaiisediminis</i> XPDH WP144449381.1)	43
1-34a-e	1-1	76	119	33 (<i>Neobacillus cucumis</i> XHPD WP_198313751.1)	43
1-35a-d	1-1	77	120	34	43
1-36a-e	1-1	78	121	35	43
1-37a-d	1-1	79	122	36	43
1-38a-f	1-1	80	123	37	43
1-39a-f	1-1	81	124	38	43
1-40a-e	1-1	82	125	39	43
1-41a-e	1-1	83	127	41	43
1-42a-f	1-1	84	128	42	43

Example 5-Shake Flask Fermentation Assay

[0083] Strains 1-1, 1-35a-d, 1-37a-d, 1-38a-f, 1-39a-f, 1-42a-f, 1-13a-f, and 1-15a-f (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0084] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0085] A 250 ml non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 72 and 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 6 and FIGS. 6 and 7.

TABLE 5

Production Medium	
Component	Concentration (units)
Glucose	300 (g/L)
KH ₂ PO ₄	1.27 (g/L)
(NH ₄) ₂ HPO ₄	0.13 (g/L)

TABLE 5-continued

Production Medium	
Component	Concentration (units)
(NH4)2SO4	1.80 (g/L)
Urea	2.85 (g/L)
Citric Acid	200.0 (mg/L)
MgSO4—7H2O	690.2 (mg/L)
FeSO4—7H2O	32.35 (mg/L)
ZnSO4—7H2O	4.40 (mg/L)
MnSO4—H2O	0.53 (mg/L)
CuSO4—5H2O	0.45 (mg/L)
CaCl2—2H2O	91.60 (mg/L)
Na2SO4	154.50 (mg/L)
Thiamine-HCl	10 (mg/L)
Choline-Cl	100 (mg/L)
Antifoam CF-32	1.00 (g/L)

[0086] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the sequence was not correctly integrated at the ER1 locus. Further analysis indicated that strains 1-35a, 1-37a-d, 1-38a-c, 1-39d-f, 1-42a-b, 1-42d, 1-13a-b, 1-13d-e, 1-15b-c, and 1-15e-f include the transformed polynucleotide sequence, but it is not integrated at the ER1 locus.

TABLE 6

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-1	54.2270	0.5672	0.1156	56.9108	14.4101	45.2717
1-35a	19.0690	77.8018	5.5078	14.1262	11.6922	45.6143
1-35b	22.4343	69.5363	2.6051	18.4159	6.7667	51.1619
1-35c	31.4030	66.5826	2.4193	17.3705	7.5255	49.2545
1-35d	28.2827	77.4169	5.2674	13.9265	7.1954	49.1733
1-37a	42.3156	58.8415	1.1308	17.6218	8.1394	44.6216
1-37b	92.0525	30.1794	0.5249	29.1345	17.4800	28.4251
1-37c	77.9520	25.8442	0.3365	37.6779	28.5369	33.4608
1-37d	62.2726	0.9251	0.2062	50.0038	18.7245	45.7655
1-38a	19.5123	89.5643	4.5079	13.9258	6.0135	46.2436
1-38b	15.4952	97.2570	7.8962	13.6124	6.7302	35.1221
1-38c	9.1530	100.4624	8.8382	14.4767	6.4634	41.8789
1-38d	3.5262	93.3559	5.1135	14.0524	7.8424	46.9099
1-38e	5.8345	91.9564	4.7706	13.7265	7.3560	48.8018
1-38f	5.6278	90.6644	4.9442	13.6917	7.7348	47.5447
1-39a	18.6244	72.8649	2.8000	19.1466	8.1946	40.2287
1-39b	2.3933	100.5639	10.0874	14.2583	6.9797	44.4668
1-39c	31.7163	67.5513	2.4932	17.8548	7.1724	47.5427
1-39d	3.3290	101.3044	9.6496	13.9511	6.9558	46.4375
1-39e	38.9049	66.5485	2.4033	16.8345	7.1407	48.7270
1-39f	0.8569	91.5653	11.1004	24.3341	6.1712	37.6166
1-42a	53.2303	1.3617	0.1586	62.1601	16.1305	44.7835
1-42b	70.3640	0.5390	0.1319	52.0362	13.1695	44.2371
1-42c	54.3271	0.6740	0.1317	61.0264	17.1397	44.9797
1-42d	55.3052	0.7063	0.1484	63.2909	12.3513	43.1762
1-42e	61.0003	0.5738	0.1502	56.8237	13.3096	39.6208
1-42f	64.0496	0.5451	0.1352	55.2579	13.6874	42.7263
1-13a	58.2322	3.0713	39.0606	24.5223	9.9512	43.5850
1-13b	64.5326	2.8987	38.2994	23.4455	9.2364	42.1528
1-13c	62.8811	2.8168	38.3638	23.0771	9.2614	43.8806
1-13d	47.8236	3.3178	40.0543	25.7124	9.9847	40.8596
1-13e	59.3667	4.8671	28.8414	30.2839	9.5968	42.2960
1-13f	66.1336	2.8202	38.0642	22.4811	8.8765	42.3954
1-15a	58.6182	8.6323	38.8703	21.1152	9.3006	43.9871
1-15b	78.7837	1.2617	12.2069	34.2608	11.5806	43.9409
1-15c	55.9848	1.1212	10.2043	47.6605	12.0438	44.2130
1-15d	54.0847	8.9671	38.7047	21.4173	9.3022	44.1645

TABLE 6-continued

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-15e	60.7735	1.9279	16.0149	43.7036	11.1713	46.0607
1-15f	102.6556	8.2515	30.3843	19.3150	13.0177	32.7172

Example 6-Shake Flask Fermentation Assay

[0087] Strains 1-13c, 1-29a-e, 1-33a-e, and 1-34a-e (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0088] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0089] A 250 ml non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 7 and FIG. 8.

[0090] As seen in FIG. 8, while sister strains 1-34c and 1-34d produced 15.8 and 18.6 g/L xylitol, respectively, strains 1-34a, 1-34b, and 1-34e did not produce significantly more xylitol than wild-type (strain 1-1, FIG. 6). While strains 1-34a, 1-34b, and 1-34e were initially PCR verified, it was later determined that the integrated polynucleotide, which should encode the *N. cucumis* XPDH homolog, contained a frameshift mutation and no functional XPDH was expressed. Therefore, while the results appear varied, they are in fact consistent given that strains 1-34a, 1-34b, and 1-34e did not contain a polynucleotide that encoded a functional XPDH.

TABLE 7

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-13c	104.7609	2.4041	25.3431	20.3872	8.3483	28.8597
1-29a	107.59	2.7375	23.9236	21.2592	8.327	29.7606
1-29b	117.0722	2.8833	21.624	21.9028	7.5941	30.1983
1-29c	92.8027	3.0122	25.0203	22.3197	8.607	34.1865
1-29d	113.7418	2.654	22.973	21.1195	7.992	30.2967
1-29e	106.1983	2.7505	24.6828	21.1105	8.3573	30.0138
1-33a	102.5508	0.4847	0.2175	48.9982	9.7203	31.4812
1-33b	101.4479	0.507	0.0751	50.0129	9.8179	31.3012

TABLE 7-continued

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-33c	231.4215	0.2896	0.0353	6.9913	16.2002	6.2899
1-33d	107.2532	0.5626	0.0542	48.3365	10.36	31.982
1-33e	89.76	0.5258	0.0847	54.4229	10.6124	27.6685
1-34a	97.4677	0.487	0.0849	49.4197	10.4031	31.8369
1-34b	99.6607	0.4724	0.0682	49.2765	9.7575	31.2798
1-34c	125.3303	1.7402	15.7865	23.7361	8.5658	28.3016
1-34d	109.6517	2.4549	18.5615	26.2014	8.1791	30.7102
1-34e	92.4856	0.434	0.22	47.7303	9.2374	32.5853

Example 7-Shake Flask Fermentation Assay

[0091] Strains 1-13c, 1-17a-e, 1-18a-e, 19a-e, 1-21a-e, 1-22a-e, 1-23a-e, 1-24a-e, 1-25a-e, and 1-27a-d (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0092] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0093] A 250 ml non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 8.

TABLE 8

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-13a	98.6422	2.6387	24.7712	25.2307	13.8005	28.8903
1-17a	74.915	0.5991	0.2543	60.8569	13.0048	33.3547
1-17b	79.2438	0.577	0.0725	58.9302	12.8216	35.3409
1-17c	151.0953	0.4237	0.0975	24.7655	21.0532	20.2202
1-17d	90.0013	1.4266	0.0645	55.808	18.8406	27.8706
1-17e	84.2381	0.7091	0.0428	59.7786	15.6082	29.9387
1-18a	187.0194	0.3163	0.1069	21.6446	15.8366	10.9522
1-18b	83.2939	0.6768	0.0524	63.3983	14.7387	31.6461
1-18c	88.21	0.6902	0.0625	59.74	14.7915	31.5115
1-18d	76.4267	0.8183	0.0692	57.1505	16.2282	34.0999
1-18e	89.0621	0.7336	0.0841	42.5495	26.7038	25.7627
1-19a	94.518	0.6166	0.0801	54.4468	18.8522	28.9871
1-19b	83.559	0.6632	0.0659	60.2352	14.9726	31.2065
1-19c	93.3168	0.7412	0.0646	60.6521	14.3042	29.9464
1-19d	165.946	0.4311	0.1236	24.1998	15.9581	19.0694
1-19e	78.3357	0.675	0.0637	61.3653	15.847	29.361

TABLE 8-continued

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-21a	77.8724	0.882	0.0789	65.2592	16.0186	31.302
1-21b	85.2124	0.6777	0.0619	56.1455	15.4518	33.3606
1-21c	85.008	1.0769	0.0802	58.6289	14.8934	32.8785
1-21d	83.4408	1.9855	0.0373	56.0976	14.2115	34.3856
1-21e	60.5121	2.039	0.0418	63.3641	15.5567	31.3166
1-22a	96.2241	0.8385	0.0644	62.4003	13.1106	29.8987
1-22b	90.5783	0.5506	0.0722	54.306	13.9584	34.1293
1-22c	85.3285	0.6442	0.0595	60.2244	15.0443	31.2101
1-22d	87.0015	0.6859	0.0824	59.9578	14.9889	31.3812
1-22e	78.5983	0.5432	0.0718	58.3192	14.9655	30.6604
1-23a	84.0794	0.4999	0.109	52.3855	13.872	35.266
1-23b	84.7042	0.5799	0.0696	58.9566	15.0869	33.9501
1-23c	87.9261	0.7983	0.0538	55.5429	17.3164	30.1523
1-23d	87.2081	0.6745	0.0623	59.4328	14.8259	31.3395
1-23e	82.9585	1.0568	0.0795	62.242	18.8816	23.5949
1-24a	86.7546	0.534	0.0725	53.0638	13.0757	34.4192
1-24b	104.4802	0.583	0.0821	48.7903	14.9732	29.8021
1-24c	104.5435	0.563	0.0872	48.1204	14.5086	31.0036
1-24d	169.2148	0.425	0.1078	25.2327	14.0784	19.6567
1-24e	78.1939	0.5172	0.082	55.4618	15.2743	36.423
1-25a	85.5913	0.8201	0.0711	57.2974	18.9442	29.1298
1-25b	83.3265	0.6891	0.0801	60.4083	15.034	31.4877
1-25c	139.0213	0.6194	0.1079	28.2606	22.9621	22.0197
1-25d	137.1956	0.6387	0.1254	28.3686	22.7144	22.2391
1-25e	78.0893	0.6945	0.0792	61.3866	15.5101	31.7923
1-27a	54.0049	0.8002	0.0653	73.3789	15.1766	36.3078
1-27b	88.2258	0.5067	0.0543	53.4474	13.5761	34.0372
1-27c	114.4683	0.7334	0.0729	43.7274	17.9455	26.3899
1-27d	76.682	0.5437	0.0552	58.8882	12.8684	38.8414

Example 8-Shake Flask Fermentation Assay

[0094] Strains 1-13c, 1-3a-e, 1-10a-e, 1-11a-e, 1-12a-e, 1-14a-e, 1-16a-e, 1-28a-e, and 1-2a-e (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0095] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0096] A 250 ml non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 72 and 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 9 and FIG. 9.

[0097] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the

sequence was not correctly integrated at the ER1 locus. Further analysis indicated that strain 1-16b-e includes the transformed polynucleotide sequence, but it is not at the ER1 locus. Further analysis was inconclusive on the integration location in strains 1-2c and 1-2d.

tometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0100] A 250 ml non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production

TABLE 9

96-hour Shake Flask Results							
Fermentation Broth Analyte (g/L)							
Strain	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol	Arabitol
1-13c	52.365	3.1859	40.2435	26.0209	11.1145	37.6398	
1-3a	58.7394	0.6129	0.3235	59.7362	15.12	40.891	
1-3b	40.836	0.4677	0.1051	49.0415	15.6583	51.2491	
1-3c	55.8652	0.5887	0.1253	59.2426	14.1511	42.761	
1-3d	47.4122	0.5038	0.1279	50.8589	16.8177	41.1038	
1-3e	45.8606	0.5127	0.1267	50.1914	16.3743	48.2382	
1-10a	58.5369	0.5891	0.1201	57.9712	15.3885	40.4663	
1-10b	58.9318	0.6075	0.1492	57.9368	14.8489	41.9797	
1-10c	67.992	0.4588	0.142	49.0939	14.307	40.394	
1-10d	46.0003	0.6906	0.0995	62.4692	19.849	32.9163	
1-10e	55.0568	0.655	0.136	61.0313	15.9185	38.2694	
1-11a	51.9376	0.429	0.0526	60.4417	14.8425	40.9479	
1-11b	45.6503	0.554	0.1326	61.3534	18.1726	44.3344	
1-11c	5.6815	1.5471	0.0801	86.3607	27.719	41.9823	
1-11d	40.3772	0.5489	0.1304	61.8324	18.171	41.9493	
1-11e	53.0558	0.6124	0.115	60.6504	15.489	39.7766	
1-12a	30.456	n.a.	n.a.	41.1173	9.6187	44.7618	46.2065
1-12b	28.2127	n.a.	n.a.	41.7725	9.3555	45.1461	47.4416
1-12c	45.2228	n.a.	n.a.	41.1422	9.0341	39.2179	42.4228
1-12d	30.6334	n.a.	n.a.	36.381	10.0025	39.6166	44.7009
1-12e	81.9896	n.a.	n.a.	34.348	12.8173	30.9024	30.2901
1-14a	36.1578	60.3639	19.8883	14.677	6.9789	39.9443	
1-14b	29.7316	53.6709	14.9948	11.3645	12.08	43.9787	
1-14c	27.9273	59.5839	17.4099	13.471	7.8201	45.0742	
1-14d	13.2601	64.6881	20.7494	15.3596	7.6729	41.5184	
1-14e	24.8568	64.1917	20.7387	15.176	7.2937	39.5752	
1-16a	53.8198	4.4367	36.058	32.5055	10.8652	39.1211	
1-16b	52.0735	4.9579	38.9526	23.8416	10.1467	41.0993	
1-16c	55.1727	1.6863	24.425	36.6549	11.7111	41.8186	
1-16d	45.5429	5.6345	22.6578	36.7881	11.2977	41.0565	
1-16e	41.996	2.3924	19.2866	43.8223	17.3556	40.5198	
1-28a	57.3853	0.6449	0.2041	58.1115	15.5452	41.2408	
1-28b	58.7477	0.5753	0.1016	56.9917	14.4135	41.1243	
1-28c	59.1807	0.5957	0.1202	57.6647	14.5907	42.3829	
1-28d	32.0943	0.6578	0.1353	62.5783	18.1666	40.946	
1-28e	56.5966	0.5859	0.1193	57.5414	14.0459	41.5865	
1-2a	50.0384	0.7556	0.1162	61.3047	15.3111	41.1167	
1-2b	34.3286	0.79	0.1848	58.4062	19.6802	48.8752	
1-2c	43.167	0.7455	0.0948	70.8234	11.4465	42.3114	
1-2d	41.842	0.6423	0.1082	63.7359	15.0666	39.6665	
1-2e	47.5807	0.6122	0.1415	60.543	13.8641	43.2135	

Example 9-Shake Flask Fermentation Assay

[0098] Strains 1-13c, 1-8a-d, 1-26a-e, 1-36a-e, 1-41a-e, 1-40a-e, and 1-20a-e (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0099] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectropho-

culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 10 and FIG. 10.

[0101] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the sequence was not correctly integrated at the ER1 locus. Further analysis indicated that strains 1-8c, 1-8d and 1-41c include the transformed polynucleotide sequence, but it is not integrated at the ER1 locus. Further analysis was inconclusive on integration locus in strains 1-36a, 1-41b, 1-41e, and 1-20a-e.

TABLE 10

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-13c	43.7126	3.1176	38.8613	24.5197	9.666	42.859
1-8a	46.3518	0.537	0.3381	57.0411	14.4597	46.2915
1-8b	47.5195	0.5334	0.1431	56.9461	13.7069	47.0977
1-8c	48.0894	0.5594	0.1063	55.9984	13.6091	47.8657
1-8d	37.4186	0.6121	0.1105	56.5403	13.8863	51.6109
1-26a	44.7559	0.6392	0.1211	58.7843	13.0302	45.1907
1-26b	26.8789	0.7454	0.1365	65.6702	14.7237	48.8557
1-26c	42.4506	0.7303	0.1018	65.269	12.7456	45.1127
1-26d	54.3955	0.5664	0.0934	50.543	16.7051	47.0098
1-26e	36.6369	0.7275	0.1162	70.0149	10.7288	48.273
1-36a	43.9038	23.4607	0.4003	37.8737	12.3842	43.023
1-36b	46.2296	17.4734	0.2509	28.3498	17.1992	51.7303
1-36c	33.5473	22.7172	0.384	32.9889	10.9137	53.8241
1-36d	30.3483	21.9108	0.3459	32.6397	12.4463	55.6903
1-36e	34.2625	22.4675	0.4111	33.3163	10.3248	53.7075
1-41a	50.84	0.6102	0.1238	53.2111	12.3563	47.0405
1-41b	27.0994	0.9407	0.0998	67.672	11.9654	54.0057
1-41c	54.6053	0.5387	0.1112	52.0424	14.0354	47.3747
1-41d	58.9086	0.6276	0.1701	51.36	21.0093	44.2126
1-41e	48.8262	0.5743	0.1143	57.4473	13.8819	45.8719
1-40a	38.7776	39.7726	0.2401	21.5304	12.246	44.721
1-40b	44.7092	49.927	0.3738	22.8583	10.4058	44.314
1-40c	43.6888	31.47	0.2197	29.7888	10.3401	50.5487
1-40d	43.0144	32.1802	0.1935	28.5121	13.9717	49.4035
1-40e	38.1811	32.5265	0.2004	31.8564	10.647	51.5381
1-20a	43.2545	0.7591	0.0973	60.0985	13.8425	44.7316
1-20b	44.7893	0.7386	0.1007	69.3394	13.0988	46.5606
1-20c	29.207	0.673	0.1105	63.3536	16.4597	49.4451
1-20d	39.0349	0.612	0.1224	57.5853	14.8878	49.7504
1-20e	48.061	0.7296	0.1017	61.7944	10.9017	47.5667

Example 10-Shake Flask Fermentation Assay

[0102] Strains 1-13c, 1-30a-e, 1-31a-e, 1-32a-e, 1-4a-e, 1-5a-e, 1-6a-e, 1-7a-e, and 1-9a-e (outlined in Table 4 above), were run in shake flasks to assess glucose consumption as well as ribitol, xylitol, glycerol, and ethanol production.

[0103] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL non-baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0104] A 250 ml non-baffled flask containing production medium (Table 5) was inoculated with 0.8 mL of the seed culture to form the production culture. The production culture was incubated at 35° C. and 250 rpm. Samples were

taken from the production culture after 72 and 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Table 11 and FIG. 11.

[0105] While PCR verification indicated that the transformed polynucleotide sequence was present in the indicated strains, further analysis indicated that in some strains, the sequence was not correctly integrated at the ER1 locus. Further analysis indicated that strains 1-30c and 1-30d include the transformed polynucleotide sequence, but it is not integrated at the ER1 locus. Further analysis was inconclusive on the integration locus in strain 1-6c.

TABLE 11

96-hour Shake Flask Results						
Strain	Fermentation Broth Analyte (g/L)					
	Glucose	Ribitol	Xylitol	Erythritol	Glycerol	Ethanol
1-13c	65.2534	2.6255	37.1061	24.3306	10.3659	40.8187
1-13c	63.7146	2.5391	36.7742	22.9396	9.9483	41.7289
1-30a	64.53	2.2035	28.9854	29.1288	11.3525	39.7976
1-30b	58.571	3.5146	29.9395	25.1729	12.5878	42.2656
1-30c	61.7808	1.8694	26.2541	26.3438	15.1573	43.1151
1-30d	57.4209	1.9473	26.054	27.0466	13.4411	44.8634
1-30e	74.7224	2.7504	28.386	28.8879	14.2582	37.7887
1-31a	65.2966	1.7919	23.8031	38.9002	11.2516	37.7702
1-31b	58.0672	1.528	25.7531	34.4798	12.434	40.5374
1-31c	60.9593	1.4068	24.8131	31.8421	11.6425	41.8413
1-31d	60.5991	1.4246	25.033	31.5056	11.5117	42.3126
1-31e	65.0496	1.7865	25.7844	33.8874	12.8272	40.8807
1-32a	67.1283	2.236	38.4541	23.5687	11.1949	38.707
1-32b	70.2815	2.1497	38.3141	22.7553	10.5468	38.8618
1-32c	68.0248	2.2063	38.5515	22.7602	10.3843	38.5577
1-32d	66.3494	2.1502	38.3931	21.9976	11.0603	40.7174
1-32e	63.8648	2.5133	38.8442	23.9287	10.8515	39.9071
1-4a	70.2783	0.4834	0.3673	46.1316	14.32	43.4293
1-4b	72.2855	0.7763	0.1831	57.21	19.3193	35.128
1-4c	56.1849	0.5101	0.1291	47.8537	15.7451	47.9659
1-4d	77.2722	0.5838	0.1901	50.3863	14.1389	39.9826
1-4e	70.9407	0.4671	0.1525	46.2662	15.3398	44.2176
1-5a	56.7359	0.6505	0.1281	69.7951	13.2652	40.3786
1-5b	66.4853	0.4758	0.1138	47.5548	17.2347	44.8984
1-5c	56.2946	0.4561	0.1072	46.0599	15.9891	48.8712
1-5d	63.8624	0.5188	0.1223	55.0207	15.5579	41.5538
1-5e	53.5706	0.4438	0.0978	46.4767	15.7478	49.9892
1-6a	53.5487	0.4466	0.0857	46.0621	14.3516	50.2479
1-6b	56.3098	0.5689	0.0987	59.4119	14.2449	44.3129
1-6c	62.2098	0.4975	0.1159	50.7104	13.912	46.5117
1-6d	88.3493	0.392	0.0963	46.7061	15.7906	39.0682
1-6e	66.3903	0.513	0.1236	54.7569	15.6941	42.6935
1-7a	84.3102	0.7204	0.0924	52.2635	19.827	33.7597
1-7b	69.0519	0.4897	0.1187	51.9057	14.6164	43.9185
1-7c	67.7409	0.5193	0.1426	53.5386	14.7419	42.8362
1-7d	61.1123	0.5279	0.1161	54.4933	14.502	43.378
1-7e	58.0693	0.4778	0.1356	50.1206	18.9325	46.1013
1-9a	52.6189	0.5986	0.1156	66.6905	13.1314	43.3405
1-9b	59.3034	0.464	0.1102	58.2455	14.7562	43.3825
1-9c	53.516	0.4302	0.1246	51.9753	15.0032	48.9462
1-9d	53.9501	0.4221	0.0876	46.6648	15.398	49.7159
1-9e	54.575	0.4136	0.0957	46.5254	15.3139	49.828

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 VKLPAKHVIP LPNVVNYDEA ALIEPASVVA HGFYRTNIKP GASVAIMGVG SIGLLAVQWA 180
 KIFGATTVFA IDIDEQKLV ANQLGADVLI SSLQRPAHKQ ILEYTNGIGV DVAVESAGTP 240
 STSAQVFALP KKGGEVFLG IPYADVQIER FYFEKIVRNE LHVYGSWNAL SSPFPKG EWA 300
 TTIHYMSSGQ LNVAPMISYR LPLAKGPETF QQIAGKELKP TKVLFYPEKL SERK 354

SEQ ID NO: 16 moltype = AA length = 351
 FEATURE Location/Qualifiers
 source 1..351
 mol_type = protein
 note = Lachnospiraceae bacterium
 organism = unidentified

SEQUENCE: 16
 MKVLSLHKPG DIRFEDAPMP EPGEGEVLLK VMACGVCSSD EARIFKDG AH IMPIVPGHEF 60
 CGQVVKTAG VDES LIGRKA SVFPMLPCFE CPSCRGDYA TCSNYKYYGS RNDGGMG EYR 120
 AIAWNLNLL DDSVDYCVGA LSEPSAVGHH ATEVGGVKAG DKV FVIGSGT IGLLIGFFAQ 180
 AKGAEVWIGA RRKESNDFAA SLGFKTIDTN NTEDEVAKAT DNGMDVVFE AVGSNKCLIE 240
 AIGSVTAGGT LVTGPNVPGD MNLPRDVYWR ILRKQLTVRG TWNSSYS DKA NDWHEVAALM 300
 KEKKIPFEKL ITRKYKLSQC DEAFDFFRDH SIQKSRVMFV MHEKESEAYN G 351

SEQ ID NO: 17 moltype = AA length = 336
 FEATURE Location/Qualifiers
 source 1..336
 mol_type = protein
 note = Faecalibacterium sp. OF03-6AC
 organism = Faecalibacterium sp.

SEQUENCE: 17
 MVNMMKALVF TAPHTFEYRD VPRPEPAPDE VLVEVKACAI CGSDVHGSSG ENGRRIPLV 60
 MGHEASGTIA QLGEAVQSWA VGDRTVFDST EYCGYCW FCT HGMHNLC AQR RIVGVACTEY 120
 RKNGAMAQYI AVKARTLYAL PPAVSFEQGC LVEPPAVGMH AVLRSGLAAG QTAAVIGDGT 180
 IGLMTLIAAK SQGAQVCISG HHTRPMQLAA QLGA LDCAQL PQLTAERGAD VVYDAVGSQS 240
 SFTQAMELVR KGG RIMVGN HARTIDFPLQ RCITEQIDVQ FSYSSAGEYP VCLQAIAEGR 300
 ADLTGLWQSF PLAEGAQVFD RLVRHDES LT KAVLIP 336

SEQ ID NO: 18 moltype = AA length = 357

-continued

FEATURE
source Location/Qualifiers
1..357
mol_type = protein
organism = Agathobacter ruminis

SEQUENCE: 18

MKAYVLDGIG	QLNQHEYVLR	DLQPGEALVQ	VAAAGICGSD	LPRIFETGT	YHFPTIPGHEF	60
AGVVKELYAD	ADEEMLRQWM	GKRVAVFPLI	PCRTCRCAN	HQYEMCSNYN	YLGSRDGGF	120
AEYCIVPVWN	LLQLPDEVSF	AQAAMMEPAA	VALHAIRRG	LHPDHPETGI	GKRVLMGLG	180
TIGGLIAQWL	SQCGVTELT	TGHHAPQEEL	MHRYANENYR	FINTNQTAWT	DDENTYDLIF	240
DCINDSESVQ	QALRYVKPGG	QIVAVGNPKK	DLTLQKDIYW	KLLRKQVCMT	GTWNSSFSDH	300
SNDDWQILLR	YLQEQLHPQ	ALITHRLPFA	QLPKGLEIMR	LREEFYNKIM	IDLTVCS	357

SEQ ID NO: 19 moltype = AA length = 342

FEATURE
source Location/Qualifiers
1..342
mol_type = protein
organism = Alicyclobacillus acidoterrestris

SEQUENCE: 19

MKAVTYTPDQ	GFALVEVETP	QLGPDEVLLQ	VEACGVCSD	RQVVAGESVP	TGTAFPLVMG	60
HEIAGRIAL	GETVTDWHVG	DSVIVHPPFA	CGTCAPCVHG	QPNLCVRQVC	IGYQRPGGFA	120
EQVAVPARQL	VRRPAELAPA	AAALVVDAFA	TPYHAMRAAG	VESGQTVLII	GTGGLGLAAL	180
MLAMGFHVR	LGAVTRRQEG	IAIAEAHGAQ	MVVVTDQDER	TIARQLRRWS	GASGIDVVID	240
TVATADTITL	AAESVRPGGT	IAIVGMSADT	ALYPVAKTVR	RGLTFVASYG	SVREDVEQLV	300
SWVAEGRLLD	VRLVAGALPL	HEAARAFAPD	RPAGRWDIDP	SE		342

SEQ ID NO: 20 moltype = AA length = 357

FEATURE
source Location/Qualifiers
1..357
mol_type = protein
note = Ruminococcus sp. KGMB03662
organism = Ruminococcus sp.

SEQUENCE: 20

MKSLINIEPK	MKALVLYGVG	DLRVTEVDVP	KLEKGTVLLK	IKACGICSSD	MPRCFVTGTY	60
HFPTIPGHEF	SGQIVAVGDD	VDESLLGKRS	CVFPMLPCKH	CKACKMEEYA	QCSGYSYFGS	120
RQDGAWAEYL	VVPVNNLVFP	DDTLDYKTAA	LCEPAAVSMH	AVNIGDIKEG	QNVVIIGTGT	180
IGFMIALFAK	ERTKTGKVVI	AGRSKNKLEY	AKKLGFDVVD	TSVDDLPAQV	KKIIGIDGAD	240
VTFEAVGSNE	AIASAIESTG	ALGSIVLVGN	PSTDLLLPKN	IYWSILRKQI	TVKGSWNSSY	300
NSRVNDWKA	LEVFEHSDVN	FADLITHFTT	IEENEKAFSA	IRNTGEFTLK	VMFTFDD	357

SEQ ID NO: 21 moltype = AA length = 348

FEATURE
source Location/Qualifiers
1..348
mol_type = protein
note = Clostridiaceae bacterium
organism = unidentified

SEQUENCE: 21

MKAVVLHGKN	DICYEWEPEP	ILKTGTVKIH	VKACGICGSD	IPRVLGDSA	YYPILGHEF	60
SGCVTEVADD	VDKIQVGDHV	AGVPLLPCKM	CDDCQQGNYS	QCKHYSFIGS	REQGGFADYV	120
VIPADNVVKV	DPTLPYEQVA	LFEPCTVALH	ALKVSSYKGG	KRVAILGGGT	IGLFALQWAK	180
IYGASQVTVF	GRSKEHLKIA	AALGADTTIS	TLDNFMTEA	MRVTEKKGFD	YIYESAGSTI	240
TMKLAFELAA	NHAKICFIGT	PTENLTFSPK	LWENMNRKEF	QLMGSWMSCS	APFPGDEWKE	300
TAHFNTGQL	RYVPEMFPAQ	YPMDDVKKAF	EQFHNKNKVK	GRILLKNK		348

SEQ ID NO: 22 moltype = AA length = 337

FEATURE
source Location/Qualifiers
1..337
mol_type = protein
note = Thermoplasmata archaeon
organism = unidentified

SEQUENCE: 22

MKVAVVYGDK	NIVLEDRPIP	SIGEKDVLVR	VKSSSICGAD	LHAYKGLSKR	RKPLVMGHE	60
FSGMVERLGD	ETKGLKVGRD	VVVEPIISCG	KCKACKKGKT	NVCENRKLIG	LNLDGAFAYE	120
VSVPYEKCHE	LPRDLSFDEG	AIIEPLAVAV	HAVNITPISS	NDNVLIIISG	VIGLLITQLL	180
RSRTSGDILV	TGRRDYLLEK	AREFGATDAI	NVIDCDAVKT	LLKKHGRVDV	VFEAVGSNET	240
VNQSLVLES	GGSTILVGM	DREMTVDMME	VVAREITLRG	DYGYTTDDFK	EAEELASSGK	300
VDLKLITHV	LPLDEIKKGF	DILAERRENV	IKVVINP			337

SEQ ID NO: 23 moltype = AA length = 343

FEATURE
source Location/Qualifiers
1..343
mol_type = protein
note = Lachnospiraceae bacterium
organism = unidentified

SEQUENCE: 23

MKALVLEEIG	KLTLKEVETP	VPGRGEVLLR	IHACGICSSD	KARLFKNGTH	TMPLIPGHEI	60
SGEIVAGGPG	IREELIGRRA	AVFPLIPCRQ	CDSCRIGKYA	QCEHYDYFGS	RRDGGFAEYL	120

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AVPVFNINLL	PDDMDYHTAA	LCEPATVASH	ALKRAKMKSG	DIVTIIGTGT	VGLLAAGIAR	180
IRRASKIIV	GRSKEKIMNA	AHFADHVINA	EGWDVNEAVA	SLTEGHMSDV	VLETGGNSDA	240
CTYALMAAGK	GTTVMFIGNP	EEPIVFNRI	YWKILRNELT	LKGTWNISGYA	PEDNDWNDVI	300
RYMADGQIPA	DKIITDSFPL	EDYNKAFELL	LSDRFYTKIM	FDI		343

SEQ ID NO: 24 moltype = AA length = 348
 FEATURE Location/Qualifiers
 source 1..348
 mol_type = protein
 note = Prevotella marshii DSM 16973 = JCM 13450
 organism = Prevotella marshii

SEQUENCE: 24

MKANVLYHIG	DLRYTDVALP	ELHDGEALVR	VRACGICGSD	VARVFTTGT	HFPTIIGHEF	60
SGEVVETYS	ADAEWVGRV	SVFPLIPCRH	CDACHHGQYE	MCTCYNLYGS	RTDGGFAEYV	120
AVPVNLFRL	PDSVSYEEGA	MMEPAAVAMH	ALKIARVGMG	DTVAITGPGT	IGMILTQLIQ	180
RIGGTKAVLI	GRTQAKLDYA	RRIGIRHVCN	SGTEHVADRV	GQLTGGRGAD	IAIEGTGASE	240
PMNLCLNLVR	SEGRIVAMGN	PQGDMLFRKD	AYWKLRLKQL	TLRGTWNSTY	GLGRDDWQDI	300
SALLAEGALN	LRGLITHRLP	LALBQQGLSL	MRDAAVYTNK	VMIVDEQG		348

SEQ ID NO: 25 moltype = AA length = 337
 FEATURE Location/Qualifiers
 source 1..337
 mol_type = protein
 note = Blautia sp. OM05-6
 organism = Blautia sp.

SEQUENCE: 25

MRAYELHGVN	DLRREDIEKP	EIPSGWVLVQ	VKASGICSSD	IPRIFTNGTY	HFPTIPGHEF	60
SGVVAAYGEG	VPEERVGKRV	GIFPLIPCR	CPQCRQKKYE	MCEHYDYLGS	RRDGGFAEYV	120
AVPDWNLLEL	PENVSFREAA	MLEPLSVALH	AVKRSVGVKPG	DTAAVIGTGM	IGFAAAAWAK	180
ALGATVTVFI	GRGEAKRAIA	EQIPGISYVT	EEESKAIQAD	VVVEAVGTPQ	AVSSAVLLAR	240
PGGSIVLMGN	PSGLAMEKN	VYWRILRKQL	NLMGTWNSSY	EKDEACDWTE	ALDALSKKKI	300
PAQALITHCF	PSEKLMGDL	LMKNHKEPYC	KVMVSWE			337

SEQ ID NO: 26 moltype = AA length = 354
 FEATURE Location/Qualifiers
 source 1..354
 mol_type = protein
 note = Candidatus Bathyarchaeota archaeon
 organism = unidentified

SEQUENCE: 26

MKRMRAAILY	GPKDIRIENI	EMPKVKPEW	LIKVRATGIC	GSDLHFYKEK	SFPIISDVG	60
KGRYVPGHEF	SGEVYELGSK	VKGLKKGDRV	GVEPIIGCGK	CQWCKVGCYN	LCEHYKLIGF	120
HYIGGMAEYC	VAPAEFCFL	PNNVSFESAA	MLDCIAVAIH	AVNKAIEIRGG	DSVLIVGAGA	180
IGLFALQAAL	IQGASDVYV	GKYEFLQLEK	RKLGAATVIN	AKNESFSEKI	MEITSGKVD	240
KVIEAVGGAS	NVISDVLDVT	RKKGITVTTG	IFTRPLSVDF	FKILTKEITI	TSSWGYCYFD	300
RRKEFGTALD	FLAKEKINTQ	ELITHKYPLE	KVSEAFETAL	NKNKEKSIKV	EIIF	354

SEQ ID NO: 27 moltype = AA length = 350
 FEATURE Location/Qualifiers
 source 1..350
 mol_type = protein
 note = Lachnospiraceae bacterium
 organism = unidentified

SEQUENCE: 27

MKAUWLHGIG	DIQYEEVKEP	QSAGDEVIMK	VMSVGICGSD	IPRIYKTGT	HFPTIPGHEF	60
SGIVADVAG	VNPWKWKRV	GVFPLIPCHV	CASCKKGQYE	MCRNYGYLGS	RQNGGFAEYA	120
AVPVKNLIEL	PDSVSMQEA	MLEPMAVAH	AMRRARPMTG	DRIVVCGLGT	IGMLLVSFRL	180
ETGIRDIFVI	GNKAFQRITI	LKTGLQENCY	CDSRVQDVEK	WLMERTEGKG	ADVFFECVGR	240
NETISQAVNL	TAPGGSVCLV	GNPCSDMTLE	REVYWKILRN	QLTITGTWNS	SFTHEIEDDW	300
HYVLDRLQEG	KISPGDLISH	KFPLKNLESG	LHIMRDKTQD	YIKIMAEINV		350

SEQ ID NO: 28 moltype = AA length = 348
 FEATURE Location/Qualifiers
 source 1..348
 mol_type = protein
 organism = Alkalihalobacillus ligniniphilus

SEQUENCE: 28

MKAUWLHSEN	LFTYEEVEID	PCKDDEVKIK	VMAAGICGSD	THKMTSKWKY	PLPTIMGHEF	60
SGFIVEIGEK	VTNASIGDRV	VAIPFLPCND	CEYCKRGQFS	LCEDHGMVGS	KSPGGFAEYV	120
NTKSTNVLP	GDMDFEEAAM	IEPLAVALHG	VLNIHPRIQD	IAAVFGIGTI	GILTIQWLKI	180
AGVKEVIAID	ISDNKLADAH	AMGCERTINP	FKENLLEKID	EYTNRGVVDI	AFECAGSKVT	240
QEQCLLVTA	RGKVGYLGLA	YSDVHLSEQA	FENIFRREL	LKGFWNISYS	PPGGEWTKS	300
IDFINKGRK	LKELISHRFP	LQDTQKAFDM	MINREETYGK	VLILPNKE		348

SEQ ID NO: 29 moltype = AA length = 348
 FEATURE Location/Qualifiers

-continued

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source                1..348
                      mol_type = protein
                      organism = Jeotgalibacillus soli

SEQUENCE: 29
MKAAYLHSEN HFIYEEIDIA PCQDDEVKIK VIATGICGSD THKMTGKWKY PLPAVMGHEF 60
SGYIVEKGKN VTNAAGIDRV VAIPFLPCND CEYCKRGQFS LCEDHGMVGS KSPGGFAEYV 120
NIESSNVLP I GDMEFEAAAM IEPLAVALHG VLNIHPRVGD VAAVFGVGTI GMLTIQWLKI 180
AGAKETIIAD LSDEKLADAR AMGCEITINP LNEDLLEKIN EYTNGMGVDI AFECAGSKIT 240
QEQCCLITRK QGKIGYLGIA YSDVHLSEKA FENIFRRELK LKGFWNSYSA PFPGEWTKS 300
IEFIKNGRIK LKELISHRFK LEDTQKAFDM MINREEKYGK VLILPNEE 348

SEQ ID NO: 30         moltype = AA length = 350
FEATURE              Location/Qualifiers
source                1..350
                      mol_type = protein
                      organism = Heyndrickxia sporothermodurans

SEQUENCE: 30
MKAAYLHSEN EIIYQDIEIS ECQPNEVKVK VMAAGICGSD THKMVSKWKY PLPAIMGHEF 60
AGYVVEKGKD VTNVEIGDRV AAPFIPPCNE CEYCLRGQFS LCENHGMIGS KSPGAFAEYV 120
NIPAKNVLP I GEMSFEDAAM IEPSAVSLHG VLNIHPQLGD VVAVFGVGTI GILAIQWLRV 180
AGVKEIIAID ISDKKLQDAM DLGCEKSINP LKEDLIKKIY EYTNGLGVDI AFECAGSKIT 240
QEQCCLITRK QGKVGYLGLA YSDVTLSEKA FENIFRRELK LKGFWNSYSA PFPGEWMNT 300
IEFVQGGKIK LKELVSHRYK LEDTQKAFNM MINREEEYNK VMIIPNEELE 350

SEQ ID NO: 31         moltype = AA length = 346
FEATURE              Location/Qualifiers
source                1..346
                      mol_type = protein
                      organism = Clostridium fungisolvans

SEQUENCE: 31
MKAAYLHSVG DIRYENIEIK PYGDDEVKIK VKAAGICGSD PPRALKKWKY PVPAILGHEF 60
SGYIVEVGKN VKGPKVGDV VAVPLEPCHQ CEYCKKGQFS LCESYDMLGA TSPGGFAEYA 120
NVKATNVLP I DDIDYEEAAM IEPLAVALHG VLNIHPQLGD TVAVLGAGTI GQLTIQWLKV 180
SGVEKIIAIV ISDKKISEAM SLGADIGINA LKENPVEKIM ELTNGFGVDI CIECAGSKIT 240
QEQCCLLVTKK KGKIGYQGIG HAGIELSEVA FEGIPRKELN IQGFWNSYSA PFPGEWFPS 300
IEYVQKKIK LKQLISHRFS LEDTAKAFEM IGERKEEYNK IMIVQD 346

SEQ ID NO: 32         moltype = AA length = 348
FEATURE              Location/Qualifiers
source                1..348
                      mol_type = protein
                      organism = Alkalihalobacillus nanhaiisediminis

SEQUENCE: 32
MKAAYLHSEN HFIYQDMESF ECQNDDEVKIK VMAAGICGSD THKMVSKWKY PLPAIMGHEF 60
SGYVVEKGSD VANVEIGDRV VAIPFLPCHT CEYCKSGVYS LCEDHGMIGS KSYGGFAEYV 120
NLKATNVLP I GNMDFEAAAM IEPLAVALHG VLHIQPRLDG VAAVFGVGTI GMLAIQWLKI 180
AGVKEIIAID ISDEKLAEAK AMGCEITINP LQDNLEKIVH EYTNGLGVDV ALECAGSKIT 240
QEQCCLITRK KGEIGYLGIA YSNVVLSEQA FENIFRKELT LKGFWNCYSA PFPGEWTKS 300
IDFILKGR I K LKELISHRFP LEDTQKAFDL MINRKETYGK VMIPNKE 348

SEQ ID NO: 33         moltype = AA length = 348
FEATURE              Location/Qualifiers
source                1..348
                      mol_type = protein
                      organism = Neobacillus cucumis

SEQUENCE: 33
MKAAYLHSEN QIVFQEF EIG ELKPDEVKVK VMAAGICGSD THKMVSKWKY PLPAIMGHEF 60
AGYIVEKGKD VVHVEVGDRV AGIPFVPCND CEYCFRGQFS LCENHGMVGS KSPGAFAEYV 120
NIPAKNVLP I GDMDFEDAAM IEPSAVALHG VLNIHPQIGD VVAVFGVGTI GILALQWLKV 180
AGVKEIIAID ISDKKLQDAI ALGCERTINP LKENLLEKIH EYTNGLGVDI AIECAGSKIT 240
QEQCCLITRK QGKVGYLGLA YSDINLSEKA FENIFRRELK INGFWNSYSA PFPGEWTNT 300
VEYVQGGKIK LKELISHRFP LEDTQQAQNM IINREEEYNK VMIVPNGE 348

SEQ ID NO: 34         moltype = AA length = 341
FEATURE              Location/Qualifiers
source                1..341
                      mol_type = protein
                      organism = Staphylococcus aureus

SEQUENCE: 34
MINQVYQLVA PRQFDVTYNN VDIYGNHVIV RPLYLSICAA DQRYTGRRD ENVLRKKLPM 60
SLVHEAVGEV VFDSKGVEFEK GTKVVMVPNT PTEQHIIAE NYLASSYFRS SGYDGFMDQY 120
VVMADHRIVP LPNDIDLSTI SYTELVSVSY HAIQRFERKS IPLKNSPGIW GDGNLGYITA 180
ILLRKLYPEA KIYVFGKTDY KLSHFSFVDD IFTVNQIPDD LKIDHAFECV GKGGSQVALQ 240
QIVEHISPEG SIALLGVSSEL PVEVNTLRLV EKGLTLIGSS RSGSKDFEQV VDLRYKYPDI 300
VEKLALLKGH EINVTQLQDI VQAFEMDLST SWGKTVLKWT I 341

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SEQ ID NO: 35 moltype = AA length = 350
 FEATURE Location/Qualifiers
 source 1..350
 mol_type = protein
 note = Staphylococcus aureus subsp. aureus 71193
 organism = Staphylococcus aureus

SEQUENCE: 35
 MLLFEVVLPM INQVYQLVAP RQFEVYNNV DIYSDYVIVR PLYMSICAAD QRYYTGSRDE 60
 NVLSQKLPMs LIHEGVGEV FDSKGVFNKG TKVVMVPNTP TEKDDVIAEN YLKSSYFRSS 120
 GHDGFMQDFV LLNHDRAPL PDDIDLSIIS YTELVTVSLH AIRRFEKSI SNKNTPGIWG 180
 DGNLGYITAI LLRKLYPESK IYVFGKTDYK LSHFSFVDDV PFINKIPEGL TFDHAFECVG 240
 GRGSQSAINQ MIDYISPEGS IALLGVSEFP VEVNTRLVLE KGLTLIGSSR SGSKDFQDVV 300
 DLYIQPDIIV DKLALLKGQE FEIATINDLT EAFEADLSTS WGKTVLKWIM 350

SEQ ID NO: 36 moltype = AA length = 341
 FEATURE Location/Qualifiers
 source 1..341
 mol_type = protein
 organism = Eubacterium ventriosum

SEQUENCE: 36
 MINTVYQLVA PRRFEIKYSD IDLNNNKVIV RPTYLSICNA DQRYYQGLRD AEILRKKLPM 60
 ALIHEAIGEV VRDPGNGFKT GELVVMVPNT PVEEDDIAE NYLRSSKFRA SGFDGFMQEV 120
 VEITPDRLVR LPQNIDRTVA AFTEIVSVSV HAIGRFDKIA HKRRNVVGIW GDGNLGYITA 180
 VFVKTMFPET KLYIFGVNPD KLQDFTFADA TFTVEHIPED VKIDHAFECV GGAGSGKAIN 240
 QIIDYINPEG TISILGVSEY PVPINTRMVL EKGLRIFGSS RSGVADFEXT VALYKSNPEI 300
 INYLSNIVGA VVPVRTIEDM KKAFEMDIQK AFGKTIMMWD K 341

SEQ ID NO: 37 moltype = AA length = 343
 FEATURE Location/Qualifiers
 source 1..343
 mol_type = protein
 note = Pradoshia sp. D12
 organism = Pradoshia sp.

SEQUENCE: 37
 MINQVYRLVS PRQFEVAYND RSLQYHEDFV VVRPTHLSIC AADQRYYTGS RGKEAMAKKL 60
 PMALIHEAIG KVVYDPKGEI KNGTMVVMIP NTPTEKDEVI GENYLRSKF RSGYDGFQM 120
 DYVFMGRDRI VELPEGIDPY VVAFTELMTI TTHVLSRFES RSHKRRNVFG VWGDGNLGF 180
 TTLFLKKMYK DCKVIFGKGT GYKLDNFSFV DEVYTIDEIP EDLRIDHAFE CVGGKGSQYA 240
 IDQMIDHLHP EGSIALMGVS EPFVEVNTRM ILEKGITLIG SSRSGREDFV RTVNFLKENP 300
 DVVEYGLTLV GEVQEVKTIS DILSAFEKDL TSSWGKTVME WKI 343

SEQ ID NO: 38 moltype = AA length = 341
 FEATURE Location/Qualifiers
 source 1..341
 mol_type = protein
 note = Lactobacillus plantarum EGD-AQ4
 organism = Lactobacillus plantarum

SEQUENCE: 38
 MLNQVYRLVD PRQIEVQTV AEEITDNDIV RPKYLSVCHA DTRYFTGQRP QATLRQKLPM 60
 ALIHEGVGEV VKDPQNKFKP GTLVAMVPNT PFETDPIIE NYLPTSKFRS SGYDGFQM 120
 VTLHRDRAIV VPDNFDHOMS AFTEMVSVGV HALTQLEAVM DADRKIIGIW GDGNLGFITA 180
 TLVKQIFPDS QLMIFGRHRS KLDYFSFADK TYLVDDIPSD LQVSALECT GGRGSESAIA 240
 QIIQHIRPMG TAILMGVSED PVGIDTRSVL AEGTLRGVS RSGRADFQRA IDILTKSPVT 300
 RERLQNLVGF TRKVSTIQDI TDFFEGALTN YWGKAVMEWD V 341

SEQ ID NO: 39 moltype = AA length = 337
 FEATURE Location/Qualifiers
 source 1..337
 mol_type = protein
 note = uncultured Ruminococcus sp.
 organism = Ruminococcus sp.

SEQUENCE: 39
 MINYVYQLVS PHVFSKYEE VDFKKEVIIR PKCMAVCHAD QRYFLGRRDA KILRQKLPM 60
 LIHECCGQVV WDGTSFQRG QKVVLIPNVP GRDDGVTYEN YAEGRFLSS GCDGFMREYV 120
 NLSPDRVVPF EQVPLPVAIA TEFVSVAVHA AGRFMRAAHT RREAGVWGD GSLAYVCCV 180
 LRRLLPESRI FVVGKHERKL LQFSFADRTF LFDLSPEDFH IDHAFECTGG EGSYYAIDDI 240
 ITAVRPQGA VMLMGVSEN RVTRTRDILEK GLTFTGCSRS GREDFVKAVE LMEQPQFQKR 300
 LSAIYEDRP VHTVADLSRV FETDLNTPFK TVFRWEI 337

SEQ ID NO: 40 moltype = AA length = 338
 FEATURE Location/Qualifiers
 source 1..338
 mol_type = protein
 organism = Kandleria vitulina

SEQUENCE: 40
 MINTVYRLIA PKSIRADFVD LKLDEERIII RPTYLSICAA DQRYYTGSRG RVVMKQKLPM 60

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ALIHEAVGEV	VYDPKGELPV	GTHVVMIPNT	PNEKDDIIKE	NYLRSSQFRA	SGFDGFMQNV	120
VSMRRDLVIP	FDYEPEVGV	LELMVGMNA	VSTFKKHAHD	RKDAIGIWGN	GNVGFVTALI	180
LRKEFPDAKI	YVFGVEEHKN	RYFSFADDTF	IINDIPEDLQ	LDHAFECVGG	RGSEAAISQI	240
IDYIKPQGTI	NLMGVSENPV	PINTRMVLEK	GLQLLGNSRS	NYDDFKKSVD	MMANNKDVRE	300
YLSTIVSEMV	VVNTIDDMIR	AFEDDTYNDF	KTMVKWEI			338

SEQ ID NO: 41 moltype = AA length = 341
 FEATURE Location/Qualifiers
 source 1..341
 mol_type = protein
 organism = Lactiplantibacillus plantarum

SEQUENCE: 41

MLNQVYRLVS	PRQFEVETVA	EEVDADDLIV	RPKFLSICHA	DTRYFFGQRS	QAALRAKLPM	60
ALIHEAVGEV	VYDPRNEYKP	GTLVAMVPNT	PVKKDSIIQE	NYLPSSRFRS	SGYDGFMEY	120
VCLRRDRAT	VPSDFDPHMA	AFIEMVSVGV	QALTKLESTM	DEDQRVIGIW	GDGNLGYITA	180
TLVKQIFPDS	QLLVFGRHQS	KLDYFSFADK	TYLVDDIPAD	VKVSQLLECT	GGGSEAAIQ	240
QIIQHIQPMG	TAILMGVSEN	PVGIDTRSVL	AEGLTLRGVS	RSSREDFKRA	VDILSSSPVT	300
RERLQNLVGY	VQKVSSIQDI	TNFFDSALTN	YWGKEVMEWD	V		341

SEQ ID NO: 42 moltype = AA length = 342
 FEATURE Location/Qualifiers
 source 1..342
 mol_type = protein
 note = Lachnospiraceae bacterium
 organism = unidentified

SEQUENCE: 42

MINTLYKLTA	PLIVESFYEN	IDCHSDNIIV	RPEKLSICKA	DIRYFFGMRN	AKVLKEKLPL	60
ALIHEACGSV	VYDITGKFKA	GDQVVMLPNI	PGINMQYHEN	YRRDSLFRSS	GADGFMQELV	120
SIPQSQVIKF	EKMQDNIAAF	TEFISVAIHA	VSSYLDRSAG	KRNGIKRKIG	VWGSAGMGYA	180
VSCVLKYFIP	SAETITVIGGS	KAKLDMFSFV	NHTCSLSELP	DDAEFDCCFE	CVGGAQSGNA	240
INQIIDHILP	EGLLLLVGS	EEAVPVNTRM	VLEKGITLLG	RSRERSDFI	KTIDILDITNE	300
NIRRRISNLI	SGEVEINEVN	DIHKAFNQSK	IADYKIILNW	RI		342

SEQ ID NO: 43 moltype = DNA length = 1599
 FEATURE Location/Qualifiers
 source 1..1599
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 43

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SEQ ID NO: 44 moltype = DNA length = 4618
 FEATURE Location/Qualifiers
 source 1..4618
 mol_type = other DNA
 organism = synthetic construct

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FEATURE	Location/Qualifiers
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SEQ ID NO: 46 moltype = DNA length = 4594
 FEATURE Location/Qualifiers
 source 1..4594
 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 49 moltype = DNA length = 4627
 FEATURE Location/Qualifiers
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 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 49

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SEQ ID NO: 50 moltype = DNA length = 4594
 FEATURE Location/Qualifiers
 source 1..4594
 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 51 moltype = DNA length = 4624
 FEATURE Location/Qualifiers
 source 1..4624
 mol_type = other DNA
 organism = synthetic construct

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organism = synthetic construct

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aaagactgaa	agtcocaaag	attgctcaaa	ggccctcgag	gtctgagcct	ctgagcgcgc	3000
tcttcgtcaa	aattcgccgg	ggccatcggt	catcctctcg	agggcagcat	cttctcgctt	3060

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gaagactgag tccgtctcag ataacttcgt ataatgtatg ctatacgaag ttatcctgca 3120
ggagctcaga ccttgcatgt tagccttgac cttgcctcta tgaccactaa ttcgacacaag 3180
attgcacgtt accattgtca tcttgccatg tatccaatgt ttctgtttg ccatagtacg 3240
tagtagatac atcgatcatc gatgcagatg acgcagatgg cattgggttcg cttgccccta 3300
tgcattgcatc atgcatcatg catcatgcgt catgcgtcac tatccatatt aatcattgcc 3360
ctctgtacta ggccatgaca cctgcagacc cttcccacag gatgcttctt gtaatagcca 3420
atatactaca caactttacc gattacgata gagatcgatc tcaactatgt ttgtcgaact 3480
tcgatgcaca tcgaggggtgc ggaaaaaagc tgacggaagc ggaagtgggt gaattgggta 3540
atattagttag attggcaggg ttgttcccaa gattcaacgc aagctgagaa agaaaacaga 3600
aacacgaggg ataaggccag tcagaaaatg cattacgagg ggagtgaag tgactgtggg 3660
ccagtgcctg ggtcgctcatg tgcaaatctt ttcttcggtt agactgcgcg agcatgcagc 3720
aactctgccg gccctatata actagtgcct catcttcttg tcattcttgt cttcttacca 3780
acacaccggt actctttcac catgggtaag gaaaagacac acgtatcgcg accccgactc 3840
aactccaca tggcgcgaga cctgtacgga tacaatggg ctcgagataa tgcggacacg 3900
tcgggtgcga cgtatctacag actctatgga aaaccggacg ccccagactc ctttcttaaa 3960
cacggcaagg gtacggttgc aaacgacgta accgacgaaa tggtcctgtc gaattgggta 4020
actgagttca tgcacactcc caccatcaag cacttcaccc gtacacctga cgtgcgtgg 4080
ctgttgacta cggccatccc cggtaagacc gcttttcagg tcctcgagga ataccctgac 4140
tcgggcgaga acatcgtgga tgccttggtc gtgtttttgc gccgccttca ctctattccg 4200
gtctgcaact gccctttcaa cagcgaccgt gtttttaggc ttgcccagcg acaatcacgc 4260
atgaacaatg gtctagtcca cgtagtgtat ttcatgacg agcgcaacgg ctggccgggtg 4320
gaacaagtgt ggaagagatg gcataagctc ctgccctttt cccagactc tgtcgtgacg 4380
catggtgatt tctcgttga caacctcatc tttagcagg gcaagctcat tggctgcatt 4440
gacgtcggcc gggttgccat cgcgatcgc taccaagatc ttgcgatcct ctggaattgt 4500
ctgggcgagt tctcccctag cctccagaag cgtctgttcc agaaatatgg gatcg 4555

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SEQ ID NO: 85      moltype = DNA length = 500
FEATURE           Location/Qualifiers
source            1..500
                  mol_type = genomic DNA
                  organism = Moniliella pollinis

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SEQUENCE: 85
tcagagcct tgcaaatgat gaaagaccaa tgggaattgat cattgattat tgatgatgca 60
tgatgaaaga gatagaaaca gatgcaagat gggatggcgc gatgatcaaa tgataatgaa 120
cgttcaaccc ttcaaaattg aaactctcag gctgagcggc tgagcaagct ctgagaatgt 180
gggtcttcga agactttcgc ttgagcatgg tgcattgtgca aagagatgga atgtggaatt 240
gttgtaggta ggaacacagc ccggtagagc ggcgcagtggt ctcagcgaag ctgggagaaac 300
gactgaacgc gtcttgagat tcacatctc agcgtgcgtc agcgcgtcag cttcagcgtt 360
tgagaatggc acggactgac ctgtcgtaca tcccaaggag gcccggtgcg tatactcgcc 420
ttgtggccta atctgcgata aacctatcct caacgaaagc gcaagtacgg actaagcaca 480
ctatctttcg ggtcggtgcg

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SEQ ID NO: 86      moltype = DNA length = 1002
FEATURE           Location/Qualifiers
source            1..1002
                  mol_type = other DNA
                  organism = Moniliella pollinis

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SEQUENCE: 86
tttagctacg atggtgtgcg tgtaactgat cagtacaccc gctgtatcaa aactctatct 60
tcaatccgct aagaatcgtc accatcagta catcagtaca tcagtgggct aagatctaatt 120
ggtgtcgctt gacggcgcgc ggcactagcc acgagccaag agcgggtacag ttgctgcgaa 180
atggtgggca gccagacagg ttcatgttga ataggatgaa aacgccatga agaacagaga 240
agctgggcaat tgacgatagt ggggcgagcg cacgctgctt gctgggtccga gcgtaggacg 300
aacgacaaac catgggcggg gccaccagcg cccaggggagg gcgtcactgg ttggatgtcg 360
tctgttaggc tagagctcag gcgctcaggc cctctggccc aagcctcgga cgcaagcctc 420
tgacgagcga gcgcgcgctg gggcgggtct gcgcaggaaac ggcaggactg ccgagttggt 480
tgaatattga atggttggtg gattggtttg ctgggttttg ttttgacacg caccggcacag 540
gacaagttag tttggaggtt ggatttgcgc aggtagaaag ggcgctcggc ctcagggcct 600
cagcgtcagc agacgagctc gccagtcagc gtcagtaccc agtgttcgtt acttcccacc 660
gtccagagtt tgctgtgccc tctcagcttg attcgtgtcg gagaagtaga aatggtcaca 720
gcgtaccaca cactgtcccc atgcctcgct cctgctccac gagtgtttag ccacaaaaat 780
ttatggacaa ccttactgag cctgtgctgg ggggtgcttt agcaccagca ctggcgccat 840
ggcgtcaagg ttaattccac tgtcattcca caacctcaa ctgattgggg caactgggca 900
caaaaccttg gatataaac caactccaac tcccacgggt ttttaccttt tgctcttctc 960
cccaggactc actacttcta ctctctatt catcttcacc aa 1002

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SEQ ID NO: 87      moltype = DNA length = 1038
FEATURE           Location/Qualifiers
source            1..1038
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 87
atgaaagccc tcgttttggg cgaatacaat caattgaatt accgtgattt ccccgagccc 60
gaaatccact ccggagaggt gctgatcgag atcaaggcgt gcggtatttt tggctctgat 120
gtgcacggta tcgacggatc gaccggacgt cgaaaaccac cgtgggttat gggtcacgaa 180
gcctctggcg tcattatcga aaccggtagc gatgttaaga actggaaaac aggtgaccga 240

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gtcacgcttg atagcacctg gtaccacatt aatgactggt tcaactctgaa aggccactac 300
aacctcagca acaatcgaca agtggtggga gtgtctcctg gcgagtatcg cegtcattggt 360
gcggtttgccg aacgacttgc ggttccgcaa catatcctgc accgaatccc agagaacggt 420
agcttcgtgg aggctgcgat ggttgagccc atcgctgtgg cagctcacgc ggtgaatggt 480
tctggattga agccgggcaa ttccgctatg gttgcgggcg cgggcatggt cggagtgttt 540
gttgccaac tgctgaagct ttttgagct aacctgttg ttggtattga catcgatcac 600
aagaaactga agcttgctca cgaattcggt gctgacttta cattcaatcc taaagaaaag 660
gatattgacg aacgaatctt gcaactcaca aaatcgcgcg gttttgacct tacatttgaa 720
gtcgtcggca tcacgcgagc ggtccagacc tgcattacca acacacgtaa ggggtggcacg 780
gtggttcttg ttgaaaactt gtcccccga atccaatttc ccctgcaaaa agtcgttact 840
tctgaaatca ctgtgcaggg ctcttgtgcc atcaagggtg aatacgagct ggcactggac 900
ctcttgcccg caaagaaggt tgacgtctcg aaactcggtt cggctgttgc accgctgtct 960
gagggagcgg aatggtttaa gcgactttac gagaaggaac cgggattgaa taaagtgtac 1020
ctcattccag ataagtaa 1038

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SEQ ID NO: 88      moltype = DNA length = 978
FEATURE           Location/Qualifiers
source            1..978
                  mol_type = other DNA
                  note = Candidatus Dormibacteraeota bacterium
                  organism = synthetic construct

```

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SEQUENCE: 88
atgaaagcta tgggtgtggt tggaccggaa cgaatggaag tgcgagatct tectgcgccc 60
cgaccgggtg caggtgaagt gctgggtcag acggaagctg ctggaatttg tggtctgtaa 120
gtggaggcct atttgggtcg catggctaata cgaacaccgc cgttgggtgat gggtcacgaa 180
tatgctggac ttgtcaccga ggtcggagag ggagtcgacc gcgtctggtc cegtcacgc 240
gttgccgtta acccgattgt tctgtgagct cgttgcgaa actgcgcgcg cgggtgacga 300
aaactgtgtc ctgactgtca cctcttggc attgcagcgc caggaggatt cgcagcgttc 360
tcgggtggtc ccgaacgatg cctcttttct ctgctgaag gaatggatgc gcgactgggc 420
gcgctcacca agccattggc caacggtgtg catcttattc gcaaaagcgg cgcgggacga 480
ggccgagttg tgggttatcg gcgagggaact attgactcg gatgtttgca agctgctttg 540
ttgcacggtg ctgagcaagt tgacgtcgtt gaacgccatc cgggacgacg ccgtcacgca 600
cttgccctgg gagcacgcca ggcggtggcg gaattggcag atgtcgaacc ttggtctgat 660
cttgtggttg atgcgcgagg atccgaagcg acacgcgcgc cggccgtgga attggttggc 720
cggcgggaaa cggcctcttt tgttggactc cactccgacg aaactgtcct gccttggcat 780
cgcatattcc gagccaatat tgcagttcaa ggaacatttg gctacgcaga tctgtatttc 840
cagcaagctc ttgactgggt ggcagcggga aaggcagcca ttccattgtc tgaactccgc 900
cccttgagcg atggcccagc ggccttcgag cgccttgctc gtggaccgat cgacgaaatc 960
aaagtctttt tggcttaa 978

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SEQ ID NO: 89      moltype = DNA length = 1014
FEATURE           Location/Qualifiers
source            1..1014
                  mol_type = other DNA
                  note = Candidatus Falkowbacteria bacterium
                  organism = synthetic construct

```

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SEQUENCE: 89
atgaagtcgt tgatttacta tggcccgcac gatctgcgat atggagattc tcctaagcca 60
aagatccgat ctggagaagt cctcctccga atcaaaagcg tcggtatctg cggcactgac 120
ctccacatct ataacggagg catgaaaaat ctgcccgttg ctctgattat gggtcacgag 180
ttcagcggag ttatttgaaga gcttgcgcca gacgtgaaga actttaaggt ttgagatcga 240
gtgatcgctg aacacgttat caattgttga aaatgttcgt attgtcttag cggcaagccg 300
aatctgtgtg ctaactctcg cattatttga gttcacccgc ccggagcgct cgcgcaatac 360
ttggtgttcc ccgcggaact ggtgtacaag atcccggcga aactgtcttt tgaagagggc 420
gcaatgatcg aacctcttcc tatgcctatg atgcagctcg tagcatctcg 480
ggtaaaaagg tggcggtcat cggccaagga cccatcgccc tgctgcttga ccaaactctc 540
cgcgacgcag gcgctgaagt cattggcatt gacatcgacg attggcgatt gaaactcgcg 600
aagaaaaaga agtgggcaga tcatacaatc aactccagcc agaaaacctg gctcaagaag 660
atccagaaga tcgcaccaga aggcgtggac acagtgtttg aggttgttgg ccgtgaggaa 720
actgcccaga ttgctattga aatcgcgcggt cgagatgggt acgtttttat cctcggtgtc 780
tttgaaagcg ccgcatcttt gaacatgatg aatattgtta agaaggaact caacgtgttc 840
ggaagctgga cctgtgcggt ttctttcccc ccttcgattg ctctcgccgt tgcgaagaaa 900
gttgatttga aatccttgat cacgaatcga tatgacgtga aggacggtgc cctggcattt 960
aaggagtctg cctcttataa aggtaaagca atcaagactg ttatcaactt ttaa 1014

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SEQ ID NO: 90      moltype = DNA length = 1044
FEATURE           Location/Qualifiers
source            1..1044
                  mol_type = other DNA
                  note = Acidobacteria bacterium
                  organism = synthetic construct

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SEQUENCE: 90
atgaaggcaa tgggtgcttag ccagtatcga cgactggaaa ttgccgagtt gccgacaccg 60
atccctggtc gtggcgaact cttgatccat gtcgcggcat ttggtatttg cgggaagcgat 120
gtgcacggat acgatggcag cagcggacga cgtatccctc caatcgtagt gggccatgag 180
gctgcaggac gtgttgcggc actcggtcgc caggtcacgc gctggtcgga aggtgatcgc 240

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gtcacattcg atagcaccat ctctcgccgc acttgccctt actgctcccg cggcgagatt 300
aacctctgcg accgtcgta agtgcttgga gtttcgtgcg gtgactatcg ttgcgctgga 360
gcttttgcgg agtatgttat tgtgccgcaa cgcattgctt ataaattgcc ggatgccttg 420
agcttctcgg aggcgcgaac ttggaagcg gtccggttg ctgtgcacgc tgttagcctt 480
tcccaggttt cgcaggttac tcaggcactg gttattggcg ctggtatgat tggcctctc 540
accttgcaag cgctgcgcgc agccggttgc agctccgtgg tgatccggga cgttgatgcg 600
tcccgtctgc gtttggttaa agaggttggc gcaaccgaaa cgcttaatgc gacaggagat 660
gaatttggtt ccgaaatctt caaacgcaca aacggagcgg gcacagacct tgcgctgaa 720
gctgtcggta tcgatgttac cgtccgaacg gccattaaact ctgtccgaaa aggcggtaca 780
gttaccctgg tcggcaatat cacgcggag gtcaactctt cgcttcagaa ggtggtgaca 840
cgccagattc gaatccaagg ttcttgcgca tccgctgggt aatacccccg agctatccag 900
ctccttacgt ctggcgctat taaagtcaaa cctctcatta ccgctgtggc cccgcttgaa 960
gacggtcctc agtgggttga gcgcctttac agccgcgagc ctaatcttat gaaggtggtc 1020
cttaccgccg gcagcggtgc ttaa 1044

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SEQ ID NO: 91      moltype = DNA length = 1035
FEATURE           Location/Qualifiers
source            1..1035
                  mol_type = other DNA
                  note = Spirochaetaceae bacterium
                  organism = synthetic construct

```

```

SEQUENCE: 91
atgaaagcga tgggtgtgaa agaataaag tatctggaag tcgaagacat tctatttcg 60
cacatcgaa cagcggacga actgttggtc cgcgtgaagg ctaccgccat ctgcggttcg 120
gatatccatg gttttgatgg ttccaccgga cgcgtaaac cgccattggt catgggacac 180
gaggcatcgg gtgtgtgcga aagcgtcgga agcgaagtta aatcgctgaa aatcggtgac 240
cgtgtcacct tcgactccac agtttgggtg ggcgaatgct acttttgcg acagggtcag 300
gttaacctgt cgcgcaaacg acgtgtgctt ggtgtgctgt gcgacgaata ccgcaaggac 360
ggtacctttg cggaaatagt catttgccc gagcgtattg tgataaagt gccggataat 420
gttgcttcg aagaagcagc attgactgag ccggtttctg tcgcatgca cgcatttgat 480
atcccggaat tgaagccagg agaaaatgcg gccgttatcg gtacgggtct taccggactc 540
ctcctgattc aaattatccg tgettattcg cccacttga tcatcgctt cgatacacag 600
gaaaaacgcc gaaaagcagc tctcgaagct ggagccgata ttgattgga cccccaat 660
cctgagtcgc ataaaactgt cctgagcatg actaataacc gaggtgtcga ccgagcattt 720
gaggctgtgg cgcgcctcgg agcgatcggt acagctattg aatccgtccg aaaaggctcg 780
tccgtcactc tgatcggcaa tattagcaat aaagctgaga tcccgctcca atctgtggtt 840
acgcgccaaa ttgccccttt gggttcgtgc gctatttctg gagagtatcc tgtctctttg 900
gcttgatgg cgaatcgcac tattgacgtg cgtcaagtta tctccgcaat cgcctctttg 960
gaggagggtc ctctgtggtt cgaacgctt tataaacgcg aacctggtct ttgaaaggtt 1020
gttctcacgc cgtaa 1035

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SEQ ID NO: 92      moltype = DNA length = 1047
FEATURE           Location/Qualifiers
source            1..1047
                  mol_type = other DNA
                  note = Candidatus Hydrogenedentes bacterium
                  organism = synthetic construct

```

```

SEQUENCE: 92
atgaaggcac tccgcctgta cgctccaggt gacttgcgtt tggaggaaat cccaattcct 60
ttgccacgtc ccgatgaggt gttgggtcgc gtgcttgcgt gcggtatctg cggttcggat 120
atccccctgt tttttgaaca cggtaacgtat aaatacccat tgggtgccgg ccgatgagttt 180
tccggtattg ttgaaaaagt ggctccgaa acgggaaaag attgggttgg tcgtcacgtt 240
gtcgtctatc cctcctatcc gtgtcgacaa tgcctctctt gctactctgg tctttatacc 300
caatgcctca actacgacta ctctgggatc cgcctgacg gaggatttgc agagtatgtg 360
actgtccctg tgaataatat tatcccggtc cctaattgaa tctctggaga gatctcggcg 420
atcactgagc catgtgcggt cgccttcat gccgttcgca agggcccaagc ctgcgtgggt 480
gacacggtcc ttattacggg agctggaacc attggtatct tgatcgccct ctgggcgcga 540
gcagcaggtg ccgaagctt tatgactgac attgtggtgg gaaagttgaa actggctcgt 600
tcgttgggtt ttcagcatct gctggacgcc aaacaaacaa atgtttctga atgggttaag 660
caaaagtgtg aacacggagt tgatctgtg ttcgaaacca cgggacatcc cgcgggactc 720
gaaacgtcga ttgactcttg ttctattaca ggcaaaatta ttctcgtggg caatcccagc 780
ttcgaagaga tgaaggttcc ggctaataaa tatggaatga ttctgcgcaa ggagatccag 840
attactgaa tttggaattc gtcctcaaac gaataccaa aaaacgagtg gcaagttgtg 900
ctgcaagagt tcaagaagga gtcgtccttg tactcgaaga ttatcacgca taaggttccc 960
attgagggcc ttaagggaatt gtttgtagc atttacaaca aacaggtctt tccggtgaa 1020
gctatgttta tcaataagga gatctaa 1047

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SEQ ID NO: 93      moltype = DNA length = 1014
FEATURE           Location/Qualifiers
source            1..1014
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 93
atgaaggcag tctgttacac gcaaccagac gcagtgccg tccttgagcg tgaatgcct 60
gcactgggtc caggagaggt gcttcttcga attgatgcca gcggaatttg cggttccgat 120
ttgcacgcat atcatggaca tgaccacgt cgtaaacctg gacttgtgtt gggtcacgag 180

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tttgcaggca ccggtggttc ttctagcgat gaaagccgcc tccctgtggg tcgacgcgtc 240
acggctaata cctcogttac atgcggacac tgtgattact gccttcaggg ccgtagcaat 300
ctgtgtgcga atcgtggcat ggtgggtatg tctcgccctg gtgcctcttc tgagtatctt 360
gccattccca tgcgatgcgc tattgctatt cccgaagatc tgccgagcgt gtctgctgcg 420
cttaccgaac ctgctgcgac ggcactgcac gcagtcgaatc ttacaatgcg agcgatgagc 480
cgacctctcc cggaggcgcg tgttttgatt atcggagggtg gagcgatcgg catgttgacc 540
ggagcacttg tcggtgcgta cggtgtcgcg tcccttgccg tcgcggaaac caacgccttg 600
cgccgtgcgt ccgtccgacg tcactcttga tgcgagggtt tggacctac agcgaaacgc 660
cgggctgagt cttcgttcga cgtggtcatt gacgcagtg gtgcggcacc aacccgccat 720
ctggcactgg cggcaacgcg accaggaggt gttatcatgc acattggact ccaggcatgg 780
tccagcgaaa tcgatatgcg taagtcact ctgtctgaaa tcaactctct cggtacgtat 840
acatacacga tggcggatat gcatgcgacc gtgcgagcac ttgcctctgg aagcttcgga 900
tctctggaat gggtcgagac gcgagccctg acagatgcgc catctgcttt tgcagacctt 960
gctgcgggtc gctcgcggc gggcaagggt gtgcttctcc cgcaacagga ctaa 1014

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SEQ ID NO: 94      moltype = DNA length = 1044
FEATURE           Location/Qualifiers
source            1..1044
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 94
atgaagtcgc tcgtatttca cgcgaagga gatgtccgtg tggaaagagcg acccattccc 60
catttgcaaa cggatgaaga tgttttggtg aaagtcgtgt cttccggact ttgtggcagc 120
gacattcccc gcattctcaa tcgtggcgct cattattacc caatcacact tggccatgaa 180
ttctcgggct atgtcagtcg ctatggagcg ggtgtcactg acttgcaacc aggtgatgct 240
gtcgcttggt tgcgcgtctc cccgtgtttt cactgcccc agtgcgagcg aggttacttc 300
tctctgtgta agcagtatca atttgcggt tctcgacgag agggaggaaa cgcagagtat 360
gtcgttatga aacgagcgaa tcttttccga ctcccgtccg atatgccaat tgaagacgga 420
gctttcattg agcccatcac cgtcggctcg catgccttcc accttgccga gggttgcgag 480
ggaaagaatg tcattatcgt ggtgtcagcg accattggcc tccctgcctt gcaatgtgca 540
cgagaactcg gagccaaatc gattaccgca atcgacatca acccacagaa gcttgcatgt 600
gcaaggagcg tgggcgcgac ttactgtgtt aattcccagc aaatgtcggc gaccgaaatt 660
cactctgatt tggaaaatgt gcagttcgac caacttattc tcgaaacggc tggaaacgct 720
cagactgtca cgcctcccat cgaaattgcg ggtccccgag ccaattggc gcttgttgg 780
acgttgacac agcatttggc gcttgcgtcg ggtgtcttgg gtcaaatctt gcgcaaaag 840
cttactatcc tcggatcttg gatgaactac agctctccct ggccaggcga agaattgggtc 900
actgcggcac gtctgtcgac ggagaaacgt ttgcaattgg aaccacttat tgcgcactcc 960
ggagatgcgc aatcctttgc gtctgcagtg aagtctctca atggcgcacc tatgcaaggt 1020
aaaaattctgc tcaagctttc gtaa 1044

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SEQ ID NO: 95      moltype = DNA length = 1041
FEATURE           Location/Qualifiers
source            1..1041
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 95
atgaaatctg ttgtgaacga tacggacggt attgttctga ttgcagagtc ggtgattccc 60
gaaattaaac accaagatca agtgcgtgtt aagatgcgct cttccggcct ttgtggttct 120
gacctgcgcg gcatttttaa gaacggagca cactattatc ccatcactct cggacacgaa 180
ttttcgggat acatcgatgc agttggctcc ggtgtcgatg atcttcatcc tggtagcgc 240
gtggcatgtg ttcccctctc gccctgtttt acctgtccag aatgtctgaa aggcctttac 300
agcgaatgtg ccaagtacga ttttattgga tcccgtcgtg atggtggttt cgcggagtat 360
atcgtggtta agcgaaaaaa cgtgtttgct ttgcccactg acatgtctat tgaagatgga 420
gcctttatcg aaccaattac ggttggcctt caccggttcc acctcgcgca aggttgcgaa 480
aataagaatg tcattattat tggcgccggt actatcgccc tgccttgcga ccagtgtgcc 540
gtcgtctcgc gtgccaagtc tgttaccgcc atcgacatct cgtctgaaaa actcgcctg 600
gccaagtctt ttggagcgat gttgacgttc aattccttgg agatgtccgc cctcagatg 660
caaggagttc tccgagaatt gcgattcaat cagttgattc tggaaacgcg tggcgtgcct 720
aagaccgtgg aactggcggt cgagattgct ggaccccacg ctcaactggc cttggttggc 780
actttgcacc aagacttgca tcttacgagc accacttttg gaaaaatcct gcgcaaaag 840
cttactgtca tcggatcttg gatgaattat tctgtcccct ggccgtgtca agaattggag 900
actgcactcc gtctcctcac cgagcgaaa tgtctctcgc aaccgttgat tgcctcatcg 960
ggcagctttg aatcgttcgc tcaggctgtc cgagacattg ctcgcaatgc catgccagga 1020
aaagtgtctc tgattcctta a 1041

```

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SEQ ID NO: 96      moltype = DNA length = 1020
FEATURE           Location/Qualifiers
source            1..1020
                  mol_type = other DNA
                  organism = synthetic construct
                  strain = K12

```

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SEQUENCE: 96
atgaaatcta ttctgattga gaaaccaaac caattggcca ttgttgaaac tgaaatccca 60
actccatccg ccggtgaagt tcgtgtgaaa gtgaagctgg cgggaatctg cggatccgat 120
tcccatattt accgtgtgta taaccattt gcaaaatata ctctgtgtga cggacacgaa 180
ttttttggtg tcattgacgc agtcggagaa ggtgttgaaa gcgcccgcgt tggatgaacga 240

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gttgccagtgg atccccgtggt gtcgtgcccgc cattgctatc cctgctctat cggaaaagccc 300
aatgttttgta ctaccttggtg ggtgctggggc gtgcatgccc acggaggctt ctctgagtac 360
gcggtggttcc cggccaagaa cgcgtggaag attccccagg cagttgcccga tcaatatgcc 420
gtgatgattg agccgtttac aatcgccgcg aacgtgacgg gacacgggtca gccaactgaa 480
aatgacactg ttctgggtgta tggcgctgga cccattggcc tcacaattgt tcaagtctg 540
aaaggagtgat ataattgtcaa aaacgtgatt gtccgcccgc gaattgacga ggcacttgag 600
aaggccaagg aatccggagc tgaactgggct attaataata gccagacacc gctgggcccag 660
attttcactg aaaaaggcat taagccgacg ctgattattg atgcccgttg tcaccctcgc 720
attctgaagg aggcgggtgac gctggccttc ccagcggcac gtattgttct gatgggattc 780
tcgtctgaac catctgaagt tattcagcaa ggaattacgg gcaaggagct ctctatttcc 840
tccagccgcg ttaacgcaaa caagttcccc atcgtgatcg attggctctc caagggcctg 900
attaaccag aaaagctcat tacgcacacg ttgatttttc aacatgttgc ggacgctatc 960
agcctttttg agcaagacca gaaacactgc tgtaagggtc tgctcacttt ctccgagtaa 1020

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SEQ ID NO: 97      moltype = DNA length = 1074
FEATURE
source            Location/Qualifiers
                  1..1074
                  mol_type = other DNA
                  organism = synthetic construct
                  strain = cp400

```

```

SEQUENCE: 97
atgaacattc cgaccaagat gaaggcagtc gttaaatcgg aggcaggtta tgacaatatg 60
tccctgcaag agattctctgt ccagaaagtt aagggaagaa aggtgttgat gaaagtcgcc 120
tatacgggta tctgcccgtac ggacattcat acccttaaaag gcgagtagcc caatgccgtg 180
accccaactg tccctggccca tgaattttct ggcaaggctcg ttgcccgtcg agatgaagtt 240
accaagggtc aaccgggtgga ccgagttacc tccgaaacaa catttgatac atcgcgagag 300
tgtgtgtact gtaaaaaataa ggaatataac ctgtgcccga accgttccggg tattgtgtacg 360
aaagccaatg gatcgatggc taattatgtg cttacgcgag aggagctctg gcatgtcctt 420
cctgacaaatg ttctcctacaa acttgcccgc atgtctgagc cgtgggcttc ctgtgtgcac 480
gcaatgtacc agaaaacccc gtccaccctt catgatacag ttctcattat cggcccccgga 540
cctatggggt tgctctctct gcagattgca aaggaaattg gagcctttac catcgtctcc 600
ggtattacga aagacagcga ccgcccggaa atcgcaaaag aacttgagc cgacatcggt 660
gttgacacgc agaagggaag cttggcgaag atcgtcatgg aggtactata cggagtgagg 720
gttaccaaaag tctatgattg tagccgcccgt acaccagcgg tgaatgctgc cctcccctt 780
ctgaagaagg gaggtacttt tcagcagggt ggccctgttg caaagcctct gaacgaattg 840
gatgaacgta caattattca gcacgaaatt acatacgtgg gatcccgctc tcagaatccg 900
tatgattggc ctatcgccat tcacctggag gccaaagggt ctattaatga agataaaatg 960
gttacgaaag tcttcgatgt ggatcattgg cgtgaggcat tcgaggctat gatggctgga 1020
aaagaactca aggttcttat tgccagcaat ccagacgacg aaactcttaa ttaa 1074

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SEQ ID NO: 98      moltype = DNA length = 1053
FEATURE
source            Location/Qualifiers
                  1..1053
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 98
atgaaagccg cggtctctca tggaaacgaac gacatgcgct tcgaggatat tgaatcaag 60
ccgtgcccga ggcacgaggt caagatcaag gtcatggccc ctggcatttg tggatctgat 120
cccccccag tgctgaaaca ttggaagtat cctgttccgg ctattcctgg acatgagttc 180
tctggtgtta tcgcccaggt cggaaaaagat gtgaaaaatg tcaagggtgg tgaccagatt 240
gtggcgatcc ccttatttcc gtgtaatgaa tgtgagtagt gtaaacgagg tctttttagc 300
ctttgtgacg atcacggaat cctgggtgccc aaatcctttg gagcatttgc cgatgatgtg 360
aacattaaag cgactaatgt ccttcccacg ggccgatatg atttcgaaga cgcagcgatg 420
attgagcccc ttgcccgtgc catgcacgga gtgttgaaca tcggagttca agtgggagat 480
acggtcgccc tcatgggctc gggcacaaat ggtcaacttg tgattcaggg tcttaaaatt 540
gctggtgctg gcaactatcat tgcgtgtgac atttctgata acaaaactgc agagtccaag 600
gaattgggcg ctgacattat cattaacgca aaggacatta accctgttga aaaaatcaaa 660
gagcttacag cgggaaaagg agtggacatc gctctggaat gtgcccgttc caaaattacc 720
caagagcaat gccctctgat cactaaaaaa aagtccaaga tcggtttcct ggggaattgcg 780
tactctgaca ttaccttgtc ggaggaggcg ttccagaaac tctttccgaa agagctcgaa 840
cttaaaagat cttggaacag ctattcggca cctttccctg gccaaagatg gaccaagggc 900
attaatcttg ttaacgaggg aaaaattaag ctgaaggaaa tgggtttcga ccgcttttct 960
cttgaggata cgtataaagc gtccagatg atccgagatc gcaagggaag gttaacaaa 1020
atcctgatcc tccccaggg cgtggaaaaa taa 1053

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SEQ ID NO: 99      moltype = DNA length = 1053
FEATURE
source            Location/Qualifiers
                  1..1053
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 99
atgaagtcgg tccgatttta tggaaattcgt gatacacgag tcgaagacgt tgacgtgccc 60
aagattcttg aaaaggatga cgtcattatc aaggtcaagg tggcgggcat ctgtgttct 120
gacattagca aatactcgaa aacaggcccg cacatggttg gcgagatttt gggacacgag 180
ttctccggcg aggtcgctca ggtcggcaag gaagttcgct ccttcaagat cggagatcga 240
gtcgcagttt gccctgcgat gccctgcttc gagtgtgatg agtgcaaaaa aggtttgtat 300

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tcccgttgca ataatgtcgc tatcatcgga aataaggagc tcggaggatg ttttgcggag 360
tacacgaaa tttaagagcg caatctgatt aaaattccag atgagattag ctacgaaacc 420
gctgcggccc ttgaaccgct ctgtatcgca ggacatggtc ttttccgctc tgaagccaag 480
gtcggagata cggctgttgt gctgggaacg ggtcctatcg gattgttcag cattcagtgg 540
gccaaaatct ttggttccac gaagattatc gcggttgacg tgtttgacga aaaactggac 600
ttggcgaaa agctcgtgtg ggatatttgt attaatgcta aggagaaaa tatcgttgaa 660
gagatcaaac gactgacaga cggagatggt gcagatattg tcattgagtc tgccggtacc 720
ccgcttacat gtggccaggt gcttctgctg gcgaaaaaag gtggtactgt cttgatatga 780
ggcgccccat acggcgacgt cgccttact cgagaacaat ttgagaaaa cgttcgaagc 840
gagcttacgg tgaagggaac ttggtttggc aattctttcc cattccggg aaaggaatgg 900
tctgctggcc tgtaaccatc gcagaaagga gacatgaacg tggaaaagct tgtaactcac 960
cgtatcaatc tggaaagagc tcagacatac ttcgagaagg tttataaacg tgatatcttc 1020
tttgtaaga ttatgattaa tattgacaat taa 1053

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SEQ ID NO: 100      moltype = DNA length = 1050
FEATURE            Location/Qualifiers
source              1..1050
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 100
atgaaagctt ccatgctgga ggacttgaac aaattcagcg tcaaagagat cgacatcccg 60
tccccaaaa aggatgaggt cgtcgtgaag gtgatggcgg caggcacgtg tggatccgat 120
tcgcataaaa tgattagcgg atggaaatat ggctatcccg cggtgatggg acacgaattt 180
tctggaattg tgacacaaat gggagaaaaac gtgtccaacg tcacgctcgg tcaacacgtc 240
gcctggctcc cttcatttcc ctgcttcaag tgtcattatt gtcaaatcgg actcttcacg 300
atgtgtgaaa attattccat gctcggtcag caaaaatttg gaggtttcga gcaatacgtt 360
tccgtcccg cactgaacgt tctggacatc ggcaagatgt cgttcgagga aggtgccctt 420
attgaaccga tggcagttgc ggcccacgct gttatgggaa ttaaacgga gctgggtgat 480
acagttgcgg tcttcggcct cggtacagtc ggagatcttg ttgtccgtct tcttatttcc 540
tcgggagcga caaacgtcat tggcattgac attgacgacc agaagcttga aaagggactg 600
gatgaggggt gcacacacgt gtcaactctc gccaaagagt ctcttgaaga aaagatcatg 660
gaatacacag acggtctggg tctcgatatt tccatggaa gcgcggcgag caaaatcact 720
gaggagcaaa cactgctcgt tacgaaacgc cgtgaaaagg ttgattttgt cggattttgt 780
tatagcgacg ttctgtctca tcagaaagca tttgagaaca tcttcgccca cgagcttacg 840
gtgactggat tttggaattc ctattccgca ccatttctcg gacgcgagtg gaccaactcg 900
atccaattgg ttaatcgtgg tcgtatcaag atcaaaagatc tcattactca tcgattttgag 960
cttgaagaca tgcataaaagc attcaatatg atcaccactc gttcgaatc gtttaataag 1020
gtcatgtttt tccccaaagg catcaattaa 1050

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SEQ ID NO: 101      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 101
atgaaggctc tcaattttgta cggtatccag gacctccgct ttgaggaaac acccgctccc 60
tctatcgaa atgatgacga tatcattatc aaagtcaagg ctgtcggaat ctgcggatcc 120
gatttgctcg gttataagaa actgggacct tatgtccctg gaatgacttt tggacacgaa 180
ttcgcgggag agtctgtcaa aattggtcgt tccgtgacgg gattttctat cggtgatcgt 240
gtcgtgtctt gccctactta tacttgcggc cagtgcctgt actgtcaact cggcgagcgg 300
acacgatgtg agcgactgtc ttgtgatcgg gcacgccacc caggtgcata cgcagaatac 360
gtcaaaactc ccgcgaagca tgtcatcccc ttgcccaacg ttgttaacta tgacgaggcc 420
gcccctcatt aaccggcttc ggttgtttgc cagcgctttt atcgtactaa cattaagcca 480
ggtgcaagcg tcgctattat gggcgttggc tcgatcggtc tgctcgcgt ccaatgggca 540
aaaaatcttc gtcgcaaac ggtctttgcc attgatattg acgaacagaa acttaacgtc 600
gctaatacac ttggagcaga tgtcttgatc tccagcctcc aacgaccgc acacaagcaa 660
atccttgaat ataccaacgg aatcgtgtgt gacgtggctg tggagtctgc tggtaacct 720
tctacctcgc ccaggtgtt tgcgctgcgg aaaaaggtg gtgaagtgt ctttcttggt 780
attccttatg ccgatgtcca gattgaacga ttttacttcg aaaaaattgt tcgtaatgaa 840
ctgcacgttt acggatcttg gaatgcactc tctcccccac tcccggttaa agagtgggac 900
acgactatcc attatattgc gtccggacaa ttgaatgtcg cgcctcatgat ttcctaccga 960
ctccctctcg ctaagggccc cgagacgttc cagcagatcg caaagggtag actcaagccc 1020
acaaaagtc tttctatcc tgagaagttg tctgagcgca agtaa 1065

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SEQ ID NO: 102      moltype = DNA length = 1056
FEATURE            Location/Qualifiers
source              1..1056
                    mol_type = other DNA
                    note = Lachnospiraceae bacterium
                    organism = synthetic construct

```

```

SEQUENCE: 102
atgaaagtcc ttgactgca taagcctggc gatattcgat ttgaggacgc tccaatgcct 60
gagccaggag agggcgaagt gctgctgaaa gttatggcat gcggtgctg ttcgtcggac 120
gaggctcgaa ttttcaagga cgggtcccat attatgccaa ttgttctctg ccacgagttc 180
tgtggccaag tgggaaaacg gggagcagga gtggacgaaa gcctcatcgg ccgaaaagcg 240
agcgtcttcc caatgcttcc ctgtttcgaa tgcccttctt gtaaagcagg tgactacgca 300

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acctgttcta actataaata ctatgggtcc cgcaacgatg gtggaatggg agaataccga 360
gcaatcaagg cgtggaacct gaaccttctg gatgattccg tcgattattg cgtcggcgca 420
ctgtccgagc catccgcggt tggccatcac gccactgaag tgggaggagt gaaagccgga 480
gacaagggtt tcgtcattgg tagcggaacc atcggaactt tgatcggtt cttegtcag 540
gctaagggtg cagagggttg gatcggtgcg cgtcgtaaag agagcaacga ttccgcgcg 600
agcctgggtt ttaaaacgat cgacaccaat aataccggaag acgaggttgc aaaggccacg 660
gatggtaacg gcatggatgt tgtcttcgaa gcagttggta gcaataagtg tcttattgaa 720
gctattggat ctgtcacggc gggaggcaca cttgtcaccg tgggcaatcc tgtgggagac 780
atgaatcttc cccgagatgt gtactggcgc atcctccgaa agcaactcac ggtccgaggt 840
acttggaact ctactactc tgacaaggcc aatgactggc acgaagtggc cgcgttgatg 900
aaagaaaaaa aaatcccggt cgaaaaactc atcacccgta agtacaacct ttctcaatgc 960
gatgaggcct tcgatttttt ccgtgaccac tctattcaga agtcccgcgt gatgtttgtt 1020
atgcacgaaa aagaatccga agcttacaac ggataa 1056

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SEQ ID NO: 103      moltype = DNA length = 1011
FEATURE            Location/Qualifiers
source              1..1011
                    mol_type = other DNA
                    note = Faecalibacterium sp. OF03-6AC
                    organism = synthetic construct

```

```

SEQUENCE: 103
atggtcaaca tgaagaaagc cctcgtgttt actgcgcccc atacgttcga gtatcgtgac 60
gttccccgac ccgaaccgac ccccgacgag gtccctgggtg aggtgaaagc atgcgcaatt 120
tgtgttctcg atgttcacgg tagctcgggc gaaaacggtc gacgcattcc tcccttgggtg 180
atgggcccag aggcgaagcgg cactatcgca caattggcgc aagctgtgca gtcgtgggca 240
gtgggagatc gcgttacatt tgactctact gagtactgcg gctactgttg gttttgcacg 300
cacggtatgc acaatttctg cgcacgagca cgtatcgtgg gcgttgcggt cacagagtac 360
cgtaaaaaatg gtgcaatggc tcaatacatt gcagtgaag cccgtaccct ttatgctttg 420
ccgcggcgcg tctcctttga acagggttgc cttgtcgaa cctttgctgt gggatatgcat 480
gctgtcttgc gatctggact ggcagcgagg cagacggccg ctgtgatttg agatggaaca 540
atcggtctga tgactcttat tgcggccaag agccaggagg cacaagtctg tatcagcgga 600
caccacaccg gtcccatgca gctcgcagcc cagctcggag cccctggattg tgcaccaactg 660
ccccagctga cagctgagcg aggtgcggac gttgtttacg acgctgttgg ctctcaatcc 720
agctttacgc aagcgaatga gcttgcctt aaggagggcc gaattatggt tatcggcaat 780
catgcacgca cgatcgatgt tcccttgcaa cgttgtatca cggaaacagat cgacgttcag 840
ttttcgtaca gctctgcggg tgagtatcct gtttgtcttc aagccattgc tgagggacgc 900
gcagatttga caggagtgtg gcaatctttc ccaactggctg agggcgctca agttttcgac 960
cgctgggttc accgagatga gtccttgacg aaagccggtt tgatcccata a 1011

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SEQ ID NO: 104      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
source              1..1074
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 104
atgaaggcct atgtttttgga tggatttggc cagctgaatc aacatgagta tgtcctccga 60
gacttgcagc ctggagaagc cctcgttcaa gtgggtgcgg ctggtatctg cggtagcgac 120
ctgccccgca tctttgagac ttgaaacttat catttccaa ctatcccagg ccatgaattc 180
gcaggtgttg tcaagaacct ttatgctgat gccgacgaag aatgctccg acaattggatg 240
ggtaagcgcg ttgcgggtgtt tccactgatt cccgtgcgaa cctgtcgaag ctgcgcgaat 300
catcagtagc agatgtgttc taactataat tacctcggct cgcgtacgga cggaggcttc 360
gctgaatact gtatcgtgcc ttgttggaa cttgttcagt tgcgggacga ggttagcttc 420
gcacaggccg ctatgatgga accggcagct gtggcattgc atgcaatccg acgtggcgcc 480
ctgcaccccg accacccgga gaccggtatc ggcaagcgctg ttctggttat gggacttggc 540
actattggcg gattgtgcgc ccagtggcct agccaatgtg gcgttacgga gctcacgcc 600
actggtcacc acgcccccca ggaagagctt atgcaccgat atgctaacga gaactaccga 660
ttcattaata ctaatcaaac tgcttggacc gatgacgaaa acacctacga tcttattttt 720
gactgcatta atgacagcga atccgttcag caggcggttc gctacgtcaa acccggtggt 780
cagattgttg ctgttggcaa cccgaaaaaa gatcttactc ttcaaaagga catctattgg 840
aaactcctgc gaaaaacaag gtgtatgacc ggaacatgga actcgagctt cagccatgat 900
tcgaacgatg actggcagat cctccttcga taccttcagg agcaaaagct gcatcccgag 960
gcacttatca cacatcgctt tccgtttgcc caacttccca agggcttggg gatcatgcgt 1020
cttcgtgaag agttttacaa caaaattatg atcgacttga ctgtctgttc ttaa 1074

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SEQ ID NO: 105      moltype = DNA length = 1029
FEATURE            Location/Qualifiers
source              1..1029
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 105
atgaaggccg tgacttacac tcccgaccag ggttttgcac tggtcgaggt tgagacaccc 60
cagcttggac tgcaggaagt tcttctccag gtggaggcct gtggcgctg cggaagcgac 120
cgtaagtcg ttgcaggcga atcggttcct accggtaccg cctccctct ggtgatgggc 180
cacgagattg ccggacgatc cgcgcggttg ggcgaaactg tgacagattg gcatgttggg 240
gactctgtga ctgtccatcc cctcgtggca tgtggtacgt gcgcacctg tgttcacggc 300
cagcctaact tgtgcgtccg acaggtgtgt attggttatc aacgacctgg aggattttga 360

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gagcaagttg ctgtcccgcc tctgcaactg gttcgacgtc cagcagaatt ggcaccggcc 420
gccgcagcat tgggttggtg tgcgtttgcc acgccatacc acgcgatgcg agcggcggtt 480
gttgagagcg gccaaaccgt tttgattatt ggaaccgcgc gacttggtct cgcggcactt 540
atgctggcca tgggcttcca tgtcggccgt ctggcgccgc tcacgcgacg ccaagagggt 600
attgcaatcg cggagggtca tggcgcgag atgggtgtcg tcacggacca ggatgagcgt 660
acaattgcgc gccagcttcg ccgctgggtct ggccatcccg gcattgacgt cgtcatcgat 720
acggctcgcta ctgctgacac tattacattg gcagctgaaa gcgttcgccc aggaggcact 780
atcgctattg tcggaattgc ggccgacacc gccctctatc ctgtcgctaa gaccgttcgc 840
cgaggtctga cattcggtgc gagctacgga tccgtccgag aggatgtcga gcagttggtc 900
agctgggtgg cagaaggctg actggaccgc gttcgactcg tcgcgggagc cctccactc 960
catgaagcgc ctcgctgcat cgcgccggac cgaccgcgcg gacgttggtt tattgatcca 1020
tctgaataa 1029

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```

SEQ ID NO: 106      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
source              1..1074
                    mol_type = other DNA
                    organism = synthetic construct
                    strain = KGBM03662

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```

SEQUENCE: 106
atgaaaagct tgattaatat cccggaaaaa atgaaagccc tcgttttgta tgggtgttgg 60
gatttgctg tcaactgaagt ggacgttccg aagctcgaaa aaggaacggt cctccttaaa 120
attaaagctt cggcgcactg ttctccgac atgcctcggt gtttcgttac tggcacttac 180
cattttccga ccattccagg acacgagttc tctggacaaa tcgttcggtt ggggtgatgat 240
gttgatgaat cctgtttggg aaagcgttcg tgcgttttcc ctatgcttcc ctgcaaacac 300
tgcaaaagctt gcaagatgga ggaatatgct cagtgtagcg gctattctta cttcggtctct 360
cgccaaagatg gtgcctgggc ggaataacct gtgggtcccg tgtggaacct ggtgcccttt 420
gacgatacac tcgattataa aactgcagcc ctgtgcgaac cagctgcggt ctctatgcac 480
gccgttaata ttggagacat taagaaggtt cagaatgttg tgatcatcgg aacgggtacc 540
atcggtattca tgattgctct ctttgccaaa gaacgtacga aaacaggaaa ggttgtgatc 600
gcgggacgct cgaataataa gctggaatac gcgaaaaagt tgggcttcga tgcgttgac 660
acctcggttg acgatttgcc tccccaaagt aagaaaaatc tcggtattga tggagctgac 720
gtcacattcg aagcggtcgg ctccaatgag gccattgcct ctgctattga atctacgggc 780
gctttgggtt cgattgttct tgcggtaat ccttccactg atctcatctt gccaaagaat 840
attttatggg ccatcctccg aaagcaaat acagttaag gttcctggaa ctgctcttat 900
aactctcgcg ttaatgactg gaagaaagcg ttggagggtt ttgagcatag cgacgtgaat 960
tttgacagct tgatcacaca cacattcact atcgaaagaa acgagaaagc tttttctgcc 1020
atccgtaaca ctggagaatt cactctcaaa gtcattgtta ccttcgatga ttaa 1074

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```

SEQ ID NO: 107      moltype = DNA length = 1047
FEATURE            Location/Qualifiers
source              1..1047
                    mol_type = other DNA
                    note = Clostridiaceae bacterium
                    organism = synthetic construct

```

```

SEQUENCE: 107
atgaaagctg tcgtctcgca tggtaaaaac gacatttgct atgaagaatg gcctgaaccc 60
atcctgaaaa caggaacagt taagattcat gtcaaggctt gtggcatttg cggctctgac 120
attcctcgcg ttctcggaga ttctgcccac tactaccgca tcctcctcgg ccattgagttc 180
tccggttggt tcaactgagg ggtgatgac gttgacaaga tccaggttgg cgatcatggt 240
gcaggagtc ccctctctcc gtgcgatgaa tgcgatgatt gtcaacaggg aaactattct 300
cagtgtaaac attactctt tatcggcagc cgcgagcaag gtggatttgc agattatggt 360
gtgatcccg ccgataacgt ggttaaagtc gacccaactc ttcttacga acaggtcgca 420
ctcttcgagc ctgtgactgt tgccttgcat gcaactcaag tctcctcgta taaaggagga 480
aaacgagtg cgatcctggg tggcggcagc attggtcttt tgcactcca gtgggctaaa 540
atctatggcg cgtcccaagt cacagtcttc ggtcgcagca aggaacactt gaagatcgct 600
gctgctctgg gtgcgacac aactatttcc actctggacg ataattttat gacagaggct 660
atgcgagtc cagagaaaaa gggttttgac tacatttacg agtcgcgcgg ttccacaatc 720
accatgaagc ttgcttttga actggcagcc aaccacgcga aaatctgttt cattggcacg 780
cccacggaaa acctgacttt cagccctaag ctgtgggaga acatgaatcg aaaggagttc 840
cagcttatgg gttcctggat gtctgttcc gcgcggttcc ccgcgatga atggaaggag 900
acagcccat tctttaacac cggacaactt cgatatgtgc ctgagatggt ttgcgcacag 960
tatcccatgg acgatgttaa aaaagcggtc gagcagttcc ataacaagaa taaagttaa 1020
ggacgtatcc tgcataaaaa caagtaa 1047

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SEQ ID NO: 108      moltype = DNA length = 1014
FEATURE            Location/Qualifiers
source              1..1014
                    mol_type = other DNA
                    note = Thermoplasmata archaeon
                    organism = synthetic construct

```

```

SEQUENCE: 108
atgaaagtgg cggtttggtg tggtgacaaa aatattgtct tggaggaccg tccgatccct 60
tctatcggtg aaaaggacgt cttggtccgt gtcaagtcgt cttccatctg cggagctgac 120
cttcacgctt aaaaaggcct ccttaagcgc cgcaagcctc cattgggtgat gggacatgaa 180
ttttccggtg tggttgaacg cctcgcgcat gagacaaagg gccttaaggt cggcgatcgt 240

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gtcgtggtg agccattat tagctgcgga aagtgcagg catgcaaaaa gggaaaaacg 300
aatgtctgcg aaaatcgtaa gctgacggg ctgaacctcg acggtgcttt cgctgaatat 360
gtttctgtcc cttatgaaaa gtgccatgaa ctcccccggt acctttcggt cgatgagggt 420
gcgattatcg agccgttggc cgtgcgcgtt catgcgggta atattactcc catctcgctc 480
aacgataaac tgcttatcat tggtagcggt gtgattggcc tcctcatcac acaattgctt 540
cgaagccgca cttcgggcga cattttggtt accggacgtc gcgattacct ccttgaaaag 600
gccccgtgaat ttggtgccac agatgcaatc aacgtgatcg actgcgatgc tgtgaaaacg 660
ctgctcaaaa aacacgggtc cgtcgatgtt gtcttcgaag cagtgggctc taacgagact 720
gtgaaccagt ccatttcctt ggtcgaatct ggaggttcta tcaactcctg tggcatgctc 780
gatcgcgaga tgaccgttga tatgatggag gttgttggcc gtgaaatcac tctgcgagga 840
gattatgggt atacaactga cgactttaag gagggcatcg agctcgcttc ttcgggtaaa 900
gttgatctca aaggcctcat cacgcatgtt ctcccgctcg atgagatcaa aaagggtctc 960
gatactctgg cagaacgccg agaaaatgtg atcaaagtgt ttattaaccc ataa 1014

```

```

SEQ ID NO: 109      moltype = DNA length = 1032
FEATURE
source             Location/Qualifiers
                   1..1032
                   mol_type = other DNA
                   note = Lachnospiraceae bacterium
                   organism = synthetic construct

```

```

SEQUENCE: 109
atgaaagcac tgggtgttga ggaaatcggt aaattgacct ttaaagaggt tgagacgccc 60
gtgccaggac gccgagaggt ttgtgtgcgc atccatgcct gtggtatttg ctcttcggac 120
aaagcccgct tcttcaagaa cggtaactcac accatgcgcg ttatcccagg tcacgagatc 180
tcgggtgaga ttgttgccag aggtcccgcc attcgagaag agcttatcgg ccgacgcgct 240
gcggtcttcc cgttgatttc ttgtgcgcaa tgcgactcct gccgaatcgg caagtacgct 300
caatgcgagc attacgacta ttccgatcgc cgtcgagacg gaggtcttgc tgagtacctc 360
gctgtccccc tgtttaatat caatcttctt cccgatgata tggattatca tacggctgca 420
ttgtgcgaac cagccacgggt tgcactctcat gcaactaagc gcgccaaaat gaaatctggt 480
gatatcggtt cgatecattg cactggcacg gtcggcctcc tggctgctgg catcgctcgt 540
attcgacgag catctaaaat cattatcgtc ggacgaagca aagaaaaaat catgaatgca 600
gctcatttcg ctgatcatgt gatcaatgca gagggatggg acgtgaacga ggcgctggct 660
tcgctgacg ggggtcacat ggcgatgtg gtcttgagga ctggcggcaa cagcgatgct 720
tgacgctacg cgttatggc tgcgggcaaa ggtacgactg tgatgtttat tggcaatccc 780
gaagaaccca ttgtttttaa ccgtgatatt tattggaaga tcctccgaaa cgaactgaca 840
ctcaagggtt cttggaattc tggatgatgt cctgaggata atgactggaa tgacgtcatc 900
cgttatatgg cataaggaca gattccggca gataaaaatta tcaccgatag ctttcattg 960
gaagattata aaaggcatt cgagcttctt ctctcggatc gctttttcac caaaatcatg 1020
tttgacatct aa 1032

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SEQ ID NO: 110      moltype = DNA length = 1047
FEATURE
source             Location/Qualifiers
                   1..1047
                   mol_type = other DNA
                   organism = synthetic construct
                   strain = DSM 16973 = JCM 13450

```

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SEQUENCE: 110
atgaaagcca atgttctgta ccacattggt gacctgcgat acactgacgt tgccttgccg 60
gaactccacg acggtgaggg cctcgtgcga gttcgtgcat gcggtatttg tggtagcgat 120
gttgcgccag tcttcaccac tggtaactgt catttcccca ccattatttg ccatgagttc 180
tcgggccaag tcgttgaaac atattctgcg gctgacgcgg agtgggtggg tcgccgcgtg 240
agcgtttttt ccctcatccc ttgtcgccat tgtgatgcct gtcactatgg tcagtacgaa 300
atgtgcatat gttacaacta tctcggcagc cgtaccgatg gaggtctcgc agagtacgtg 360
gctgtcccggt tctggaatct ctttcgtctg ccagattccg tgtcttacga ggaggcgca 420
atgatggaac cggccgctgt cgtatgcac gcgctcaaga ttgcccgtgt gggatatgggt 480
gacactgtgg caattacagg tcccggaaat attggtatga ttttgaccca attgatccaa 540
cgtattggcg gtacaaaagc ggttctgac ggacgcaccc aagcaaaatt ggattacgca 600
cgacgcacgc gcatccgtca tgtgtgcaat tcgggaaccg agcacgtcgc ggaccgcgtt 660
ggacaactta caggcggagc tggagccgac atcgcaattg aaggaaaccg tgcgtcggaa 720
cccatgaatc tgtgcttgaa tttggtgcgc tctgaaggac gtattgtcgc tatgggaaac 780
cccaaagggt acatgctggt ccgaaaggat gcgtattgga agcttctgcg caagcagctg 840
acccttctgt gaacctggaa ttgcacatat ggattggggc gagatgattg gcaggatatt 900
tcgccctctc ttgctgaggg agcattgaac ttgcgcggcc ttatcacaca ccgactccca 960
ctggccgagc tgcagcaggg actcagcctc atgcgagacg cagcagtcta cactaacaaa 1020
gttatgctg tcgacgagca aggttaa 1047

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SEQ ID NO: 111      moltype = DNA length = 1014
FEATURE
source             Location/Qualifiers
                   1..1014
                   mol_type = other DNA
                   organism = synthetic construct
                   strain = OM05-6

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```

SEQUENCE: 111
atgcgcgcct atgagcttca cggagtcaac gacttgcgtc gcgaggacat cgaaaaaact 60
gagattccct ctggatgggt tcttgtgcaa gtcaaaagca gcggcatttg cagctccgac 120
atccctcgta tcttcacgaa cggcacttat cactttccta ccattccggg ccacgaattc 180

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tctggagtcg tggccgccta tggcgaaggt gtcccgaag aacgcgttg caaacgagt 240
ggatattttc cgctgatccc gtgtcgtaca tgtcccaat gccgccagaa gaaatacga 300
atgtgcgagc actatgatta ccttggtctc gtctcgtgac gcggattcgc cgaatacgtt 360
gcagtcgccg actggaacct cctggaattg ccggagaacg ttctgttcgc cgaggcggcg 420
atgctcgaa cctcgtcggt ggcacttcac gccgtgaaac gcagcgcggt gaaacctggg 480
gatacagccg ctgtgatcgg cacaggaaat atcggttttg ccgcagctgc ttgggccaaa 540
gcactgggag cagaacgggt gtctgtgatt ggacgcggcg aggccaaacg cgctattgct 600
gaacaaattc ccggaatttc ctatgtgact gaagaggaat ccaaggccat tcaggcgagc 660
gtggtcgtcg aagcagttgg tacaccacag gctgtttcct ccgcggtggt gcttgacaga 720
ccgggaggtt ccattgttct gatgggcaac ccacccggtg atcttgcgat ggaaaaaaac 780
gtctactggc gtatcctcgc taagcaactt aacttgatgg gaacgtggaa tagctcgat 840
gaaaaagacg aagcctgtga ctggaccgaa gcgctcgacg cattgtccaa gaaaaagatt 900
cctgcccagg cacttatcac tcattgtttt cccagcgaaa aacttatgga tggactggaa 960
ttgatgaaga atcataagga accatactgt aaagtattgg tcagctggga gtaa 1014

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SEQ ID NO: 112      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
source              1..1065
                    mol_type = other DNA
                    note = Candidatus Bathyarchaeota archaeon
                    organism = synthetic construct

```

```

SEQUENCE: 112
atgaagcgta tgcgagctgc catcttgtat ggacccaaag acatccgaat tgaaaacatt 60
gagatgcccc aggttaagcc ggagtggttg cttattaagg ttccgcgaac aggtatctgc 120
ggttctgata tgcactttta caaagagaag agctttattc ctattattag cgagctgggc 180
aaaggacgct acgtcccggt tcatgaattt tccggcgagg tctacgagt gggttcgaag 240
gtcaaaagcc tgaagaagg agatcgcgtc gccgtcgaa ccatcattgg ttccggcaaa 300
tgtcagtggt gcaaggtcgg atgttacaac ttgtgtgagc actacaaatt gattggcttt 360
cactatattg aggttatggc tgagtattgt gttgctcctg ctgagaattg ttttaagctc 420
ccaaacaatg tttcgttcga aagcgcggca atgcttgact gtatcgccgt ggctattcac 480
gcagtgaata aagcagaat tccgcggcgt gacagcgttc tcattgtcgg cgccggtgcc 540
atcgactctc tgcacctcca gccagcgctc atccaggag cgagcgacgt gtacgtcgct 600
ggaaaatacg aattccagtt ggagaaagct cgcgaagtgg gagcgacagc tgtgattaac 660
gctaaaaacg aatctttctc cgaaaaaatc atggaaatta cttcgggtaa aggagtogat 720
aaggctcatt aggcgcgtcg cggagcctcc aatgttattt ctgacgtcct tgacgtgact 780
cgaaaaaagg gtactatcgt gaccactggt atctttacac gaccactctc ggctcgattt 840
ttcaaaattc tcacaaaaga gttgacgac acttcgtcgt ggggatactg ttacttcgac 900
cgtcgcaagg agtttggcac gccgctcgac tttctggcaa aagaaaagat caacactcag 960
gaacttatta cacataagta tctttggag aaggtttccg aagccttga gacggcactc 1020
aacaaaaata aggagaagag catcaagggt gagattatct tctaa 1065

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SEQ ID NO: 113      moltype = DNA length = 1053
FEATURE            Location/Qualifiers
source              1..1053
                    mol_type = other DNA
                    note = Lachnospiraceae bacterium
                    organism = synthetic construct

```

```

SEQUENCE: 113
atgaaagcct ggggtcgtga cggtatcgga gatatccaat acgaggaagt taaggagccg 60
cagtcggctg gcgacgaggt tattatgaaa gtgatgagcg tgggaatctg tggatctgac 120
atccctcgca ttataaaac cggcacctat catttcccaa caatccccgg ccacgaattc 180
tccggtattg tggccgatgt ttgtgccgga gtcaaccgga aatggaagaa taagcgcgtc 240
ggcgtgtttc catgatccc atgtcacgtt tgtgcttcgt gtaaaaaagg tcaatatgaa 300
atgtgtcgca actacggata tctgggtagc cgccaaaatg gtggttttgc agagtatgcg 360
gcggttcctg tcaagaatct catcgaaact ccggaactct tctctatgca ggaagctgcc 420
atgcttgagc cgatggccgt ggcagtcctt gctatgcgac gagcccgctc catgaccgga 480
gatcgatttg tcgtgtgcgg acttggaaat atcggcatgc ttctcgtgct ctttttgcca 540
gagacgggaa ttcgagacat ctttgtcatc ggcaataagg cattccagcg aatcaccatc 600
ctcaaaaactg gccctcgagga aaactgttat tgtgattctc gcgtgcagga tgtcgagaag 660
tggctcatgg agcgaaactga aggaaagggc gcagacgtgt tctttgagtg tgttggccga 720
aacgaaacca tgcaccaagc atccaacctg acggcacccg gaggttcggg ctgtctggtt 780
ggtaatcctt gttcggatat gacgcttgag cgtgaggtct actggaaaat tctccgtaac 840
cagcttacta ttaactggcac ctggaattcc tcgttcaccc acgagatcga ggacgactgg 900
cattacgtgt tggatcgctt ggaacaaggt aagatctcgc ctggtgatct catttctcac 960
aaattccgcg tcaagaactt ggaatctggt ctgcatatta tgcgtgataa gactcaagat 1020
tatatcaaaa tcatggccga gattaatgtg taa 1053

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SEQ ID NO: 114      moltype = DNA length = 1047
FEATURE            Location/Qualifiers
source              1..1047
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 114
atgaaggccg cagtgtccca ttccgaaaat ctgtttacgt acgaagaagt ggaatcgat 60
ccctgcaaag acgacgaggt taagatcaag gtgatggctg cgggtatctg tggttccgat 120
acacataaga tgacgagcaa atggaaatat ccccttccga ccatcatggg ccatgagttt 180

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tcgggcttta tcgtcgagat tggtgaaaag gtcactaacg cgtcgattgg cgatcgagtt 240
gtggcgatcc ctttcttgcc gtgcaatgac tgcgaatact gtaagcgtgg acaattctcg 300
ttgtgcgagg atcacggcat ggttggatcc aaaagctttg gtggtttcgc ggagtagctc 360
aatatcaaaa gcaactaatgt tcttccgatt ggtgacatgg attttgagga agcagccatg 420
attgaacctc tcgcagttgc gcttccggc gtgttgaata ttcacccgcg tattggagat 480
attgcagccg tctttggtat tggcacaatc ggaatccctc caatccaatg gcttaaaatt 540
gcaggagtga aagaggttat tgcgatcgac atttccgata acaagctcgc agacgccccat 600
gctatgggct gtgagcgtag tatcaatcca tttaaagaga acctccttga gaaaaatcgac 660
gaataatact acggtcgccg tgttgacatc gcctttgagt gtgctggatc taaagtcaca 720
caagaacagt gcctgctcgt gacagccaag cgaggaaaag tcggctattt gggtagctgt 780
tactctgatg tccattttgc ggagcaagct tttgaaaaa ttttccgcgc cgagctcaca 840
cttaaggat tttggaactc ttactccgca ccattcccgc gagaggaatg gacccaaagc 900
attgacttca ttaataaggg tcgtattaaa ctgaaagagc tgattagcca ccgatttaag 960
ctccaagata ctcaagaagc gtttgacatg atgatcaacc gagaagagac atacggcgaag 1020
tgctttattt tgcccaacaa agagtaa 1047

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```

SEQ ID NO: 115      moltype = DNA length = 1047
FEATURE            Location/Qualifiers
source             1..1047
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 115
atgaaggcgg cggtgcttca ttctgagaat cacttcattt atgaagaaat tgatattgct 60
ccttgctcaag atgatgaggt taaaatcaaaa gtcattgcga ctggaatttg tgatcggac 120
acgcacaaga tagcaggcaa atggaatat cctctccccg ccgtgatggg acacgagttc 180
tccgataata tcggttgagaa aggaaagaac gtcactaatg cagccatcgg tgaccgcgtc 240
gtggcgatcc ctttccctcc gtgtaacgac tgtgagtatt gtaaacgtgg ccaatttagc 300
ctttgcgaag accacggtat ggtcggctcg aagtccttcg gaggttttgc tgaagtacgtt 360
aatattgagt ccttcacatg ggagatatgg aatttgaaga ggcgggccatg 420
attgagccgc tcgcggtggc ccttcacatg gtcttgaata ttcacccgcg tgttggcgac 480
gtcgcggccg ggttcggtgt cggtagcatt ggtatgctga ccattccagt gctgaaaatt 540
gcgggtgcca aggagatcat cgcaatcgac ttgtcggatg aaaaacttgc tgacgcgcgt 600
gctatggggt gtgaacgtac gatcaatccc ctgaaacgaag accttcttga aaaaatcaac 660
gaatacacca acggaatggg agttgacatc gccttcgagt gtgctggctc gaagattaca 720
caagaacagt gccttcttat caccgcgaag caaggcaaaa ttggatacct gggcattgct 780
tattccgatg tccacctgtc tgagaaggct ttcgagaaca tttttcgacg cgagcttacg 840
ttgaaagggt ttggaattc gtattccgct ccatttccgc gtgaagaatg gacaaaatcg 900
attgaattca ttaaaaacgg ccgcattaag ttgaaggaac tgatctctca ccgatttaag 960
ctggaggata cacaaaaagc atttgatatg atgatcaacc gtgaagagaa atattgtaag 1020
gttttgattc ttccgaacga ggagtaa 1047

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SEQ ID NO: 116      moltype = DNA length = 1053
FEATURE            Location/Qualifiers
source             1..1053
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 116
atgaaggcgg cggtcctgca ctccgaaaat gagattattt accaggacat tgaattttcc 60
gagtggtcaac caaacgaagt gaaagttaag gtcattggcgg ctggaatctg tggttccgat 120
acccataaaa tgggtgagcaa atggaataac cccctccctg cgatcatggg tcacgaattt 180
gccggctacg tcggtgaaaa aggcacaagc gtgacgaacg tcgaaatttg cgatcgtggt 240
gccgctatcc cggtcattcc atgcaacgag tgtgagtatt gcctgcgagg tcagttoagc 300
ctttgcgaaa atcacggaat gatcggcagc aagtcctttg gcgcatttgc cgagtttgtt 360
aatattccag ctaagaacgt tctcccgtat ggcgaatgt ctttcgaaga tgcagctatg 420
atcgaaacgt ctgcggtcag cctgcattga gtgctcaaca ttcaccaca gctcggcgat 480
gtcgttgctg tcttcggtgt gggaaactatt ggaatcctcg caattcagtg gttgcgtgtc 540
gccggcgtga aagagattat tgctattgac atttccgaca agaagctcca ggacgctatg 600
gatctggggt gtgagaagtc gattaaacct ttgaaggaaag acttgattaa gaaaatttat 660
gaatacacga atggcctcgg cgttgacatc gcatttgaat gcgctggatc taaaattacg 720
caagaacagt gccgtgcat taccgcgtaaa caaggaaaagg ttggttattt ggggaattgcg 780
tacagcgatg ttaactcttc tgaaaaagcc tttgaaaata ttttccgcgc agagctcact 840
ctcaagggtt tctggaattc gtattccgcg ccgtttccag gaaaggagtg gatgaacacg 900
atcgagttcg tccaacaggg caaaattaag cttaagaat tggtgtcgca tcgctacaag 960
ctcgaagaca cccagaagcg gtttaacatg atgatcaacc gagaagaaga atacaataag 1020
gtgatgatca tcccaacga ggaactggag taa 1053

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SEQ ID NO: 117      moltype = DNA length = 1041
FEATURE            Location/Qualifiers
source             1..1041
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 117
atgaaggcag cggtgcttca ttccggtcga gacatccgct acgaaaacat cgagatcaag 60
ccttacggag acgatgaagt gaaaatcaaaa gttaaagccg ccggtatctg tgatctgac 120
ccgccacgtg cactcaaaaa gtggaaatat cccgtgccag ccattttggg tcacgagttc 180
tcgggttata tcgttgaagt gggaaagaac gtgaagggtt taaagtggg agatcgtgtc 240

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gttgccgtcc ctctggaacc gtgtcaccaa tgtgaatact gcaagaaagg ccaattttcg 300
ctctgcgaaa gctacgatat gttgggtgct acttcctttg gcggttttgc agagtacgct 360
aacgtgaagg caacaaatgt tttgaaaatt gacgatatcg actatgaaga ggcggccatg 420
attgaacctc tcgcagtcgc actgcatggg gtgcttaata tcaatccgca actcgggtgat 480
actgttgctg ttttgggagc tggaacgatc ggtcaactca cgattcaatg gctcaaagggt 540
tcgggagtcg agaagattat cgcagtcgac atttcggaca aaaaaattag cgaagctatg 600
tccttgggag cagatattgg tattaatgct cttaagaga atccggtcga aaagatcatg 660
gaacttacaa acggtttcgg cgtggatata tgcattgaat gcgccggctc taagatcaact 720
caggaacaat gtctgttggt cacgaagaaa aaaggtaaga ttggatatca gggtatcggg 780
cacgcccgaat ttgagttgct tgaagttgct ttcgagggta ttttcgtaa agaattgaac 840
atccaagggt tttggaatag ctacagcgcc ccggtcccg gacaggaatg gttcaaaagc 900
atcgagtatg ttaagcagaa aaaaatcaag ctcaaacagt tgatcagcca tcgcttcagc 960
cttgaagaca cagcgaagc ttttgagatg attggtgaac gtaaggagga gtataacaag 1020
atcatgattg tcaggagcta a

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SEQ ID NO: 118      moltype = DNA length = 1047
FEATURE            Location/Qualifiers
source              1..1047
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 118
atgaaagcag cagtttttga ttcgagaat cacttcattt atcaggatat ggagagcttc 60
gagtgccaga atgacgaagt caagatcaag gttatggctg ccggcatctg cggaagcgat 120
acgcataaaa tggtttcgaa atggaaatat cccttgcctg caattatggg acatgagttc 180
tcgggttacg tcgtcgaaaaa gggtagcgat ttggcaaacg tcgagatcgg cgaccgagtg 240
gttgccattc cattcttgcc ctgtcacact tgtgagatt gcaagtcggg tgtttattct 300
ttgtgtgaag atcacggtat gatcggttcc aagagctacg ttggcttcgc ggagtatgtc 360
aatctcaagg caaccacagt cctccaatt ggaaacatgg atttgagga ggcgctatg 420
atcgaaacgc ttgcagtcgc cctccacggc gttctccata ttcagccggc tctcggtgac 480
gtggccgccc tttttggagt cggcaccatt ggcattgttg ccacccaatg gctgaagatt 540
gcgggagtg aagaaattat gcctattgat atttcggatg agaagttggc ggaagcgaaa 600
gccatgggat gcgaacgaac gattaatccc ctccaagaca atctcatcga aaaggtccac 660
gagtatacca acggtctggg cgtggatgtc gccctcgagt gcgcgggatc taagattaca 720
caggaacaat gtcttttgat cactcgcaaa aagggtgaaa ttggttactt ggaatcgcg 780
tattctaatt gttgtttgag cgaacaagct tttgagaaca tctttcgtaa agagcttacg 840
ttgaagggtt tttggaattg ttatagcgct cccttccctg gcgaggagtg gacgaaatct 900
atcgacttca tctgaagggt ccgcacaaag ctgaaggagc ttatcagcca ccgctttccc 960
ctggaagaaa cgcagaaggc attcgacctg atgattaatc gaaaagaaac ctatggtaag 1020
gtgatgatca tcccaataa ggaataa

```

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SEQ ID NO: 119      moltype = DNA length = 1047
FEATURE            Location/Qualifiers
source              1..1047
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 119
atgaaagcag cagtgtctca tttcgagaat cagatcgtgt ttcaagaatt tgagatcggc 60
gaacttaagc cagacgaagt taaggtaaaa gtgatggcag cgggcacatg tggctcggat 120
accacaaga tgggtgcgaa ttggaagtat cccttgcag caatcatggg tcatgagttt 180
gccggctata ttgtcgaaaaa gggaaaagat ttggtgcatg ttggaagtcg tgatcgtgtg 240
cgaggcatcc cttttgtgcc ttgtaatgac tgtgagtagt gttttcgcgg tcagttttcc 300
ctttgcgaga atcacggtat ggtgggttct aagtctttcg gagcctttgc ggaatatggt 360
aacattccag ccaaaaacgt gctgcccatt ggagacatgg atttgagga cgcggcgatg 420
attgagcctt cggccgttgc cctccatgga gtcttgaaca ttcaccccca aatcggtgat 480
gttggtggcg tttttggagt tggcaccatt ggaatccttg cactccagtg gttgaaagtt 540
gccggtgtta aagaaattat tgccattgac atctcgaca agaagctcca ggatgccatc 600
gccctcggat gtgagcgcac gattaatcca ttgaaagaga atttgctgga aaaaattcat 660
gagtatacga atggtcttgg agtggacatc gccattgaat gtgcgggcag caagattaca 720
caagaacagt gtcttcttat tacgcgcaag cagggcaaa gtcggctatct tggcatcgct 780
tactccgata ttaatctgtc ggaaaaagca tttgaaaaa tcttcgccg cgaaactcaaa 840
attaatggtt tctggaattc ttattcggct ccggtcccag gtaaaagatg gacaaacacc 900
gtggagtatg tgaagcaagg caagattaag ctgaaggagc ttatttccca ccgattcggg 960
ctggaagaca cgcagcaagc atttaatatg attatcaacc gagaggagga atataataaa 1020
gttatgattg tcccaaacgg cgagtaa

```

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SEQ ID NO: 120      moltype = DNA length = 1026
FEATURE            Location/Qualifiers
source              1..1026
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 120
atgattaatc aggtttacca gctggtggcg ccgcgccagt tcgacgtcac gtataataac 60
gtggacattt acggcaacca cgttatcgtc cgaccgctgt atctctcgat ctgcgcggcc 120
gatcaacgat actataccgg acgtcgtgat gagaatgtgc tgcgaaaaaa attgccaatg 180
tctctggctc atgaagccgt ttgtgaggtt gtttttgata gcaaggagtg gtttgagaag 240
ggcacaaggg ttgttatggt cccaaatcac cctacggagc aacaccatat tatcgccgag 300

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aactacctgg cgtcttcgta ttcccgctcg agcgggttac atggtttcat gcaggattat 360
gttggttatgg cgcgatgaccg tattgtcccg ctgcccacacg atattgacct gagcactatt 420
tcctatacag aattgggttc ggtagctac cagcccatc aacgtttcga gcgtaagtcc 480
attccgctca agaatagctt cggatatttg ggagacggta atttgggcta catcactgca 540
atccttcttc gcaagctcta cccggaggca aaaatttatg tcttcggaaa aacggattat 600
aaactttctc atttcagctt cgttgatgac atcttcacgg tgaaccaaat ccccgatgat 660
ttgaagattg accacgcctt cgagtgcgtt ggcggaagg gctcgcaagt tgcgttgca 720
caaatcgttg agcatatctc cctgaaggt tctattgctc tcttgggcgt gtcggaactc 780
cccgtcgaag ttaacaccgg actcgtgttg gaaaagggtc tgacctgat tggatcctct 840
cgttccggtt ccaaggactt tgagcaggtt gtcgacctct atcgaaaata tccgatatt 900
gtcgagaaac tcgcactttt gaaaggccac gaaattaatg tttgtacct tcaggatatc 960
gtgcaggcat ttgaaatgga cctctctacg tcttggggca agacagttt gaagtggacg 1020
atctaa 1026

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```

SEQ ID NO: 121      moltype = DNA length = 1053
FEATURE             Location/Qualifiers
source              1..1053
                    mol_type = other DNA
                    organism = synthetic construct
                    strain = 71193
                    sub_species = aureus

```

```

SEQUENCE: 121
atgtgtgtgt ttgaagttgt tctcccaatg atcaaccaag tttatcaact tgtcgacccc 60
cgccaatttg aggtgacgta caataacgtg gacatttaca gcgactatgt tattgtccga 120
cctctttaca ttgagcattt gcctgcccgc cagcgatact acacgggatc gcgtgacgag 180
aatgtccttt cccaaaaact gcctatgtct ctcattcacg aaggagtggt tgaggttgtt 240
tcgatttcca agggagctct taataagggc actaagggtg ttatgggtcc caatacacgg 300
accgagaaag acgacgtgat tgcagagaat tatcttaaaa gctcgtatatt tcttctctcc 360
ggccatgacg gtttcattga ggacttcgtt tctcttaact acgatcgagc tgtcccgcgt 420
ccggacgaca tcgatctctc catcatttcc tacacagagt tggtagacgt ttctctccat 480
gccatccgcc gatttgaaaa aaaatccatc agcaataaaa acaccttcgg catttgggga 540
gacggaaatc ttggttatat taccgcaatt tctctgcgca agctctaccc agaactctaa 600
atctacgtgt ttggttaaaa tgattataaa ctgagccatt tctcctcgt ggacgacgtt 660
ttcttcatca ataagatttc tgaaggactt acctttgatc acgctttga gtgcgtcgtt 720
ggacgaggtt ctcaatccgc aattaaccaa atgattgact atatttcgcc cgaaggtagc 780
atcgcttgc tcggtgtttc cgagttcccg gttgaggtca atacgcgcct cgtcctcgag 840
aagggaactg ccttgatcgg ctcttctcgc tctggttcta aagactttca agacgtggtt 900
gacctctata ttcaataccc ggatatgttc gacaaattgg cgctccttaa aggacaggag 960
ttcgaaattg ccacaattaa tgacctgaca gaggcgtttg aggtcgacct ctccacctcc 1020
tggggaaga cggttttgaa gtgattatg taa 1053

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SEQ ID NO: 122      moltype = DNA length = 1026
FEATURE             Location/Qualifiers
source              1..1026
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 122
atgattaata ccgtctacca actgggtggc cctcgacgat tcgagattaa gtattccgac 60
attgatctca acaataataa ggttatcgtc cgtcccacct acctctctat ctgcaacgca 120
gaccaacgat actaccaagg actccgcgat gccgaaatc tgcgtaagaa acttccgatg 180
gcgttgatcc atgaagccat tggagaggtc gtgcgagatc catctggaaa ctttaaaacc 240
ggcgagttgg tcgttatggt gccgaacaca cctggtgaag aagatgacat catcgctgag 300
aattatttgc gatcctcgaa atttcgtgca tcgggttttg acggcttcat gcaagaatat 360
gtggaaatca caccggatcg cctcgttcgt cttcccaga atatcgacgg tactgttgca 420
gcttttacgg aaattgtttc tgtcagcgtg catgcaatcg gtcgttttga caagatcgcg 480
cacaaacgac gaaacgttgt cggcatctgg ggagacggca acctgggata tattactcgc 540
gtttttttta agactatggt tccagagacg aagctgtata ttttggagt caaccagat 600
aaacttcaag attttacgtt tgccgacgcg acgtttacgg tggagcacat tcttgaagac 660
gttaagattg accatgcttt tgagtgcgtg ggtggcgag gctcgggcaa agccattaat 720
cagattatcg actatatcaa tcccgaaggt accatttcca ttttggcggt ttctgaatac 780
ccagtgccea ttaacaccgg catggtcctc gagaaaggac tgcgcactct tggatcgagc 840
cgtagcgggt tcgcccattt tgagaagacc gtcgcgctct acaaatccaa tcttgagatt 900
atcaattatc tttcgaatat cgtcgttgca gtcgtcccag tgcgaacgat cgaagatatg 960
aaaaaggcgt tcgaaatgga catccaaaag gcttttggtg agacgattat gatgtgggat 1020
aaataa 1026

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SEQ ID NO: 123      moltype = DNA length = 1032
FEATURE             Location/Qualifiers
source              1..1032
                    mol_type = other DNA
                    organism = synthetic construct
                    strain = D12

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SEQUENCE: 123
atgattaatac agtctatcgt tctcgtctct ccacgccaat tcgaggtcgc gtataacgat 60
cgcagcctcc agtatcaga ggactttgtt gtggttcgac ctacgcatct gagcatctgc 120
gtcgcggatc aacgatatta cactggctcg cgtggaaaag aagctatggc aaaaaagctt 180

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cccatggctc tgateccaga ggccattggt aaagtggctc acgaccccaa gggtagatc 240
aaaaatggca ctatgggtgt catgattccg aacacgccta ctgaaaagga tgaagtcatt 300
ggtagaattt atctccgata tagcaaatc cgtcgtccg gatgatggg attcatgcaa 360
gattacgtct tcatgggtcg cgaccgtatt gtggaactcc cggaaggaat tgaccggtat 420
gttgttgctt ttacogagtt gatgaccatc acaacccatg tgttgagccg atttgaatct 480
cgttcccata aacgacgtaa cgtttttggc gtttggggag acggcaacct tggtttcatt 540
acgacactct tctgaagaa gatgtataaa gattgtaaag ttatcatctt cgaaaaaact 600
ggatataaac tcgacaattt ctcttctgtt gatgaggtgt atacgattga cgaaatccct 660
gaggacctgc gcacogacca tgcctttgag tgcgttggag gtaaggggtc tcagtatgca 720
atcgacaaa tgattgacca tctccaccct gaaggttcca ttgctcttat ggggtgttagc 780
gaatttccag ttgaagttaa tactcgtatg atcctggaaa aaggaatcac actgatcgga 840
agctctcgat ccggtcgaga ggacttcgtg cgcacgggtc actttctgaa gaaaaatccc 900
gatgttggtg aatacctggg taccttctgt ggcgaggtcc aagaagtcaa gacgatctcc 960
gatatttctg tcgcatttga gaaagacctt acatcctctt ggggtaaaac cgtcatggag 1020
tggagatctt aa 1032

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SEQ ID NO: 124      moltype = DNA length = 1026
FEATURE            Location/Qualifiers
source              1..1026
                    mol_type = other DNA
                    organism = synthetic construct
                    strain = EGD-AQ4

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```

SEQUENCE: 124
atgtcaacc aagtgtatcg cctggctgat ccacgtcaga ttgaagttca aactgttgca 60
gaagagatca ctgataatga catcgtcgtc cgccccaat atctgtcggg ctgtcatgca 120
gacacgcgat actttacagg tcaacgtccc caagcaacct tgcgccagaa gctcccaatg 180
gctttgatcc atgaaggagt tggcggaggt gttaaggacc ctcaaaaataa atttaagccc 240
ggtagcttgg tcgctatggt ccccaatacg cctttgaaa cgcacctat cattaaagaa 300
aactacttgc caacgtctaa atttcgcagc agcggttacg atggtttcat gcaggataac 360
gttacacttc atcgtgatcg cgcgatcgtc gtgccagaca acttcgacca ccaaatgagc 420
gcattcattg agatgggttc cgtcggagtc cagccctca cccagctgga ggccgtgatg 480
gatgcggatc caagatcat cggaatttgg ggagatggca acttgggttt tatcacagca 540
accctcgtca agcaaatctt tcagactct caactgatga tctttggagc acatcgctcc 600
aaattggact atttcagctt tccggacaag acatacctgg tggacgacat tcctagcgat 660
cttcaagttt ctcaggctct ggaatgcacc ggaggccgag gctcggagtc tgcatttctg 720
caaattatcc aacacattcg accaatgggc accgctatcc ttatgggagt ttccgaagat 780
cccgttggta tcgatacaag cagcgtgttg gcagagggat tgacccttcg cggcgtgagc 840
cgtagcggtc gcgcggattt tcagcgcgcc atcgacattt tgacgaaatc ccccgtcaca 900
cgtgaacgtc tgcaaaatct tgtgggtttc actcgaaaag tcagcacaat ccaggatatc 960
accgatttct tcgagggtgc attgacaaac tactggggca aggcggttat ggagtgggat 1020
gttttaa 1026

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```

SEQ ID NO: 125      moltype = DNA length = 1014
FEATURE            Location/Qualifiers
source              1..1014
                    mol_type = other DNA
                    note = uncultured Ruminococcus sp.
                    organism = synthetic construct

```

```

SEQUENCE: 125
atgatcaatt acgtgtatca gctggtcagc cccacgtct tttccctcaa atatgaagaa 60
gtggacttca agaagagagt catcatccgt ccgaagtgtg tggctgtgtg ccacgcagac 120
cagcgatatt ttctgggacg cgcgcagcgt aagatcctgc gacaaaaatt gcctatggcc 180
cttatccacg aatgttgcgg tcagggtggtc tgggatggca caggttcctt ccaacagggc 240
caaaaggctc ttctcattcc aaatgtgcca ggacgcgacg acggtgtcac atacgaaaac 300
tacgctgaag gatctcgttt tcttagctcg ggctcgcagc gtttcattgc cgagtatgtc 360
aacctttccc cagatcgtgt ggttcccttc gaacagggtc cgttccgggt ggccgcgac 420
acagagttcg tgcctgtggc ggttcacgcg gccggtcgtc ttatgcgagc ggccgacaca 480
cgccgcgagg ctatcggagt gtggggagat ggctctctgg cttatgtcgt ctgctgcgtt 540
cttctcgcac tgcctccgga gtgcgctatt tttgtggtgg gtaagcatga acgcaagttg 600
ctccagttct cgtttgcgca tcgcaccttc ctttttgata gcctcccgga ggacttccat 660
attgaccacg cgttcgaatg cactggagga gagggctctt actatgctat cgatgatatt 720
attaccgcgg tgcgtcccca aggcgctggt atgcttatgg gtgtctctga aaaccgcgtt 780
ccagttcgca cccgtgacat tctggaaaaa ggcttgacgt tcaccggttg ttctcgttct 840
ggacgcgaag actttgttaa ggcagtcgag cttatggagc aaccccaatt ccaaaaaacg 900
cttagcgcaa ttatttacga ggatcgtccg gtccatactg tcgcgcatct gtcccgtgtt 960
ttcgagacgg atttgaatac gccgttcaag acagttttcc gttgggagat ttaa 1014

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```

SEQ ID NO: 126      moltype = DNA length = 1017
FEATURE            Location/Qualifiers
source              1..1017
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 126
atgattaaca cggtgtatcg ctttatcgcc ccaaagtcca ttcgagctga ctttgtggat 60
ctgaatttgg atgaagaacg cattatcatc gcacctacat atttgcgat ctgtgcagca 120
gatcaacgctt attacaccgg atcccgtggt cgcgtcgtga tgaagcaaaa acttccaatg 180

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gcacttatcc atgaggctgt cggagaggtt gtttatgacc ccaaagggtga gctcccagtg 240
ggaacgcacg ttgtcoatgat tccataact cccaatgaga aagatgatat tattaagaa 300
aattacctgc gtagctccca gttccgtgcg tccggatttg acggcttcat gcagaatgtc 360
gtgtctatgc gtcagatctt tgtcatccca ttgtattacg agccggaagt cgggtgtgctg 420
cttgaaactca tgagcgtcgg aatgaatgcg gtgagcactt tcaaaaaaca tgcccatgac 480
cgcaaacgacg ctattggaat ctggggaaat ggcaacgtgg gctttgtcac tgcaactcatt 540
ctgcgaaaag agtttcttga tgccaagatt tatgtttttg gtgtggaaga gcataaaaaat 600
cgatactttt cttttgtctga tgatacattt atcatcaatg atattccga ggacttgcaa 660
tgggaccacg cctttgagtg cgtcggaggg cgaggtctctg aagcggccat ttctcagatc 720
attgactaca tcaaacccga aggaacaatc aatttgatgg gcgtgagcga gaatccagtt 780
cccattaata cagcaatggg cctggagaaa ggccctccac ttctcggaat ttcccgaagc 840
aactacgatg attttaaaaa gtctgttgac atgatggcga acaacaaaga cgttcgtgaa 900
tacctttcca ctattgtctc tgagatgggt gtggtgaaca caattgatga tatgatccgc 960
gctttcgaag atgatactta caatgatttc aagacgtca tgaagtggga aatttaa 1017

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SEQ ID NO: 127      moltype = DNA length = 1026
FEATURE            Location/Qualifiers
source              1..1026
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 127
atgcttaacc aagtttatcg actcgtttct ccccgctcagt tcgagggtgga gacagtcgca 60
gaagaagttg atgcggatga cttgatcgtg cgtcccaagt ttctcagcat ttgccacgcg 120
gacactcgat acttttttgg acagcgctcg caagccgcac tccgagccaa acttccgatg 180
gctttgatcc acgagggcgt cgggtgaagtc gtttacgac cccgtaatga atacaagccc 240
ggaaacctgg tcgcaatggg gcctaacacc ccggtcaaga aggatagcat cattcaagaa 300
aattacctgc ccagctcccg ttccgatctc agcggttatg atggcttcat gcaagaatac 360
gtgtgtctgc gacgtgatcg agctattact gttccttccg actttgaccc gcatatggcg 420
gcgttcattg agatggtttc cgttggagtg caggcaacta ctaagttgga atccacaatg 480
gatgaggatc aacgcgttat tggcatttgg ggtgacggaa atctcggtta catcacccgc 540
accctcgtta agcaaatctt tccagactct cagctcctcg tcttcggacg ccatcagttc 600
aaactggatt atttctcggt tgcagataag acgtatctgg tggacgatat cccggccgac 660
gtcaaatgtt cgcaggccct cgaatgtacg ggcggtcagg gctccgaagc cgcgatccaa 720
cagatcatcc aacacattca accgatgggt acggcaattc tcatgggtgt ttctgagaat 780
cccgctcgaa tcgacactcg ttccggttct gctgaaggtc tgactctccg aggagtgtcc 840
cgtagctcgc ggcaggatgt taagcgtgcg gttgacatcc tgcctcttc gccagttacg 900
cgtgaacgat tgcagaatct cgtcggctat gttcagaagg ttagctccat ccaagacatt 960
accaacttct ttgatagcgc cctcaactaa tactggggca aggaggtgat ggagtgggac 1020
gtgtaa 1026

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```

SEQ ID NO: 128      moltype = DNA length = 1029
FEATURE            Location/Qualifiers
source              1..1029
                    mol_type = other DNA
                    note = Lachnospiraceae bacterium
                    organism = synthetic construct

```

```

SEQUENCE: 128
atgattaata cactgtacaa acttacggcc ccactgattg tggaaatcct ttatgaaaat 60
atcgactgtc attccgacaa tattattgtg cgccccgaaa aactttctat ctgcaaggct 120
gacatccgat actacttcgg aatgcgtaat gctaaagtgc ttaaagagaa gttgcccttc 180
gcactcattc atgaggcatg tggctctgtt gtctatgaca tcaccggcaa gtttaaggct 240
ggtgaccagg tcgttatgct tccataatatt ccgggtatta acatgcagta tcatgagaat 300
tatcgtcgtg atagcctttt ccgcagctct ggagcagacg gattcatgca ggagtgtgtg 360
tcgatccctc agtcgcaagt catcaaatgt gagaagatgg atcaaaacat tgccgcgttc 420
acggagttca tctctgtcgc tattcatgca gtcagctcgt atcttgatcg cagcgcggga 480
aagcgaaatg gtatcaagcg taaaatcggg gtgtggggta gggcgcaat gggctacgcg 540
gtgtcgtgtg tcttgaagta cttcattcca tctgcagaga tcaactgcat tggaggtagc 600
aaagcaaacg tggacatggt ttcccttctc aatcatactt gttccctgag cgagcttctc 660
gacgatgcag agttcgatga ttgcttcgaa tgtgtcggtg gagcccaatc tggcaatgca 720
atcaatcaaa ttatcgatca tattctgccc gaaggcctcc tgctgttgtt gggcgttagc 780
gaagaggccg tgcctgtcaa cactcgaatg gtgctcgaaa aaggcatcac tttgttgggt 840
cgatcgcgct ctgagcgttc ggactttatc aaaacaatcg acatccttga tacgaacgag 900
aatattcgac gccgcatttc taacttgatc tctggcgagg tggagattaa cgagtggaac 960
gacatccaca aggcctttta tcagtcacaa attgcagact ataaaatcat cttgaattgg 1020
cgtatttaa 1029

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SEQ ID NO: 129      moltype = length =
SEQUENCE: 129
000

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SEQ ID NO: 130      moltype = DNA length = 1002
FEATURE            Location/Qualifiers
source              1..1002
                    mol_type = other DNA
                    organism = Moniliella pollinis

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SEQUENCE: 130

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aacacgtaga acacagggat cgccatttgt ccgaaatgca atctttgtct tcgaagcttg 60
aagcttgaat agcttgaagc atgagtgcac gaggggctgc tatgatcttc caaccaaact 120
tgcttgggca ttgcactttg gctggatagc tatcatttag ctgaccccca aagagactgc 180
caatgatatg atatttgtgt aggtgcacga gtggatgagt ggcgcgacta cctgtcacia 240
gatagatgct cgatgcactt catgtgaagt ggctagtgga tgtccggaga ctggagtaca 300
gtaaagagtt tggagtggg aacagcgcca atgaaagagg ctacagagaga ggtaggtttg 360
atgtgacgtc aagcgggcta gtcagttagc aagggagtat ggaatgcagc gcgcaacgcg 420
cagcgcggag cccagtgcgc ctgcgctcag gtactcctac tcagtactca gtactcaggc 480
gccatgcccg agcttacatg caatcaatta accctgacgt ctggcgccgc ctgacgectg 540
aggctgagg cctgacgcgg ccaagctcca agcgcacacc caggctggct tgcgctggcg 600
ggcgcgagac gcagcagggg gcaagtgtcg cagagtccgc gatcggatct gtggaacagc 660
acaggaggcc cgtgcggtcg gctgcgcagt gcggtggaga agctggaagc tggaaactgg 720
aaggtggaac ctggaacttg gaacttgaa aggaagaaga gcgggaaggc gatgcacgat 780
gcacgatgca cgatggggtt ggattgcgta cgagtaccgc cacagcgcac agcgcacaac 840
actcacccca tgccccacg cccacgcccg tgctcgggcc atgagctgct ggcaaccccc 900
cgaggcagcc ggcctttaac tcaagttaga ttccgcgcgc gtcggttctt cttagcctcc 960
caccaccacc acctgcgcgt gcttggacgc taattacaat aa 1002

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```

SEQ ID NO: 131      moltype = DNA length = 679
FEATURE            Location/Qualifiers
source              1..679
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 131
agctcagacc ttcatgttta gccttgacct tgccctctatg accactaatt cgcacaagat 60
tgcacgttac cattgtcatc ttgccatgta tccaatgttt cctgtttgccc atagtacgta 120
gtagatatac cgatcatcga tcagatgac gcagatggca ttggttcgct ttgccctatg 180
catcatcatc gcatcatgca tcatgcatca tgctcacta tccatattaa tcattgcctc 240
ctgtactagg ccatgacacc tgacagacct tcccagggga tgcttcttgt aatagccaat 300
atactacaca actttaccga ttacgatcga gatcgatctc actatagttt gtcgaacttc 360
gatgcacatc gggggtgcgg aaaaaagctg acggaagcgg aagtggtgta attggttaat 420
ttagttagat tggcaggggt gttcccaaga gctgagaaag aaaacagaaa 480
cacgaggcat aaggccagtc agaaaatgca ttacaggggg agttgaaagt actgtgggcc 540
agtgcctggg tcgtcatgtg caaatTTTTT ctccggttag actgcgcgag catgcagcaa 600
ctctgccggc cctatataac tagtgcttca tcttcttgct attcttgctt ctttaaccaac 660
acaccggtac tctttcacc 679

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SEQ ID NO: 132      moltype = DNA length = 1002
FEATURE            Location/Qualifiers
source              1..1002
                    mol_type = other DNA
                    organism = Moniliella pollinis

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SEQUENCE: 132
catcgttcca gcttgccctt ggtgctagcc agcgcaaaag tccgacgaac agacgaattg 60
accgtagtcg atgacagtcg atgacagtcg atgacagtcg atgacagccg acgaccgacg 120
aacaacaacc aaccgcacgc agcatgaggg cgagaggctc aggtctgagcc gctaaggcgc 180
tgaagcttga agctcggttg tccagtcaga tctcatgac tcacaccgcc gcgctgctgc 240
ggctgatatg ttcaatttgc tgtgcctagc ctacgctggg tacgtcggta actagccact 300
agccactggc cattaatttg ctactgggca cactaggagg cgcactagtc catcgcaagc 360
ggtaacttac ctaccggcca ctacctgatg atttccctct tgctggcgcc cctggcggcc 420
tggcgggcga ctctttgcgc tcggacccgc aaattgcaaa ccaacaaacg actagtgtcc 480
ggtagctgtg cccaagtaac ctgggcagac atagaaccac attgatcgta gggaaaccgca 540
atggaccagt acgctattct cagtacattc tgtacacttt tgcttagatt ggaatctggt 600
gaggatcaag atcaaatttc aaaccaactc cacggcactg gttcccgaca cttggactgt 660
agtactgagg tactgaggtg ctgaggtact gaggtctaac tgaagtctaa ctgaccgtga 720
ccgcgcttgt gtgggtgcgt ggattctagt gttggatgtg cgcagtggtg tgtgcggtgg 780
tgctttcaga ttctgcggta gaacttgact attcacgtct tgtgctgcgc tgacgcttgg 840
aacctggaat gttaaacctt gggtcctgag ggtctccacc ccccaagtcg actgggttct 900
atataaagtc gggtaggcaa ttggctaggg agaattctct ttttctcttt cccagagaaca 960
cctgttcact cacttctcac ctctttactt acctacttca ca 1002

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SEQ ID NO: 133      moltype = DNA length = 1000
FEATURE            Location/Qualifiers
source              1..1000
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 133
cctcgttctt gattgcttcc aaaagtaatt taaggagaga acctggctcg ttgttgctct 60
actgaatgat cttacctata ccagcatagc acaccttgct gccctctccc ttcaccacca 120
ccatcaccac tttcctcttt ttctttgggt ttttctagtc gttgcattta tgggtgattc 180
gctctttgag aaccgtatgg ctgttatccc ttcccaactc tttgtctgtg tttactgttg 240
catccaaacg tttgggctct tcgtacctct tttttctctg tttgcctttt atttttgggt 300
tttctatccg tttgatgggt gtgactcttg ttttcttttg ggaaggtttg agctcaactt 360
ctcttcggtc cttgttggag tttgacttcc ttttcttttt cagctgtggc cgatcattac 420
gatcaataat ttgtgaatat gatctgact acatcctgaa tcttttagcg aacctatgca 480
gtttagcgtc atagatatgc atgcatgatg tacatcttac acgcaccacg gatgatgctc 540

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agcgcgcagc caccgacgag ctcagccgct cgtcgggtac gctcgggtcat gatgtgcatg 600
atgcaagcct aatgattgac gggtagcttga tctgtgtccc aattcgatcg agatcgatct 660
cactatagtt tgtcgaactt cgatgcacat cgagggtgctg gaaaaagct gacggcagcg 720
gaagtgggtg aattggttaa tttagttaga ttggcagggt tgttcccaag attcaacgca 780
agctgagaaa gaaaacagaa acacgagga taaggccagt cagaaaatgc attacgaggg 840
gagttgaagt gactgtgggc cagtgcctgg gtctcctatg gcaaatttt tcttcgggta 900
gactgccgca gcatgcagca accctgccgg ccctatataa ctagtgttc atcttcttgt 960
cattcttgtc tctttaccaa cacaccggta ctctttcacc 1000

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SEQ ID NO: 134      moltype = DNA length = 1007
FEATURE             Location/Qualifiers
source              1..1007
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

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SEQUENCE: 134
cttgacgggt tcaatattca ataactcagt ttcaagcagt ggtaggttgc tccgtggatg 60
gaacgacggg ttttaattct tccgccatgc accatgggtc atgatcggtt cagatggaca 120
gttggtatga gaaagtggta acagataaat agaatttcgt acaagttggc aagagcctgg 180
ccgatgggtc atggctcgat ggctcgatgg ctcgatggct cgatggccac acaccatgtc 240
atctactttc tgtgcttcta tgcggacgag tcggaacagc tgtcaaatcg cgtttggggg 300
atttgagcgg aatttgaaac cttgttgaaa ctacattgtc tcctcgttac ttctctggca 360
tctgtttgat cgaacggcgt catgcaatag acacgcgatg agggcccaac acggtgacgt 420
tgaacgcctg gcccacctgc tacaacaccg tagaacgtgt tgaagttgaa cgtcggacgt 480
tggacgttga gttgacgcgt gatgctggaa ccgtggatgt agcagcgaca gacaagccct 540
ctaagcaatc tagtcgattt tcaaaccaac taaacgagcc ttagttcgga caaccctagt 600
ccagtgcggg ggcagcgtgg ggtcagttgc cagcgaaagt caccocagct agctgggttg 660
gctagcttga ctggctcaga agtatcgatg ctggagcttc aggcacgtac gaaaatacac 720
tgtcgtatgt ctcttccgg agttctgggc ttctgagccc ggcggttggt catcgttgca 780
cagtgcacgg accggcgggt gggacaacaa actgtctggg cgtgaaatcg tgcgagtgtt 840
gatgcaaacg ggacgtaccc agcacgcagt tcccagtag cagacatact cagtacctca 900
gtggttgagc ctacgcacgt cctccgtatt tctacaaagc tgttgagtgc ccttttgact 960
ttttactacc accatccctt cttcttctaa accccttaca ttccaaa 1007

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SEQ ID NO: 135      moltype = DNA length = 1000
FEATURE             Location/Qualifiers
source              1..1000
                    mol_type = other DNA
                    organism = Moniliella pollinis

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SEQUENCE: 135
tcgacgacag tgcgtatagt caactatgct atctagtagc caaagtaagt ctacattccg 60
ggctttgcag gggtagctta tttcattaga caatacttac aaccttcgaa gggcttcccc 120
agaattcaat tctatattga ttcaatatcc acccatcgga cccgctctcc gtacccctcg 180
ttcgcttcgc cactacaacc atgtcaagac ctgagaatcg tagcaaatg ccccaactacg 240
gcccagtgct tgtcggctgg ttaaagtaga atgacttggg tagccagatc gagacgcctg 300
acattgatac gcagacagcg cagtcagttt caaactgaca actgacgcgt actggattga 360
caggactcct tggcgggtgc cattcgaatc gctttagtag ctcttgccag catcttgcca 420
gagaagcttg aaggctggat cgctcagcct gagctcacac ttagtagctc cactagtctc 480
tcaggtgctc agctctctct cagcgcctca agctcaagcc tcagggcctc agctgcatcg 540
agctctgggt ccagtgccag tgcagtgccc agtgccagtg ccagtgccag tgccagtgaa 600
gagagtggta tgagccaagc tatacccaaa agcaaatgat gggtattgct agcttccatg 660
caccctgggg gcgaaggaca acgaacaaca gacagagttt gaattattca ctggcctagt 720
actaccaatt taaatttggt ggttcagcgc aggaacggcg cgagtgcgag taagggcaaa 780
ctccgcacgg ggcgtgacgt cagtgctcct ctgggtgagg cactccagca ggccttggtc 840
aacgcatcaa cgtgtgtgtg tgtctaccaa gcacattgtg cgctcgtgt ccgtgcgcgg 900
aaatcgcaac agtcagaacg tctataaaac ccctcttgct tggagggtta ccgggcccgtc 960
ttttcccaac gctcgggtc actcactact ctacacaaca 1000

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SEQ ID NO: 136      moltype = DNA length = 997
FEATURE             Location/Qualifiers
source              1..997
                    mol_type = other DNA
                    organism = Moniliella pollinis

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SEQUENCE: 136
tcggacccca agctctctag cttcaagctt tcaagctagc ttccagccag ggctcatggaa 60
ccgtaggagc tgggaaacgt gacgttgtcg ctatgcctat cccatataag aattaacgtc 120
gaatcatgag gcgcatggga ggctgcattg gtggtatgat acataacggt atgattctat 180
atgagcgatg gagcgcgggc gctcgcgct agtggtgaa gttcgatacg ccagtcatt 240
tcgttcagat gacgttttgt ccagaagtac ggtacatacc gagcgcaaac gtcgaactta 300
tggccttggt cgctcgttgt cggcgctcgt cgacacttcc atcaacggag gaaaggtaga 360
gccgaacgcc agcggccaat ggcgatggcg atagtagaat gaatagtcgt cagatctagg 420
tcttaggcta caggccaccc tgccctgaggc ctgggggtct tacagcgagt ctaaaaaacg 480
aaaaagccaa cggccaactt gggcctgccc atgcacccct ccagagagcg ccaagctggg 540
ctgatgcaag cggagcaaaa tgttggaaga atagtttcaa cgatgggttg gttccagtag 600
gcacagtcga gagatgaggt actgcgcaac gagttgttgg cgacggaacc gaagatggat 660
tgaactattt gatctacagg tgatggcggt gagtcggggt tggggcgct gtgtgatag 720
tagaagtggc gctattcgag ggctgcgccc gtctggatgg atgcagaatg aagcttgtag 780

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gccgcagctg gcaggatggg cgcagaaccc ggtgaaacat agttaaagcg cggggccagt 840
gtcatatttc gcttgagcca ttctactcgt tggcagccca ggcgccgcta atttacgcca 900
actcttggtg ctcgagagct cgggcaatgt ggtatataag cttgaggttc cctacagcaa 960
agcctggcctt cttcaccaca accgcactct agcaaac 997

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SEQ ID NO: 137      moltype = DNA length = 1002
FEATURE
source              Location/Qualifiers
                    1..1002
                    mol_type = other DNA
                    organism = Moniliella pollinis

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SEQUENCE: 137
gagcgtagag agctccatcg atggcctggt tacaagttgc ctgaagacca ccaatacggc 60
ccgcacgcca acggcaggcc aagaaaggat aattgacaac cgtgaatctg aaagcttcgg 120
cgctcaggag ctcattgtgtt ttgtttaga tataaagggt tctgaatacc agttgggtat 180
tgacacgggg agttggactg aatctctcag atctcccaga tctcaagtac tgagtcccaa 240
gcgtgccata attaaactta ggttgcgcgg gacacggcac ctgagatcag ggtgtacgcg 300
cataactaat catccacca ccaaaagcaag actgagggtta ccgcctgagg ttctggctat 360
tccaacagtc acaagtcacc gcgattgccc tcaaattcca aagcgtagta ggcgctcgac 420
ggagggatga tggccagtac ccaatttcga gcactcagtt agatctgttc gtgtttagtt 480
gctcctgcag atctatctta aatgtatact gtgccaggta ggatctctcg gcacctatag 540
atgatataaa agatatgatg atatgccaaag tgccaagccg tcaaactgac aaaatgaaag 600
aaacagcagg caagctaggg aagcgtctat cgggggatta aacggatgag tcagcagcac 660
aggagcagtc aatcgacaat tcgcgctttt tcccgcgccg aaggccccag accccagacc 720
ccagccgcca ccatgtccgt atccgcgggg gttagcatga ccacggcaga ttctccgct 780
cagcttgctc ggacgactat atataagaa tcctttcctc gcagtcgctc cgagcccttc 840
ctccacccca tcggttacca ccaacgacat cacttggtct tgacgcgcgg ctgcgttttc 900
ccgcttccgc gtctccctag tgatttagct ggtgttttcc tctaccgcc actctactgt 960
ttcatcgctt gttgccaaat tttctcagtt cattctcatc aa 1002

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SEQ ID NO: 138      moltype = DNA length = 1002
FEATURE
source              Location/Qualifiers
                    1..1002
                    mol_type = other DNA
                    organism = Moniliella pollinis

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SEQUENCE: 138
tggacgtgga caagcaccct gaagtcgctc agggcttcca gatcaaggct ttgcctaccg 60
tctacggcct gtagtagggc aagcctgttc gtcagtgtaa gttttaaata atattgttct 120
tgtctgttgc ggttttacat cagtgggtga ctgacacggc tttttgtttt attgtcgcag 180
tccaagggtg acttccctgag gatatcctgc gccaatttgt ccaagatctt cccagcttgg 240
ccgcaccaaga tgcccagaac ccgcctgctg aaagttagcga ggctgccaac taagtatagt 300
ccatcctatg agctgtccta ccccaaacaa ctttttcacg cccaaagctc aaacgtcact 360
tcttattcgc gcaaaagaca gctgcttccc cactcatcca ttaccccttt ctgatccacg 420
agggtgaccc atggcccttg tacattccat gaaacaggta tccttttccc caccattaca 480
tctgaactta caaagacttt gcttctctct gtacacagct cgtcgtccat cgtccatcat 540
cgatgacgat catcctgcat tcggtatcat gcaccatcat gcagcgatgg caacgagggc 600
catgcaccat gctgcatata ctatgtgat agccaatagc caatcttcaa tctccaactt 660
cctacagaca cagcctgtgg ggaatcgctt aatcgctgtc catgcgtcca tgcctcgcgt 720
tcctcgcgct cgtcgcgggc tcgcgacgat taagtacacg tcacgcccac acagcgcgcg 780
agcgcacgag ccacatcatc cactgtatga gacaaagagt ccacttccgt aaaagcaatc 840
ttaataatc atgtaaccct agcaatgtag cctaataat ttgtttggcc ttatttttat 900
tgcccccccg cagcttttgc accccggaga gtaaaagtat tccggttgct cgtcagagca 960
tccttttctt tctccatcag caaagcgtgc gaccgccaag aa 1002

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SEQ ID NO: 139      moltype = DNA length = 921
FEATURE
source              Location/Qualifiers
                    1..921
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 139
tttgatgag gcattcaatg tctccatggt tgaagaccag agaaagtgtg ggtaacacaa 60
tagcatgga caattagcga atccgaatat gtgtactaac atttgtcctc caactgacag 120
ctggctgaat aatgatgtga atcatcaaac atatcagaga tatatcccag aaccatttaa 180
caactgttat tgtatgatg cattactggt ttacattgtg tctttccaaa tactaacatt 240
gccattatcc aatccatata tatatatata tatagattac acaacgcaaa cctcgacagt 300
cgattggttt acatcggtga agcagccttt tacggaatat gatggacaac cagatcctca 360
tagatggaga caatgtgcag atagattcca tttatgatga cattgatgtg tttttaaatg 420
caaaataatc tcctatataa gtgagttccg agtctcccca tgcttatctt atcgtgtttt 480
ttctgtaaat taggtcaaag tattcaaaac attcctcaa tccttcccat tttttgtgta 540
gcccccatat catcaaatca acgtatcaaa tttcctaatt agtagccccg tatgttaaat 600
tagacatgtg tgactgccat ccgagacagc ctacccaatg acaccgcgc accgcacata 660
ttgtgtttag tgcgcgcgtg ctgctgaagc gactccctgt ttgggaggaa ccgagggcgg 720
gttgcccgat cccctgcccc tcgctcctcc tctctggctc ccccttgacg agggacacgg 780
aggggatccc tcgtgtgaga gcttgaggt ggatgggtgt caattttctc atttgattga 840
aagatttgtt atattgaaaa ttcagtttgt ggaagtgtgt attaaaaggt tttactgttt 900
gttctgtaga cacattcaat a 921

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SEQ ID NO: 140 moltype = DNA length = 500
 FEATURE Location/Qualifiers
 source 1..500
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 140

acggatcaac	aaaaatgtgc	gccagctttg	ttgtgtgccc	ttttcctcta	gcggatccta	60
tccacatttc	gtgaatgaac	cgattaagct	ctgacgagat	ttccttgttt	tctgtagata	120
cctcttgtag	agggtgtcaa	gttatgttgt	gatcgagaga	gttacctgca	aggattattg	180
caagcggcct	gatgaggaca	ctcaactacg	tgtgagtagg	agtgcgttgc	tcagcagttg	240
acgcgtgagc	cgcttgagca	gcttgataaa	ggtcctaata	gtcttccagc	tgttggcaga	300
cgacaatgtg	ctgtagtagt	aggataggcc	cgtaaaagtg	tagaagcaat	aggttgagac	360
tgactgtaat	gggtacttgc	gttcaaagac	tgaaagtcca	aaggattgct	caaaggcccc	420
tgaggtctga	gcctctgagc	gcgctcttcg	tcaaaattcg	ccggggccat	cggtcatcct	480
ctcgagggca	gcattcttct					500

SEQ ID NO: 141 moltype = DNA length = 500
 FEATURE Location/Qualifiers
 source 1..500
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 141

gtactaacca	agttctattgt	aaagctgggtg	tctgctgctg	cttcagcaag	tgccattgct	60
accatcgcaa	gcaaatgtca	aaaaaattcg	ttcaatggtc	taatttaagt	aaaatactca	120
taatttataa	aatgccgctt	gtccttacct	tttgtcttcc	tccttattat	agtagctgta	180
gttatcgtag	tcctcaggac	ttcgtcaacg	ctcgttgatg	atcatatata	cttcacctca	240
ttcattcata	aagcattcat	taggattgta	tgatccggca	gtgcatcatg	catcatgcat	300
catgcatcag	gacatcatgc	atcatgccgt	atcggtcgat	cattgcattt	gattggataa	360
tacgatcgtg	tttaggcatt	ggcagtcgaa	acgaggataa	agaagaggtc	agtagcagtg	420
cacagcacgc	gctcgcgctc	tatgcgcact	ttgacgcttg	aaaaaaactt	ttcagccaat	480
cggatataga	cagcggaccc					500

SEQ ID NO: 142 moltype = DNA length = 493
 FEATURE Location/Qualifiers
 source 1..493
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 142

atgggtctgag	ccgccccacga	ttccttccct	catgagcgta	tgacgaccct	ttactcttagc	60
atgtgcctct	aaagaatacag	ttttgcatac	ccacatatcc	gtaaccctgt	cctctatagc	120
ttctacaggt	catagtcact	ggctgccagt	ataaacgtgt	caatatctaa	tcctcataag	180
atgcctcgct	ctgtcgtcca	caataaaatt	cagtaacgca	aggactgcgc	gactgagtga	240
ttgccgatga	tatgagaaaa	ctaatttcaa	gccatttgaa	cttcaataaa	tgcttccgct	300
aaacacacgg	atcatcgatc	atcaaccagc	ctccattaat	aggcctgccca	ccactcgcgc	360
aatcatgaat	ctgatcatat	ctgggtagcg	attctcataa	caccagaatt	acctgcccac	420
taaaagaaacc	ctattctgcc	tgtaaccccg	caccagccgg	tccgaattca	cgacagacta	480
aggctctcgc	gca					493

SEQ ID NO: 143 moltype = DNA length = 500
 FEATURE Location/Qualifiers
 source 1..500
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 143

atcagttgga	atcatcaacc	tttttttttt	gtatcgattt	cttcctaccc	tcacctccct	60
tttccccctt	aactcctcgc	tttgaacctc	cttacccgag	tgctacacct	tacctttttc	120
ttttctctct	tcccctgcct	aatattctag	gcttgctact	atactgctct	ctctaacacg	180
attttttgcc	ttgtgaagcat	ttttctcttt	ttaaattctag	gttttatcat	ttgtttgcatt	240
tatccaagtt	tttttttttt	tttgtttgta	gtggttttac	tgttttttgtg	ttttgtagct	300
tgactaggtt	cttattctat	tgtagatgta	tatgatcgga	tgcattcccc	atctaaaagg	360
aaaatgttct	cgactgatct	ttatttttac	cttattgtct	tcctaaatgg	tttgactcct	420
gccttcttcc	ttgggacgct	ttgcaacggt	ttgtacgtac	gtaaggggttc	atcgtttgga	480
taagcgacac	agttgtcaag					500

SEQ ID NO: 144 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = other DNA
 organism = Moniliella pollinis

SEQUENCE: 144

attcaccctg	gagtaactgc	gcttcacgtt	ttctttgtta	cgacaataat	accacgtctg	60
atgctttgaa	ttgagtatat	attttgatga	ctgggttcaa	tagatctttc	ctccatagcc	120
ttcgagtact	cgtgatctca	gcattatcag	atttgtgtcc	aggcccccca	ggacgatgag	180
gttttctctg	agtcactagt	tgggaagatt	tgtactgggt	gggtgtaggt	agagagaacc	240
attgatattg	gtgtagctgt	tcgttttata	aggccgagat	tacgtaatgc	acatgctgcc	300
ctagcaaatc	gcgtcgcgtc	acgc				324

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SEQ ID NO: 145      moltype = DNA length = 496
FEATURE            Location/Qualifiers
source             1..496
                   mol_type = other DNA
                   organism = Moniliella pollinis

SEQUENCE: 145
atgtgcttag attaacatac acttgacttc cttacgcgtt gatataact tggatctcca 60
taacttctat tgtatagttt aatcaatgga ctocatgttg acatcccatt atgcgatcta 120
tagtagctgg aaatagcaaa attgaagttc ggcagacgac tgcaccagga ggggtagatt 180
ctgtcccata actccctacc acaggccgtg ctctgggttc atttacctta gctgatgctg 240
ctcaatgtaa cgcgtaattc tgagtgggtt tcggtcgtcc aaggtttcag tatgcgctgt 300
gcagatagta ctgtatcgtg aaactcgacc gttgcacatg cactttatca agtgtcgtag 360
gcccagagtc gtaggccaag agtcgtaggc caagatggaa ggcctgaaga aaaagaaagc 420
tttgatcgtt ctgatctatg caatggattc atgagctaag tattagctat acaaatgatt 480
cgagcatagc agaagg                               496

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SEQ ID NO: 146      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source             1..500
                   mol_type = other DNA
                   organism = Moniliella pollinis

SEQUENCE: 146
gttgctccct ttgctgttct atgatccaaa agatgcttag tgctccactc tctactcccg 60
actacctttt acgcattctg acatgactag tagtgcaatg ttacccttgg tgttttctcg 120
caatcacttt ctgtctgcta tagcttttga tcgtcagtag ttttttggtg tagctaaagt 180
agacttttgt acctggcggt caatgtaatt ttaactcctt tttgtgcgat gcaacagatg 240
gtgtaaagat cgctgtttaa gttttggctt gatgcttgat tcggtgacgg ttgcttgggt 300
gatccttata tcgtctcctg gtgttctcat aagcccaggc tcttcttagc tatgtacaac 360
agggtccacca ctgactatgt tatgtagagg catgcattta acttacggaa agatttaaatg 420
gagaagaacg tacttagtca gatgcttgac atgaaaacgt attcctgcaa ttcacaaatc 480
agtatcctgc ctgtccactc                               500

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SEQ ID NO: 147      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source             1..500
                   mol_type = other DNA
                   organism = Moniliella pollinis

SEQUENCE: 147
gcggttcgca cttttatccc aggcgatgaaa aatgacgcac ggctgtcgca cgctacaccc 60
agtagtcgcc ttttcaggca gtctatgagg ggagcttttc attgcccata attactagcc 120
tgcaattcat gatatacagg tcaactttctc ccgatgtcta tagatttctc atagcgcttt 180
gaaaagacaa tttagctact cagtttataa gcaaatgatg tggaggtatt tattaggtac 240
aagatacgat aacgtttaca cgcggggaag caagtggcaa atcgtcgaca acaccgcag 300
gtggtgggtg aatcgtaact catcaaagct tgcctgtcac tttgggaggg accaggttga 360
ggccatcgcc cgaaggttgg tgtggccagg gtttcgatat ttcatgaatc aacgtgttga 420
tggatttga ttgggtaaga ggtcgatccg gctgttggtc agagtgtctc tcacattcgt 480
cgttgtcaag cccgagttgc                               500

```

```

SEQ ID NO: 148      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source             1..500
                   mol_type = other DNA
                   organism = Moniliella pollinis

SEQUENCE: 148
gtggcggtga ttctgtatctt attattgata tgtcctgttc ccacggttgc caagtgattg 60
ttcatgctat cctctctcca ttctgtttgt gttgcacagt cggtagttgt acgtagtagc 120
tcgtcacaaa atatgaagtg agccttgcaa tgtaggcatt cacaagctgc taggcaggct 180
atgatctaag ctaagtggca aaaattctct tgattcaatg ccaaactata ctattgttcc 240
tcaggcaaaa gcagctatat aagttgcatt caacgcaacc gccatacaag caaattagag 300
agaaacggga tataaggtgc tttatccctg aaggagtagc cagcgtctac ttggttgcta 360
cacaatacag caaacagtgt acttaatgag ccattggctg aactggagct cgcagaggtt 420
cgagaagggc gggctcatacc agctgccctc catggaagct ggggtgtcga tgatgaaaag 480
tgtcggaacg tacgctagta                               500

```

```

SEQ ID NO: 149      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source             1..500
                   mol_type = other DNA
                   organism = Moniliella pollinis

SEQUENCE: 149
acgaagggtt tcccatctgt gtgacctgat ccttttttac gtttttttgt ttgttttttt 60
cttttggtca tgatcaactt gggagtgatt cctcttggtg atgtcgaaaa ttgaactgtt 120
ataccaatat tctgttgctt ttgtttcctg tttgtagttt actacttttt acagtagtgc 180
ttgttcctgc cttgcttgcg ttgtcgaggg tttcccttta gcttagtacc aatatctagc 240
cttaattttg cttattcttt ctcaccgccc atgattgatt tcaatcatga ttaatgattt 300

```

-continued

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atgcatgaaa actcaaaata ctcaattgtc ttgtacaagg caatcacaag ctggtcagcc 360
attctcagcg cctataagag ctttcagcca ggccggcccc aaaacgatct accctgtgga 420
cctggctagg ctaaaccttg gaaagtcagt caattatgaa cgtttgaagc tcaagcgctc 480
ctaccctctc ctggagcagt                                     500

```

```

SEQ ID NO: 150      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 150
tggtgttcac cactgttgct tcccccttgt tggagcgctc gtcgttctga tggcgctctg 60
agcccaaaat atcccttctt cgcccgagag caagcctcgg agtggtcttg caatctactt 120
ttctactaaa agttatcttc ttgtggcttc tgaatttgaa tagaacgtgt tcagacttgt 180
tagctaaatt tggtgatgag agatacttga cagaatcccg ttgatgaact catctgtgtg 240
accaccgctt ggtgatggaa tctcgcgcgg aagcgtggtt gttactgact ctcaagcaat 300
tcagtagata aatttgtata acatcattgc aaggggtgct agcaatcaat tgtccaatac 360
caaacggcgg caagtgcata ggggcatag agagtttctg cgagggaagg aaggaatccg 420
tatgcctgaa aacttttgta caagtgtctt tgtaaggcgg aggtctgctc tgacatctct 480
ctgagatctg agatctgata                                     500

```

```

SEQ ID NO: 151      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 151
gtttcttggc tgttgacgaa atcttatgaa atgcacgggt ctaatatctc ctgccaatg 60
tgtaccatc tgcttttgta atgtcatagc tagctctgta gtttctaaac gaccgtctta 120
ggccgggggt tgctgattgg ggattaagga taggatactt ggccagtttg ggccgatcaa 180
tatcaatctg ctaggcgctc cttaggctcaa cttgaattat gatatatgag gtatgaggaa 240
agtgggctaa atctgtggat taggatcaga aaaaacgtgg cagaccctcg atctaggcct 300
acgtcagcaa ttctagcaat ttccagcgcct tgccccctag atagggttca gccgctcagc 360
gcgcgcaatt tcagcatgga tgagatcaaa ccaaggggca gcagttggcg accctgtagt 420
tccaatcgca tgcttagtgc ataacgcgca catcaacgtg tcagattctc agagatctca 480
acaagcccac tagaccctag                                     500

```

```

SEQ ID NO: 152      moltype = DNA length = 504
FEATURE            Location/Qualifiers
source              1..504
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 152
aatcgctctg gtagogaacg gggccagctg ggtaaggcca gcggccggta gctaggacta 60
tgaggctacc aggtctgtgg gtaactaggc agctaggcac ctaggcaact gagcaactga 120
gcaactggcg aactgagcgg tataggtttg gcttgggcac agttgatatt gatatcaacc 180
agtgcctctg tcggttcacc agaaccagct gtgtggtagc gagcgtagaa tagccaaaag 240
gttttgagct gaggcgctga ggcgcacagg gaggacatca ccgcctgagc acacaggatt 300
tcaaaactgc agacggcgct ggtgccagtg ttggaagtgg gcgaagcacc tcgcattctaa 360
gacatgtaat taaactctcg acagccatcg tccattgaca ccaacgtgca ccaacacacg 420
cgagagctgc ttctgtgtgc ctgtcgacga cattgcctcg ttccccagcc aagtcgcccc 480
accgctacac caagcgcact cgca                                     504

```

```

SEQ ID NO: 153      moltype = DNA length = 499
FEATURE            Location/Qualifiers
source              1..499
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 153
catccgaaat tgggtggttg acaaatgcac cttcttctcc agggggtcgt gcaaacgcag 60
tagccgagaa gagcgctacc ggagaggaag acccaacaac cactgatcgg ttgaaagttg 120
gatgagaaag ttgtgccaat cgccggagccg tagcaacgct acgagcgcta cctacaacga 180
ctccacgcgc aatcggtacg cgcaaaaggaa cggatgagcg accgagcagg ttgcgcgaag 240
ccatgacgat tgctctaccc ccaacgagct ggatcaagtt gggcagaaca taatttatct 300
cgcttccccg cgccacaagt agctccacca ttcttctcaa aatagcccgga ttccgctttc 360
gcgggggcaa acttcgcctc ataccgatcc gattgaaacg ccaaaaaaat taagttgtag 420
tacgcctggt ggtcgagccc gactggagca ccatttgaga cactaggaga taattttattc 480
ttttcttctc tggccccca                                     499

```

```

SEQ ID NO: 154      moltype = DNA length = 300
FEATURE            Location/Qualifiers
source              1..300
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 154
cgggtaagac caagtcggtt ggtgtttcga acttcagcac tcgtctggtc gacttggttg 60

```


-continued

```

aggaagcttc gggcgaaact cctgcggtta accagatcga agctcaccce ttgttgcaac 120
aagacgagtt gggtgctcac cataagagca agaacattgt catcactgct tacagtccct 180
tgggcaacaa tgtcgctggt aaaccacctc tgactgagaa ccccggtatt gtggatgctg 240
ctaagcggtt gaaccatact cctgctgctg tgetcattgc ttggggattt caacgcgggt 300

```

```

SEQ ID NO: 155      moltype = DNA length = 453
FEATURE            Location/Qualifiers
source              1..453
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 155
gcgaggagta gttactaggt agcgggtatg caattccgcc cataatgcat aaccgttctc 60
cctacaaaaa caaaacaatg gctacgtggc cggcggtgct ggccgacatg gtgcgggtgc 120
ggggggcggt gtacgagtgt ggggagttca ctagtctgct ggctagctga ctactgacc 180
ggcgtgtggt tgtggcagta cgcagtggga aagattaaat atgaaatatt gaatatcaaa 240
atgacgtttg ggggtggcac tgcgcgcgcc aacaacaaca agagcagcca ccaagccaaa 300
gcaccaagca ccaacccgcg acaccatgtt gttacaccgc cggcagaaat ggcgtcagcc 360
agcagtcact attactcaat attcaatagg gggtccactt gactcacttc ctcatgtata 420
tatatatcca tctaattctt ccaatacata acc 453

```

```

SEQ ID NO: 156      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 156
gcggcctatg cagacgcttg gagcttgggc acattggata ttgatattg gctgggtgcc 60
gttcgagatc ccgttgctgt cgccaccgcc accggtcacg tggtcacttg gtcacgtgga 120
ccactgcac tgctcttagc tacaagagtg catgatgcat gatgtcatgc atgatatgca 180
tgatacaagc atgaatattg atttcaatat tgaacattta gtattgatat ctcatgatgc 240
gtaaacacgtg acacgcacca tatcaatcag ccgcaccggc actaaatggt tcaatttcaa 300
catcccacac gctcgaatgc tccctcgagc taatcggatt tcggagctcc gacatggccg 360
gcctcgagcg gagtttggaa ctccgttccg ccgaccccg cttgctgcgc cgatgaaatc 420
ccgagtcaag tcatagcaag ccaagtggct ggctgactca aaagctaata aaagtcgtag 480
caagtaatac tccgcacgct 500

```

```

SEQ ID NO: 157      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 157
tcgctgtctc aaacgctggc gcgcgacacg tgccgcgacc aaaatttgtt ggtgtatggg 60
aataggggtt tctaaaattc tgttgaggat gtgagtatca ctggcatggc accagtgtac 120
tgctcatcga taatcataaa cgtatcaacc ctccggctct gcgatctgta tctgctactg 180
gagggcacag aagccttctt gcttggttgc gctctgcgct ctctgtggtg ctttgccgct 240
cgcggccaag gtccgcagcg ggcattcacc accatacggg ttaagccacc agacgagtgc 300
tgctcgccgg gccagctcgg cgcctgtcac gccccacat tggcttgatg ctccgctcct 360
tctttcatct tggaaaccggc gggcaatggc gcttatcacc actcctgggc cgggccccgc 420
gtatactctc ataactagtg cagcgtttca aataaatacc tcgctcctcg gctggccgac 480
cgacgtttac ccatccgccc 500

```

```

SEQ ID NO: 158      moltype = DNA length = 499
FEATURE            Location/Qualifiers
source              1..499
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 158
tggcatgtag gcctgtaggg gctatgccta tatgctcatg cactcatgtc gcgatgttgc 60
gatatcgaga tggaccaacc aagcaagcgt gcagcgtggg cgcttgggac aaggggtgta 120
agctaagtgt aagcactgaa gtctgaagtg gctgaagtgc tgaacgtgga acgtcaggat 180
gacacatcct ctcccttcgg ctctgtccaa gaaagggttg tcagccatct gagatctgag 240
atctgaaatg aatataatga tgaattgagt cgatgagatc tgactatctt gggagattgc 300
accaatgcgc cacggtgacc agatggtctt ggtcttagct ggtatagtgg cacctgatct 360
gccttgacag atctcatggt catacgttca tgcattcaat cttatcgtct agcgcgtctg 420
gatctgcccc actagacgac gacaaatgta attgatcctt gtaaccctgt tgctggtcgc 480
agcagcgaga tacgatgcc 499

```

```

SEQ ID NO: 159      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 159
aaatcttcat ccagcgtagc aagcatatcg cgcagtgcgc agttcctcga gtgtgcatac 60
cgaacccaaa aagccatgga tgcattgac ctgctgtatt catcaatcga gatattctta 120

```

-continued

```

ctccggagcg tacagtggca gatattatgt aggagatgat gggacaggat gcagagatac 180
ataatccatt tccagttaat agggccgcta gatacttgat attagcctgg cagcgagcga 240
gcgagggtag agtagaatag agtccagtat agatcccacc ccatgggtgcg ggaagtggat 300
gacgtaatag cgcgatatgt ccccgctccc ttgtcaccta ggccacggga ccctacgtga 360
ttgcatcggt cgtgtcgtcg cctgtcgacg cctgggtgcag gcatacgtcg cgtgcttcga 420
cgatcaacaa caccaccaac cacgactcca gttccggcca gccagtacca tggcgctcgtt 480
ctttgggttt agtgcaccgg

```

```

SEQ ID NO: 160      moltype = DNA length = 400
FEATURE            Location/Qualifiers
source              1..400
                    mol_type = other DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 160
aaccagatcg aagctcacc cttgttgcaa caagacgagt tggttgctca ccataagagc 60
aagaacattg tcatcactgc ttacagtcct ttgggcaaca atgtcgctgg taaaccacct 120
ctgactgaga acccggtgat ttgtggatgct gctaagcgtt tgaaccatac tctgtctgct 180
gtgctcattg cttgggggat tcaacgcggg tacagcgtct tggccaagtc agttacaccc 240
tcacggatca agagtaacct tgaacagatc actctgtctg atgaggaatt ccaacgggtt 300
accaacctca tcaaggagta cggtagagcg cgtacaacac ttcctttcaa ttacaagcct 360
tcgtgggtcca ttgacgtctt tggtagccag gacgaggcta

```

```

SEQ ID NO: 161      moltype = DNA length = 500
FEATURE            Location/Qualifiers
source              1..500
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 161
atcctgtgct cgtctcttac attcatcttc ctgcccataa aactttcaag ctatttcttc 60
caacgtactc aacaacctat ttttttgggc cccgctcaaa acgcacctta ttgcgtctaa 120
gtgtattcta accacgcgac ctgaattgca aagcctgtta ccccatcgc accttcgtct 180
taagatagtt gtttaacgag ccttggccag ttttagcctt gaccatgact acagatcata 240
atgggttagat gctcttgat cgataatgat atattgcagt tcgacgattg aagatttatt 300
ggatgtcgat gacgactgtc gagctcttat tcttagcaac aacttaata ttactatcgc 360
aatgcacgag acatcatgca tcgtgtctcat gaatatcact ggatcactct cgatcgagc 420
ccactaggtc agcataacac gcttgaagct gacttgaagc tttggaacaa atggcgagca 480
cagtggcggt cacggtgacc

```

```

SEQ ID NO: 162      moltype = DNA length = 486
FEATURE            Location/Qualifiers
source              1..486
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 162
attgttgcat cgtctcatca gatcgctcct cattgtagtg ataattgggt caaccttgta 60
ttagactgta tagcagaatc cctcgctata aaaagtatta tgcatacgca gttccaagca 120
ggaattagta tggtagtggg acttggctga agggctgacg acatggcgag aaaaacgtat 180
caatcggaa cacactgtgc aatgtcgatc acgttttcgg tggttcacgg taggttctct 240
agtcgatggg aaagtggtaa catctatccc ttgagcgctt gagcttaatg ttaccttggt 300
caaaagtcac ttgaagcttc acttgaagct tgaagctttg aagttttgaa gcttgaagcg 360
caaacagcag cggatcatgt atactggcgt aacctgccta cagtgggtgca tcagtacgtc 420
aagcgggtag cccgaaatct cattaaggct atttgccctc taagcttagt acctagtgat 480
ctgttc

```

```

SEQ ID NO: 163      moltype = DNA length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 163
cctgaacgtc ctgaactctg tcttattaca taccattaat ttccctgcct tccctctcta 60
cgtcatcgga attttatctc ttccctgagc gcaaagattt ggtccttgct cgggatacaa 120
cggagagtca tgattgggtt tatggaagag aggcgatagc acttgcatth tgaccattcg 180
gggattaagc gaacccccaa gtggcacaaa gagctcgtat gagggtcagt aagtatagag 240
ggacatgcaa agggagagaaa aggaacacac atattgacaa atagagacaa gtacagggtg 300
tcgcaatagc atgatgtagg ggtatgcgca agtcacaatc gttcagggcg ggcataacat 360
cagcgtaccc gagcacaatg gttaatgatt aatgagcaga ggaaaaaata atgattatgc 420
cgggatggcg aggaactaact ggcgcagctc ttcagcttcg gcaacgcgtg cacatagtgg 480
tgtagcacca cgagattgag c

```

```

SEQ ID NO: 164      moltype = DNA length = 493
FEATURE            Location/Qualifiers
source              1..493
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 164

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-continued

```

cctttgctc acgcctctcg tttgcacgct tcttcacggc ttatcaccgc ttattatate 60
catggaatcc cctccaaagt tttcttcaac atagcttcat atcttctacc ataacatcta 120
tcccaccaag tagcacatga attcaacgta gcttgagaac ttttctcgt cagctgattg 180
cgttctttag gcagtttgct tctaccccaa ttgcttctac cccaattgct tctgcttatt 240
tcaccagtcg gctcggagat cctcgacaac tctgagatct gatgatcaac ctcttggtga 300
acttgtaaaa tttgatcgat gagacgatta gggacttgaa gctttaagtt agacgcgatg 360
ggcggtaatc tatatgctta gactttgttg ctatgtttgt cctataagat ctgatgcattg 420
atgtatcatt acccagacgg gctatcgcac aagggtgcaga tgcattctat catgacagtg 480
ccacactggt ggc

```

```

SEQ ID NO: 165      moltype = DNA length = 567
FEATURE            Location/Qualifiers
source              1..567
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 165
gcttgctggt cacattattg tcagtaccta ccatgaagaa ttcaatatta tttacattg 60
tcaaccatta catggatccc atgttggtgca tagcctctat gtcacctttt ggtcttttat 120
gtctgtagcg tgctagtagt attcaagttg tagcactgct gtgaaacagt acgttcaagt 180
agtgtgcagc agtacggttca gtttagtggac tggggcagcc tcgccttcag caaagggttag 240
gtacatgagc cagttgctca ctcttgcttt ctacgaatcc cataggctac tttgccccct 300
aacgtgacgc acccctctgt tttagaagtt ttgagtcctt ttacgttacc gtgaccaagc 360
gagcagagta gaaactcttg aagttgtgta tgtaagaata caaattgaga acacaaggtt 420
acctaccttt atagttggta aacagcaagc aataccagag attttcatcg caacgtgact 480
ctatcattca ctatattaag ttgcacacgg atgtgtgaca aactacacg taaaagagag 540
aaaaatacta atgtcgatcg ctacagt

```

```

SEQ ID NO: 166      moltype = DNA length = 504
FEATURE            Location/Qualifiers
source              1..504
                    mol_type = genomic DNA
                    organism = Moniliella pollinis

```

```

SEQUENCE: 166
ttgctgtac gacctcattg atcgatgatc tccgacactt tgagtagctt tccatcctct 60
ctttactgta ctttgtattc ttgctacttt gtgtctcttc ttaatggtcc cgaaccggtg 120
attttgctca tgactctttt tttttttttt cgtcaactgc aagaagatga aaaactctact 180
gttcgatagc acctcataac cgttcagag tcccgacaac cacaaggatg tttcttagat 240
aggtgtcagc aagatccaca aaaaagtgtg ggtattagat ggtttcaagc aaggtacaat 300
attctaattc aatactcatt tgaataatca atattcaaca gcttctagcc tacaagcttc 360
aatgttctcc gtcggttaac tcagtatact tagtactcag ttcaagtcag tgttggttaa 420
ataaaatcac gtgatccaaa aacttgaaac ttttctcttt tctcctcaga tgttggtgat 480
tcgtcccata gcggttacca caac

```

```

SEQ ID NO: 167      moltype = DNA length = 1679
FEATURE            Location/Qualifiers
source              1..1679
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 167
aagacctctg tcccacaaac cttgatggtc tgtgaaacgg tcaacaacat cattggtaga 60
actgtcaatc caagaaacaa aaattgggtc tgtggtggtt cttctggtgg tgaaggtgct 120
attgttggtg tttagaggtg gtctgtactg acattggtgg ttccattaga 180
gtcccagctg ctttcaactt tttatacggg ttgagaccat ctcacggtag attgccatat 240
gctaaaaatg ctaactctat ggaaggtcaa gaaacggtc actcgcgtg ttgtcctatc 300
actcactcgg tcgaagacct gagattgttc accaaatctg tcttgggtca agaactcttg 360
aagtagcact ctaaggctcat ccccatgcca tggagacaat ctgaatctga catcattgcc 420
tctaagatta agaattggtg tttgaacatt ggttattaca atttcgacgg taacgtcttg 480
ccacaccac caattttacg tgggttcgaa actacggtg cgcctttggc caaggctggg 540
cacaccgtta ctccatggac tccatacaag catgatttct gtcatgactt gatttccac 600
atctatgctg ctgatgggtc tgcgcagctc atgagagaca tttctgcctc tggtagagcca 660
gccatcccta acatttagga ctgttgtaac ccaaatatta agcgtgttaa catgaacgaa 720
ttgtgggaca ctcatttaca aaagtggaa tatcaaatgg aatacttgga aaagtggcgt 780
gaagctgaag aaaaagctgg taagggaattg gacgctatta tcgctccaat tactcctacc 840
gtcgtgtgca gacacgatca attcagatac tacggttacg cctccgttat tagcttattg 900
gatttcacct ctgttgtcgt cccagtcact ttcgctgata agaatttgta taagaagaac 960
gaatctttta aagctgtttc cgaattggat gctttgggtc aagaagaata cgaccagag 1020
gcttatcacg gtgctcctgt tgtgtttcaa gttattggtg gaagattgtc cgaagagaga 1080
actttggcta tcgcgaaga agtcggtaaa ttgttggtga acgtgcgcac tccataatgg 1140
tgttcaccac tgttgcttcc cctttgttgg agcgtccgtc gttctgatgg cgtctgagc 1200
ccaaaatcct cctctctcgc ccgagagcaa gcctcggagt gtgcttgcaa tctacttttc 1260
ctactaaagt tatcttcttg ttgcttctga atttgaatag aacgtgttca gacttggttag 1320
ctaaatttgt tgatgcgaga tacttgacag aatcccggtg atgaactcat ctgtgtgacc 1380
accgcctggt gatggaatct cgcgcggaag ccgtggtgtt actgactctc aagcaattca 1440
gtagataaat ttgtataaca tcattgcaag ggggtgctagc aatcaattgt ccaataccaa 1500
acggcggaag gtgcgaaggg gcgatagaga gtttctgcga ggaagggaag gaatccgtat 1560
gcctgaaaac ttttgtacaa gtgtctttgt aagggcgagg tctgctctga catctctctg 1620

```

-continued

agatctgaga tctgatacct gcaggataac ttcgtataat gtatgtata cgaagttat 1679

SEQ ID NO: 168 moltype = DNA length = 876
 FEATURE Location/Qualifiers
 source 1..876
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 168

```

agatgtagcc ggcgctgtag agttttggac ggatagggtg ggtttcagtc gagacttcgt 60
tgaagatgac tttgcaggcg tctgtcgtga tgatgtgacc ctgtttatct cggccgttca 120
agatcaagtt gtccctgaca atacacttgc atgggtctgg gtacgcggat tagatgaatt 180
gtacgcggaa tggctgtaag ttgtatccac taactttcgt gacgcttcag gtctgtctat 240
gactgagatt ggtgaacaac cctgggggcg tgagttcgca ctccgagatc cagctggaaa 300
ttgtgttcat ttcgtatctg aggaacagga ttaatgggtg tcaccactgt tgettccctc 360
ttgttgagag gtccgtcggt ctgatggcgc tctgagccca aaatatccct tcttcgcccg 420
agagcaagcc tccgagtggt ctgtcaatct acttttccta ctaaagttaa cttctgtgtg 480
cttctgaatt tgaatagaac gtgttcagac ttgttagcta aatttgttga tgcgagatac 540
ttgacagaat cccgttgatg aactcatctg tgtgaccacc gcctggtgat ggaatctcgc 600
gcggaagccg ttgtgttact gactctcaag caattcagta gataaatttg tataacatca 660
ttgcaagggg tcttagcaat caattgtcca ataccaaaacg gcggcaagtg caaaggggcg 720
atagagagtt tctgcgagga aggggaaggaa tccgtatgcc tgaaaacttt tgtacaagtg 780
tctttgtaag ggcgaggtct gctctgacat ctctctgaga tctgagatct gatacctgca 840
ggataaactc gtataatgta tgcctatacga agttat 876

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SEQ ID NO: 169 moltype = DNA length = 1087
 FEATURE Location/Qualifiers
 source 1..1087
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 169

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gaagactgag tccgtctcag ataacttcgt ataatgtatg ctatacgaag ttatcctgca 60
ggagctcaga ccttgcatgt tagccttgac cttgcctcta tgaccactaa ttcgcacaag 120
attgcacgtt accattgtca tcttgccatg tatccaatgt ttctgttttg ccatagtacg 180
tagtagatag atcgatcatc gatgcagatg acgcagatgg cattgggttcg ctttgcccta 240
tgcattgcac atgcatcatg catcatgcgt catgcgtcac tatccatatt aatcattgcc 300
ctctgtacta ggcctagaca cctgcagacc cttcccccagg gatgcttctt gtaatagcca 360
atatactaca caactttacc gattacgacg gagatcgatc tcaactatag ttgtcgaaact 420
tcgatgcaca tcgaggggtgc ggaaaaaaac tgacggaagc ggaagtgggt gaattgggta 480
atttagttag attggcaggg ttgttcccaa gattcaacgc aagctgagaa agaaaacaga 540
aacacgagggc ataaggccag tcagaaaaatg cattacgagg ggagttgaag tgactgtggg 600
ccagtgcctg ggtcgtcatg tgcataattt ttcttcgggt agactgcgcg agcatgcagc 660
aactctgcgc gccctatata actagtgcct catcttcttg tcattcttgt ctctttacca 720
acacaccggt actctttcac catggcaaaag ttgacgagcg ccgtgccggg gctaaccgcg 780
agagatgtag ccggcgctgt agagttttgg acggataggt tgggtttcag tcgagacttc 840
gttgaagatg actttgcagg cgtcgtgcgt gatgatgtga ccctgtttat ctggccggtt 900
caagatcaag ttgtccctga caatacactt gcatgggtct gggtacgcgg attagatgaa 960
ttgtacgcgg aatgggtctga agttgtatcc actaaacttc gtgacgcttc aggtcctgct 1020
atgactgaga ttggtgaaca accctggggg cgtgagttcg cactccgaga tccagctgga 1080
aattgtg 1087

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SEQ ID NO: 170 moltype = DNA length = 1257
 FEATURE Location/Qualifiers
 source 1..1257
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 170

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gaagactgag tccgtctcag ataacttcgt ataatgtatg ctatacgaag ttatcctgca 60
ggagctcaga ccttgcatgt tagccttgac cttgcctcta tgaccactaa ttcgcacaag 120
attgcacgtt accattgtca tcttgccatg tatccaatgt ttctgttttg ccatagtacg 180
tagtagatag atcgatcatc gatgcagatg acgcagatgg cattgggttcg ctttgcccta 240
tgcattgcac atgcatcatg catcatgcgt catgcgtcac tatccatatt aatcattgcc 300
ctctgtacta ggcctagaca cctgcagacc cttcccccagg gatgcttctt gtaatagcca 360
atatactaca caactttacc gattacgacg gagatcgatc tcaactatag ttgtcgaaact 420
tcgatgcaca tcgaggggtgc ggaaaaaaac tgacggaagc ggaagtgggt gaattgggta 480
atttagttag attggcaggg ttgttcccaa gattcaacgc aagctgagaa agaaaacaga 540
aacacgagggc ataaggccag tcagaaaaatg cattacgagg ggagttgaag tgactgtggg 600
ccagtgcctg ggtcgtcatg tgcataattt ttcttcgggt agactgcgcg agcatgcagc 660
aactctgcgc gccctatata actagtgcct catcttcttg tcattcttgt ctctttacca 720
acacaccggt actctttcac catgggaacg acttttagatg ataccgccta ccgttatagg 780
acatcagtag cgggcgatgc agaagcaatt gaagcccttg atggatcatt caccacagac 840
acagtttttc cgtgtgacgc tactggtgac ggatttactc tccgtgaagt gctgttagat 900
ccgccattga cgaagattat tccagatgat gagagtgtat acgagagcga tgatggcgaa 960
gatggtgatc ctgactcgcg cactttcgtc gcatacgggt atgatgttga cctggccgga 1020
tttgtgttgg tgcctactc ggggtggaac cgtagattga ccgtcgaaga tatcgaagtc 1080
gcgcagaaac atcgtgggtca tggcgtggga cgagcactga tgggtctagc tacggaattc 1140
gctcgagagc gtggggcggg tcacttatgg cttgaagtta ccaatgtaaa tgctcctgct 1200

```

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attcacgctt atcgccgaat gggtttcacc ttgtgtggcc tcgacacggc attgtat 1257

SEQ ID NO: 171 moltype = DNA length = 1049
 FEATURE Location/Qualifiers
 source 1..1049
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 171
 attgaagccc ttgatggatc attcaccaca gacacagttt ttccgctgac agctactggt 60
 gacggattta ctctccgtga agtgccctga gatccgccat tgacgaaagt atttccagat 120
 gatgagagtg atgacgagag cgaatgatggc gaagatgggtg atccctgactc gcgcactttc 180
 gtccgatacgt gtgatgatgg tgacctggcc ggatttgttg ttgtgtccta ctccgggtgg 240
 aaccgtagat tgaccgtcga agatatacga gtccgcggcag aacatcgtag tcatggcgtg 300
 ggacgagcac ttatgggtct agctacggaa ttccgtcgag agcgtggggc gggtcactta 360
 tggcttgaag ttaccaatgt aaatgctcct gctatttcacg cttatcgccg aatgggtttc 420
 accctgtgtg gctcgcacac ggcattgtat gacggtagct cttctgacgg ggagcaagcg 480
 ctatacatgt ctatggcctg cccctaattg tgttcaccac tgttgcctcc cctttgttgg 540
 agcgtccgtc gttctgatgg cgtctcgagc ccaaaaatc ccttcttcgc ccgagagcaa 600
 gcctcggagt gtgcttgcaa tctacttttc ctactaaagt tatcttcttg tggcttctga 660
 atttgaatag aacgtgttca gacttgtagt ctaaaattgt tgatgcgaga tacttgacag 720
 aatcccggtg atgaactcat ctgtgtgacc accgcctggt gatggaatct ccgcggaag 780
 ccgtgggtgt actgactctc aagcaattca gtatataaat ttgtataaca tcattgcaag 840
 ggggtgctagc aatcaattgt ccaataccaa acggcggcaa gtgcaagggt gcgatagaga 900
 gtttctgcga ggaagggaa gaatccgtat gcctgaaaac ttttgtacaa gtgtctttgt 960
 aaggcgaggt tctgctctga catctctctg agatctgaga tctgatacct gcaggataac 1020
 ttcgtataat gtatgctata cgaagttat 1049

SEQ ID NO: 172 moltype = DNA length = 1199
 FEATURE Location/Qualifiers
 source 1..1199
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 172
 cttaaacacg gcaagggtag cgttgcaaac gacgtaaccg acgaaatggt ccgtctgaat 60
 tggctaactg agttcatgcc actccccacc atcaagcact tcacccgtac acctgacgat 120
 gcgtggctgt tgactacggc catccccggt aagacgcgtt ttcaggctct cgaggataac 180
 cctgactcgg gcgagaacat cgtggatgcc ttggctgtgt tttgcgcgc ccttcaactc 240
 attccggtct gcaactgccc ttccaacagc gaccgtgttt ttaggcttgc ccaggcacia 300
 tcacgcataa acaatgtgtc agtcgacgct agtgatttcg atgacgagcg caacggctgg 360
 ccggtggaac aagtgtggaa agagatgcat aagctcctgc ccttttcccc agactctgtc 420
 gtgacgcatg gtgatttctc gcttgacaac ctcatctttg acgagggcaa gctcattggc 480
 tgcattgacg tcggccgggt tggcatcgcc gatcgctacc aagatcttgc gatcctctgg 540
 aattgtctgg gcgagttctc ccttagcctc cagaagcgtc tgttcagaaa atatggggtc 600
 gataaccacg acatgaacaa gctccagttc catttgatgc tggatgaatt tttctaattg 660
 tgttcaccac ttgtgcttcc cctttgttgg agcgtccgtc gttctgatgg cgctctgagc 720
 ccaaaaatc ccttcttcgc ccgagagcaa gcctcggagt gtgcttgcaa tctacttttc 780
 ctactaaagt tatcttcttg tggcttctga atttgaatag aacgtgttca gacttgtagt 840
 ctaaaattgt tgatgcgaga tacttgacag aatcccggtg atgaactcat ctgtgtgacc 900
 accgcctggt gatggaatct ccgcgggaag ccgtgggtgt actgactctc aagcaattca 960
 gtatataaat ttgtataaca tcattgcaag ggggtgctagc aatcaattgt ccaataccaa 1020
 acggcggcaa gtgcaagggt gcgatagaga gtttctgcga ggaagggaa gaatccgtat 1080
 gcctgaaaac ttttgtacaa gtgtctttgt aaggcgaggt tctgctctga catctctctg 1140
 agatctgaga tctgatacct gcaggataac ttcgtataat gtatgctata cgaagttat 1199

SEQ ID NO: 173 moltype = DNA length = 1621
 FEATURE Location/Qualifiers
 source 1..1621
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 173
 tattctggtt cgagccatcc caaaaatgga tcatgacagc tgccaagagc caggactata 60
 agattgagat ctacagcagc gatgacctca agtcctggaa gctcgaaagc gcctttgcga 120
 acgaaggctt tcttggctat cagtatgagt gccccggcct catcgagggtg cctaccgagc 180
 aagatccctg caagagctac tgggtgatgt tcatctctat caatccggga gcaccgcgg 240
 gcggtccctt caaccagtat ttgtgggtt cgttcaacgg aacgcacttt gaagcttttg 300
 ataatacgtc aagagtcgtc gactttggca aagactacta cgccctccag accttcttta 360
 acacggatcc aacgtacggg tcggcactgg gcatcgctg ggcgtcgaa tgggagtata 420
 gcgcttttgt ccccaactaa ccttggcgct cgtcgatgtc gttagtgcgt aagttctcgc 480
 tgaacaccga gtaccaggca aatccggaga cagaactaat caaccttaag gcagagccta 540
 tcttgaacat ctctaaccgc ggaccctggt ctaggttcgc caccacacc acgctcacta 600
 aggcacaatg ttacaatgca gatctaagca attcaacggg cacactggaa ttcgagctcg 660
 tttacgcagt caacaccacc gacagcatct cgaaatccgt ctttgcagat ttgtcgtgt 720
 ggttcaagggt ccttgaggac ccggaagagt acctccgcat gggttttgaa gtctcagcgt 780
 catcgttctt cctcgaccgc ggggaattcta aggtaaagtt cgtgaaggaa aacccctact 840
 ttaccaatcg tatgtctgtg aacaatcagc cattcaaatc ggagaacgac ctgtcctact 900
 acaaggttta cggccttctc gaccaaaca tcctcgagtt gtactttaac gacggtgatg 960

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ttgtgtcgac	aaacacttat	ttcatgacaa	cgggaaacgc	tctcggatcc	gtcaatatga	1020
ctacgggtgt	cgacaatctg	ttttacattg	acaaattcca	gggtcgagaa	gtcaagtat	1080
gggtgttcacc	actgttgctt	cccccttgtt	ggagcgtccg	tcgttctgat	ggcgtctga	1140
gccccaaata	tcccttcttc	gcccagagagc	aagcctcgga	gtgtgcttgc	aatctacttt	1200
tcctactaaa	gttatcttct	tgtggcttct	gaatttgaat	agaacgtgtt	cagacttgtt	1260
agctaaattt	gttgatgcga	gatacttgac	agaatccctg	tgatgaactc	atctgtgtga	1320
ccaccgcctg	gtgatggaat	ctcgcgcgga	agccgtgggt	ttactgactc	tcaagcaatt	1380
cagtagataa	atttgatata	catcattgca	aggggtgcta	gcaatcaatt	gtccaatacc	1440
aaacggcgcc	aagtgcgaag	gggcgataga	gagtttctgc	gaggaaggga	aggaatccgt	1500
atgctgaaa	acttttgtac	aagtgtcttt	gtaaggcgca	gggtctgctc	gacatctctc	1560
tgagatctga	gatctgatac	ctgcaggata	acttcgtata	atgtatgcta	tacgaagtta	1620
t						1621

SEQ ID NO: 174 moltype = DNA length = 1903
 FEATURE Location/Qualifiers
 source 1..1903
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 174

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tagccttgac	cttgccctcta	tgaccactaa	ttcgcacaag	attgcacggt	accattgtca	120
tcttgccatg	tatccaatgt	ttcctgtttg	ccatagtagc	tagtagatag	atcgatcatc	180
gatgcagatg	acgcagatgg	catttggttcg	ctttgcccta	tgcatgcata	atgcatacatg	240
catcatgcgt	catgcgtcac	tatccatatt	aatcattgcc	ctctgtacta	ggccatgaca	300
cctgcagacc	cttccccagg	gatgcttctt	gtaatagcca	atatactaca	caactttacc	360
gattacgata	gagatcgata	tcactatagt	ttgtcgaaat	tcgatgcaca	tcgaggggtc	420
ggaaaaaagc	tgacggaagc	ggaagtgggt	gaattgggta	atttagttag	attggcaggg	480
ttgttcccaa	gattcaacgc	aagctgagaa	agaaaacaga	aacacgaggc	ataaggccag	540
tcagaaaaatg	cattacagag	ggagtgaag	tgactgtggg	ccagtgcctg	ggtcgtcatg	600
tgcaaatctt	ttcttcgggt	agactgccgc	agcatgcagc	aactctgccg	ggcctatata	660
actagtgcct	catcttcttg	tcattcttgt	ctctttacca	acacaccggg	actctttcac	720
catgccacaa	tcctgggaag	aattggccgc	cgacaaacgt	gcccgttttg	ctaaaacct	780
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gaagtccggg	attttgtctg	aggctgaatt	gaagattacc	gaagcctctg	ctgctgattt	900
ggtttccaa	ttggccgctg	gtgagttgac	ttctgttgaa	gtcactttgg	ctttttgtaa	960
gagagctgct	attgctcaac	aattaaacaa	ctgtgctcac	gaattctctc	cagatgctgc	1020
tttagctcaa	gttagagaat	tagatgaata	ctacgctaag	cataagcgtc	cagttgggtc	1080
attacacggt	ttacaacatc	ctttaaagga	ccaattgcgt	gttaagggtt	acgaaaacct	1140
catgggttac	atttccgtgt	taaacaaaata	cgatgaaggt	gattccgtct	taaccacccat	1200
gttgagaaaa	gctgggtgct	ttttctacgt	taagacctct	gtcccaaaaa	ccttgatggg	1260
ctgtgaaacc	gtcaacaaca	tcattggtag	aactgtcaat	ccaagaaaca	aaaattgggt	1320
ctgtgggtgt	tcctctgggt	gtgaagggtc	tattgttggt	attagagggt	gtgttattgg	1380
tgctcggtact	gacattgggt	gttccattag	agtcaccagc	gctttcaact	ttttatacgg	1440
tttaagacca	ttccacggta	gattggccata	tgctaaaatg	gctaactcta	tggaagggtc	1500
agaaacccgt	cactccgtcg	ttgggtcctat	cactcactcc	gtcggaagact	tgagatttgt	1560
caccaaatct	gtcttgggtg	aagaaccttg	gaagtacgac	tctaaaggta	tccccatgcc	1620
atggagacaa	cttgaatctg	acatcattgc	ctctaagatt	aagaatgggt	gtttgaacat	1680
tggttattac	aatttcgacg	gtaacgtctt	gccacaccca	ccaattttac	gtgggtgcga	1740
aaactaccgt	ggcgtcttgg	ccaaggctgg	tcacaccgtt	actccatgga	ctccatacaa	1800
gcatgatttc	tgtcatgact	tgatttccca	catctatgct	gctgatgggt	ctgccagcgt	1860
catgagagac	atttctgcct	ctgggtgagcc	agccatccct	aac		1903

SEQ ID NO: 175 moltype = DNA length = 1475
 FEATURE Location/Qualifiers
 source 1..1475
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 175

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tagccttgac	cttgccctcta	tgaccactaa	ttcgcacaag	attgcacggt	accattgtca	120
tcttgccatg	tatccaatgt	ttcctgtttg	ccatagtagc	tagtagatag	atcgatcatc	180
gatgcagatg	acgcagatgg	catttggttcg	ctttgcccta	tgcatccata	atgcatacatg	240
catcatgcgt	catgcgtcac	tatccatatt	aatcattgcc	ctctgtacta	ggccatgaca	300
cctgcagacc	cttccccagg	gatgcttctt	gtaatagcca	atatactaca	caactttacc	360
gattacgata	gagatcgata	tcactatagt	ttgtcgaaat	tcgatgcaca	tcgaggggtc	420
ggaaaaaagc	tgacggaagc	ggaagtgggt	gaattgggta	atttagttag	attggcaggg	480
ttgttcccaa	gattcaacgc	aagctgagaa	agaaaacaga	aacacgaggc	ataaggccag	540
tcagaaaaatg	cattacagag	ggagtgaag	tgactgtggg	ccagtgcctg	ggtcgtcatg	600
tgcaaatctt	ttcttcgggt	agactgccgc	agcatgcagc	aactctgccg	ggcctatata	660
actagtgcct	catcttcttg	tcattcttgt	ctctttacca	acacaccggg	actctttcac	720
catgggtaag	gaaaagacac	acgtatcgcg	accccgactc	aactccaaca	tggaagcaga	780
cctgtacgga	tacaaatggg	ctcgagataa	tgctgggacg	tcgggtgcga	cgatctacag	840
actctatgga	aaacgcgacg	cccccgagct	ctttcttaaa	cacggcaagg	gtagcgttgc	900
aaacgacgta	accgacgaaa	tggtccgtct	gaattgggta	actgagttca	tgccactccc	960
caccatcaag	cacttcacgc	gtacacctga	cgatgcgtgg	ctgttgacta	cggccatccc	1020
cggaagacc	gcttttcagg	tcctcgagga	ataccctgac	tcgggcgaga	acatcgtgga	1080

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tgccttggct gtgtttttgc gccgccttca ctctattccg gtctgcaact gccctttcaa 1140
cagcgaccgt gtttttaggc ttgcccagcg acaatcacgc atgaacaatg gtctagtcca 1200
cgctagtcat ttcatgacgc agcgcaacgc ctggcccggtg gaacaagtgt ggaaagagat 1260
gcataagctc ttgccccttt cccagactc tgtcgtgacg catgggtgatt tctcgttga 1320
caacctcatc tttgacgagg gcaagctcat tggtgcatt gacgtcggcc gggttggcat 1380
cgccgatcgc taccagatc ttgcgatcct ctggaattgt ctgggcgagt tctcccctag 1440
cctccagaag cgtctgttcc agaatatgg gatcg 1475

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SEQ ID NO: 176      moltype = DNA length = 1982
FEATURE
source              Location/Qualifiers
                    1..1982
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 176
ataacttcgt ataattgatg ctatacgaag ttatcctgca ggagctcaga ccttgcatgt 60
tagccttgac cttgctctta ttgcccagcg acaatcacgc atgaacaatg gtctagtcca 120
tcttgccatg tatccaatgt ttctgtttg ccatagtagc tagtagatc atcgatcctc 180
gatgcagatg acgcagatgg catttggttcg ctttgcccta tgcgatcctc atcgatcctc 240
catcatgctg catcggtcac tatccatatt aatcattgcc ctctgtacta ggccatgaca 300
cctgcagacc cttcccagcg gatgcttctt gtaatatgcca atatactaca caactttacc 360
gattacgacg gagatcgatc tcactatagt ttgtcgaact tcgatgcaca tcgaggggtgc 420
ggaaaaaagc tgacggaagc ggaagtgggt gaattgggta attagttag attggcaggc 480
ttgttcccaa gattcaacgc aagctgagaa agaaaacaga aacacgagcg ataaggccag 540
tcagaaaaatg cattacgagg ggagttgaag tgactgtggg ccagtgcctg ggtcgtcatg 600
tgcaaatatt ttcttcggtt agactgccgc agcatgcgc aactctgccg gccctatata 660
actagtgtct catcttcttg tcattcttgt ctctttacca acacaccggt actctttcac 720
catgttgctt caagctttcc ttttctgtct cgttggttcc gccgctaaga tttagcctag 780
catgacgaac gaaacctctg accggccggt ggtccacttc accccaaca aggggtggat 840
gaacgaccca aacggcctct ggtatgatga gaaagacgct aagtggcacc tgtacttcca 900
gtataacctt aacgatacag ttgggggtac gcccttggtt tggggccatg cgacctccga 960
cgactccacc aactgggaag atcagccctat cgcgattgct cgaagcgta acgattcagg 1020
tgcccttagt ggcagtagtg tcgtagacta caacaacacc agcgggttct tcaacgacac 1080
tatcgatcca cgacagcgct gtgttgccat atggacctac aacacgcggg agtccgagga 1140
acaatatcatt agttactcgc tggacggggg atacaccttt acagagtacc agaaaaatcc 1200
cgctctcgcc gcaaaactga ccaattccg cgatcctaag gtattctggt acgagccatc 1260
ccaaaaatgg atcatgacag ctgccaaagc ccaggactat aagattgaga tctacagcag 1320
cgatgacctc aagtctgga tcactgaggt gcctaccgag caagatccta gcaagagcta 1440
tcagtatgag tgcccccgcc ccatcgaggt gcctaccgag caagatccta gcaagagcta 1440
ctgggtgatg ttcatctcta tcaatccggg agcaccccgcg ggcggctcct tcaaccagta 1500
ttttgtgggt tcgttcaacg gaacgcactt tgaagctttt gataatcagt caagagtcgt 1560
cgactttggc aaagactact acgcccctca gaccttcttt aacacgggat caacgtacgg 1620
tcgggcactg ggcacgcgct gggcgctcga ctgggagtat agcgcccttg tccccactaa 1680
cccttgccgc tcgtcagatg cgttagtgcg taagttctcg ctgaacaccg agtaccaggc 1740
aaatccggag acagaactaa tcaaccttaa ggcagagcct atcttgaaca tctctaacgc 1800
cggaccctgg tctaggttcg ccaccaacac cagcctcact aaggccaata gttacaatgt 1860
cgatctaagc aattcaacgg gcacactgga attcgagctc gtttacgcag tcaaccacac 1920
ccagacgacg tcgaaatccg tctttgcaga tttgtcgtcg tggttcaagg gccttgagga 1980
cc 1982

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SEQ ID NO: 177      moltype = DNA length = 500
FEATURE
source              Location/Qualifiers
                    1..500
                    mol_type = other DNA
                    organism = Moniliella pollinis

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SEQUENCE: 177
tatcagatct cagatctcag agagatgtca gagcagacct cgcccttaca aagacacttg 60
tacaaaagt ttcaggcata cggattcctt ccttccctcg cagaaactct ctatcgcccc 120
tttgactctg ccgcccgttg gtattggaca attgattgct agcacccttt gcaatgatgt 180
tatacaaat tatctactga attgcttgag agtcagtaac accacggcct ccgcccgcga 240
ttccatcacc aggcgggtgg cacacagatg agttcatcaa cgggattctg tcaagtatct 300
cgcatcaaca aatttagcta acaagtctga acacgttcta tcaaatcca gaagccacaa 360
gaagataact ttagtaggaa aagtagattg caagcacact ccgaggcttg ctctcgggcg 420
aagaagggat attttgggtc cagagcgcca tcgaacgac ggacgctcca acaaaaggga 480
agcaacagtg gtgaacacca 500

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1. A genetically engineered yeast cell capable of producing xylitol, the engineered yeast cell comprising:

an exogenous polynucleotide sequence encoding a xylitol-phosphate dehydrogenase (XPDH) enzyme comprising a sequence at least 80% identical to at least one of SEQ ID NOs: 12-15, 28-31, and 33.

2. (canceled)

3. The yeast cell of claim 1, wherein the yeast cell is an osmotolerant yeast cell.

4. The yeast cell of claim 1, wherein the yeast cell is a cell of the subphylum Ustilaginomycotina.

5. The yeast cell of claim 1, wherein the yeast cell is selected from the group consisting of *Trichosporonoides megachiliensis*, *Trichosporonoides oedocephalis*, *Trichosporonoides nigrescens*, *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Moniliella*, *Ustilaginomycetes*, *Trichosporon*, *Yarrowia lipolytica*, *Penicillium*, *Torula*, *Pichia*, *Candida*, *Candida magnoliae*, and *Aureobasidium*

6. The yeast cell of claim 1, wherein the yeast cell is a yeast cell of the *Moniliella* genus.

7. (canceled)

8. The yeast cell of claim 1, wherein the cell is a *Moniliella pollinis* cell.

9. The yeast cell of claim 1, wherein the yeast cell is capable of producing xylitol at a titer of at least 20, 30, 50, 75, or 100 g/L when used in a fermentation process in the presence of dextrose at 35° for 96 hours.

10. The yeast cell of claim 1, wherein erythritol production by the yeast cell is reduced relative to erythritol production in an equivalent yeast cell lacking the exogenous polynucleotide sequence.

11. The yeast cell of claim 1, wherein the exogenous polynucleotide sequence is integrated into the genome of the yeast cell at a loci selected from the ER1 locus, the ER3 locus, the PDC1 locus, the pyrF locus, the TRP3 locus, the gpdIIA locus, and the gpdIIB locus.

12. The yeast cell of claim 1, wherein the exogenous polynucleotide sequence is operably linked to a heterologous or artificial promoter.

13. The yeast cell of claim 12, wherein the promoter is a constitutive promoter.

14. The yeast cell of claim 12, wherein the constitutive heterologous or artificial promoter is selected from the group consisting of pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:86), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:130), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO: 132), translational elongation factor 1 promoter (TEFp; SEQ ID NO:133), modified TEFp (SEQ ID NO:131), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:134), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:135), enolase promoter (ENO1p; SEQ ID NO:136), asparagine

synthetase promoter (ASNSp; SEQ ID NO:137), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:138), and RPL16B (SEQ ID NO:139).

15. The yeast cell of claim 1, wherein the XPDH enzyme has a sequence at least 85% identical to at least one of SEQ ID NOs: 14, 15, 28, or 31.

16. The yeast cell of claim 1, wherein the XPDH enzyme has a sequence at least 90% identical to at least one of SEQ ID NOs: 14, 15, 28, or 31.

17. A method for producing xylitol, the method comprising:

contacting a substrate comprising dextrose with the engineered yeast cell of claim 1, wherein fermentation of the substrate by the engineered yeast produces xylitol.

18. (canceled)

19. (canceled)

20. The method of claim 17, wherein the fermentation temperature is at or between 25° C. to 45° C. and the volumetric oxygen uptake rate (OUR) is between 0.5 to 40, mmol O₂/(L· h).

21. The method of claim 17, wherein xylitol is produced at a rate of at least 0.2 g L⁻¹ h⁻¹.

22. The method of claim 17, wherein xylitol production is at least 20 g/L when the fermentation is run at 35° C. for 96 hours.

23. The method of claim 17, wherein erythritol production is reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

24. The method of claim 17, wherein erythritol production is less than 20 g/L when the fermentation is run at 35° C. for 96 hours.

25. (canceled)

26. (canceled)

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